

EE / CE – 211

Basic Electronics

(Spring – 2025)

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**DHANANI
SCHOOL OF SCIENCE
AND ENGINEERING**

Syllabus and Course

- **Lecture Schedule**

- **Wednesday, Friday** – 10:00 am to 11:20 am in classroom C – 200

- **Office Hours & Location**

- **Monday to Thursday** – 4:00 pm to 5:00 pm
- By appointment via email – ahmad.usman@sse.habib.edu.pk
- **Location (Faculty Pod)** – C-116

- **Pre-requisite Courses**

- EE – 112 - Electric Circuit I

- **Assignments – 10%**

- No. of assignments – 3 to 6

- **Quizzes – 15%**

- No. of Quizzes – 6
- In class 15 to 20 minutes quizzes

- **Midterm Exams – 50%**

- No. of Midterm Exams - 2

- **Final Exam – 25%**

- One final Exam

Syllabus and Course

- **Text Book**

- “**Fundamentals of Microelectronics**” by **Behzad Razavi**, 2nd Edition, Wiley Publications (International Student Edition is available in Pakistan)

- **Reference Books**

- Boylestad, Robert L. “**Electronic devices and circuit theory**” Pearson Education India, 2011.
- Sedra, Adel S., Kenneth Carless Smith, Tony Chan Carusone, and Vincent Gaudet. “**Microelectronic circuits**” Oxford University Press, 2016.
- Paynter, Robert T. “**Introductory electronic devices and circuits: conventional flow version**” Prentice-Hall, Inc., 1999.
- Horowitz, Paul, and Winfield Hill. “**The art of electronics**” Cambridge Univ. Press, 1989.
- Neamen, Donald. “**Microelectronic Circuit Analysis and Design**”. McGraw-Hill Science/Engineering/Math, 2006.
- Jaeger, Richard C., and Travis N. Blalock. “**Microelectronic circuit design**”. Fifth Edition. New York: McGraw-Hill, 2015.

Introduction to Microelectronics

- **Electronics Era**

- Almost a century old devices
- Vacuum Tubes as amplifiers (for current and voltage)
- World War II Radars and Radios used these
- Small lifetime, Large, Bulky, Power Inefficient

- **Microelectronics Era**

- First transistor – Invented in 1940s
 - Inventors ?
 - **Hint:** Also won Nobel Prize
- Replaced Vacuum Tubes
- Long lifetime, Small size and volume, Better power efficiency
- Microelectronics Era started after 1960s – Invention of Integrated Circuits (ICs)
- ICs lead to foundation of Microchips

Transistor technology timeline
(summary)

Year	Technology	Organization
1947	Point contact	Bell Labs
1948	Grown junction	Bell Labs
1951	Alloy junction	General Electric
1953	Surface barrier	Philco
1953	JFET	Bell Labs
1954	Diffused base	Bell Labs
1954	Mesa	Bell Labs
1959	Planar	Fairchild
1959	MOSFET	Bell Labs

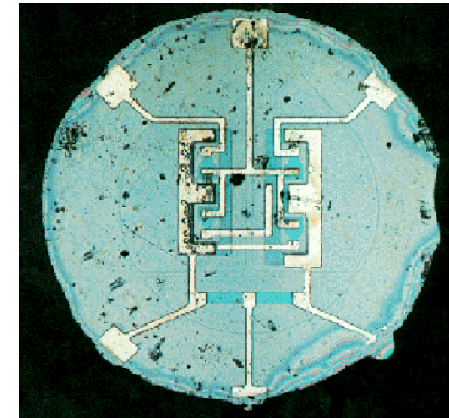


First Transistor demonstration

Introduction to Microelectronics

- **Integrated Circuits (ICs)**

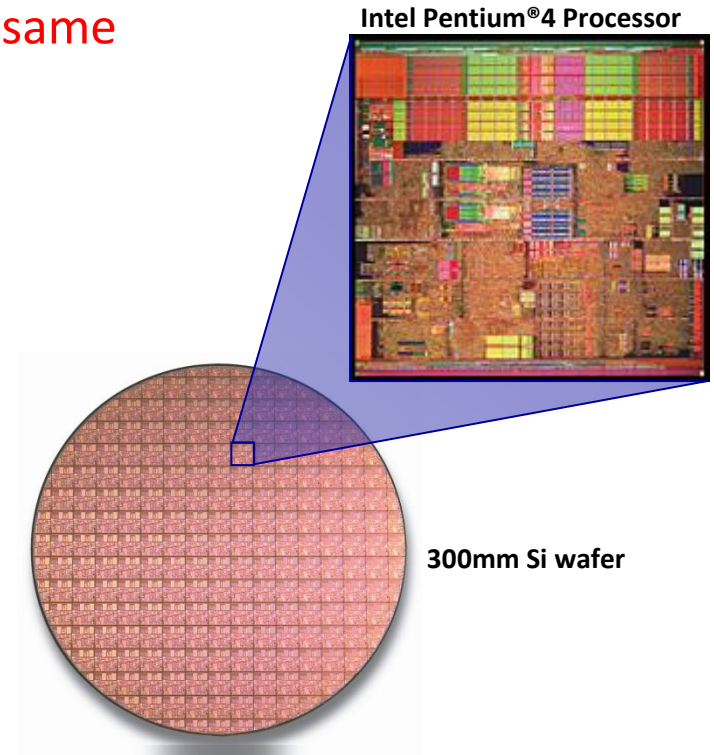
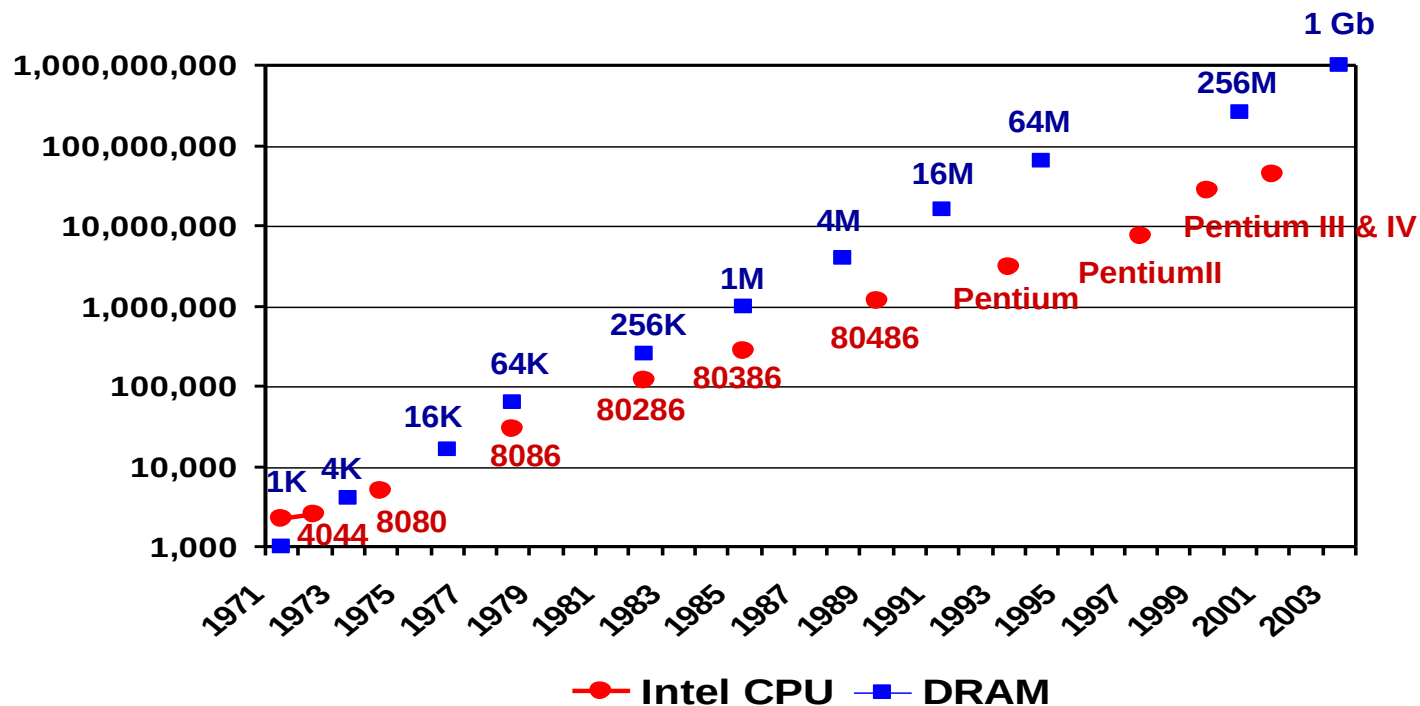
- An IC consists of interconnected electronic components in a single piece (“chip”) of semiconductor material.
- In 1958, Jack S. Kilby (*Texas Instruments*) showed that it was possible to fabricate a simple IC in germanium.
- In 1959, Robert Noyce (*Fairchild Semiconductor*) demonstrated an IC made in silicon using SiO_2 as the insulator and Al for the metallic interconnects.



The first planar IC (actual size: 0.06 in. diameter)

Introduction to Microelectronics

- By connecting a large number of components, each performing simple operations, an IC performing very complex tasks can be built.
- The degree of integration has increased at an exponential pace over the past ~40 years.
 - » The number of devices on a chip doubles every ~18 months, for the same price.

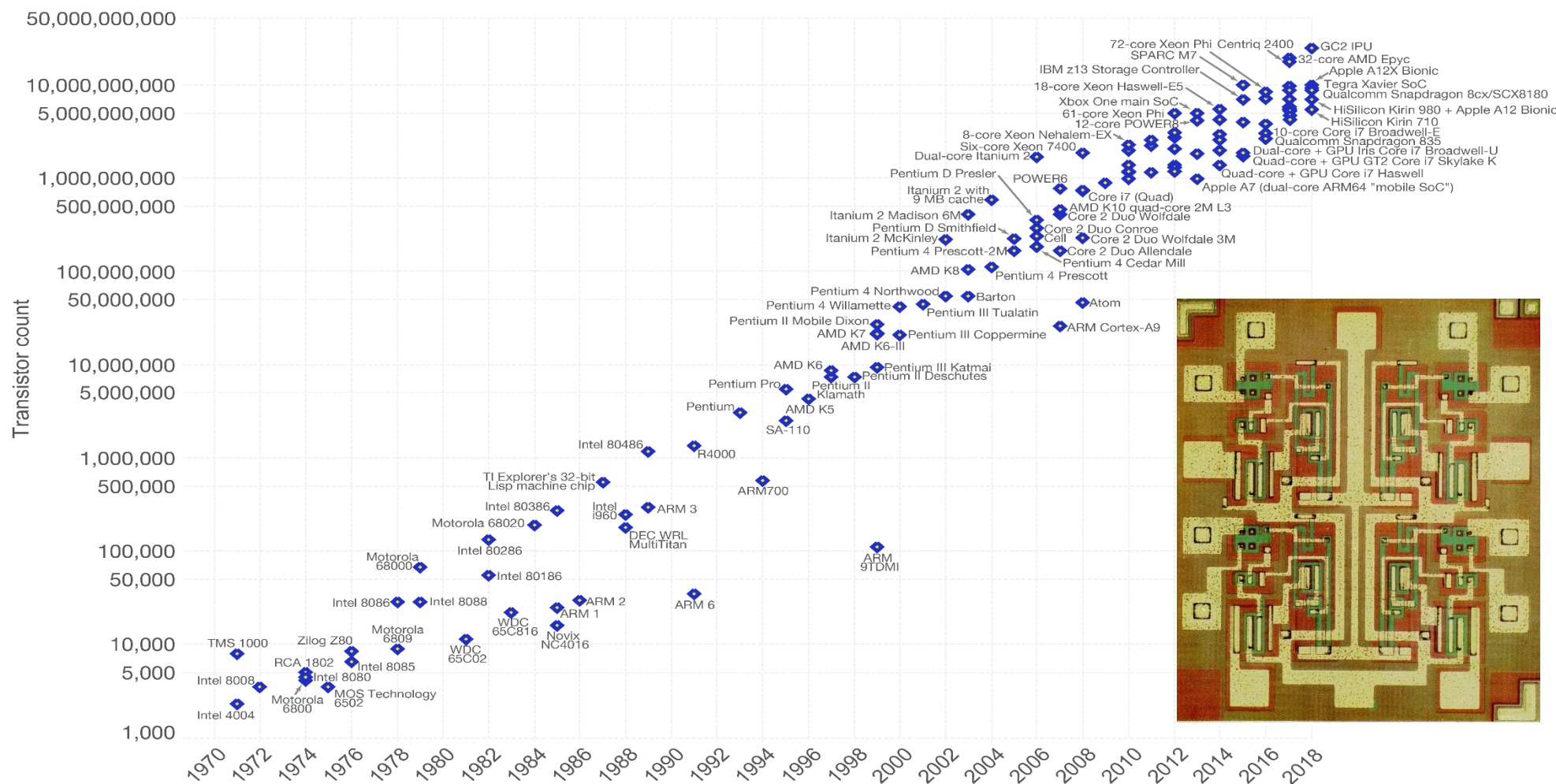


“Moore’s Law” still holds today ???

Introduction to Microelectronics

Moore's Law – The number of transistors on integrated circuit chips (1971-2018)

Moore's law describes the empirical regularity that the number of transistors on integrated circuits doubles approximately every two years. This advancement is important as other aspects of technological progress – such as processing speed or the price of electronic products – are linked to Moore's law.

Our World
in Data

Semiconductor manufacturing processes

10 μm – 1971

6 μm – 1974

3 μm – 1977

1.5 μm – 1981

1 μm – 1984

800 nm – 1987

600 nm – 1990

350 nm – 1994

250 nm – 1996

180 nm – 1999

130 nm – 2001

90 nm – 2003

65 nm – 2005

45 nm – 2007

32 nm – 2009

22 nm – 2012

14 nm – 2014

10 nm – 2016

7 nm – 2018

5 nm – 2019

3 nm – ~2021

Introduction to Microelectronics

- **Moore's law** came to be widely accepted as a goal for the industry, and it was cited by competitive semiconductor manufacturers as they strove to increase processing power.
- Rate of improvement in physical dimensions known as **Dennard scaling** (MOSFET scaling) has slowed for most semiconductor manufacturers in recent years, with much of the industry shifting in 2016 from using semiconductor scaling as a driver, to more of a focus on meeting the needs of major computing applications.
- Since the late 2010s, only two semiconductor manufacturers have been able to produce semiconductor nodes that keep pace with Moore's law, **TSMC** and **Samsung Electronics**, with **10nm**, **7nm** and **5nm** nodes in production (and plans for 3 nm nodes), whereas the pace has slowed down for **Intel** and other semiconductor manufacturers.

Samsung Electronics – Revenue: \$200.6 billion, Net Income: \$ 22.4 billion (2020)

Taiwan Semiconductor Manufacturing Corporation (TSMC) – Revenue: \$47.95 billion, Net Income: \$18.53 (2020)

Apple – Revenue: \$274.51 billion, Net Income: \$57.41 billion (2020)

Intel – Revenue: \$77.87 billion, Net Income: \$20.9 billion (2020)

***Note:** The total Forex reserves of Pakistan are \$18.9 billion (July, 2021)

Introduction to Microelectronics

- Some of the breakthroughs that have advanced integrated circuit and semiconductor device fabrication technology, allowing transistor counts to grow by more than seven orders of magnitude in less than five decades.
- **Integrated circuit (IC)** – The germanium IC was invented by Jack Kilby at Texas Instruments in 1958, followed by the invention of the silicon IC by Robert Noyce at Fairchild Semiconductor in 1959,
- **MOSFET (metal–oxide–semiconductor field-effect transistor)** – Also known as the MOS transistor, it was invented at Bell Labs in 1959. With its high scalability (MOSFET scaling and Dennard scaling), much lower power consumption and higher density than bipolar junction transistors, the MOSFET made it possible to build high-density ICs. Thus, enabling Moore's law. The MOSFET is the most widely manufactured device in history, with an estimated total of 13 sextillion MOS transistors manufactured as of 2018.
- **CMOS (complementary metal-oxide-semiconductor)** – The CMOS process, a form of MOSFET fabrication, was invented by Chih-Tang Sah and Frank Wanlass at Fairchild Semiconductor in 1963. A number of advances in CMOS technology by researchers in the semiconductor field since then have enabled the extremely dense and high-performance ICs that the industry makes today. Sub-micron semiconductor device fabrication was achieved in the mid-1980s, by Bijan Davari at IBM, and by NTT.
- **Dynamic random-access memory (DRAM)** – Bipolar DRAM was developed by Toshiba in 1965, and then MOS DRAM was developed by Robert Dennard at IBM in 1967. DRAM made it possible to fabricate single-transistor memory cells.
- **Microprocessor** – The first commercial single-chip microprocessor, the Intel 4004, released in 1971.
- **Flash memory** – Invented by Fujio Masuoka at Toshiba in 1980, this led to low-cost, high-capacity memory in diverse electronic products. NAND flash memory has since become the most rapidly scaling technology among electronic devices.

Introduction to Microelectronics

- **Chemically-amplified photoresist** – Invented at IBM in 1980, was 5-10 times more sensitive to ultraviolet light. IBM introduced chemically amplified photoresist for DRAM production in the mid-1980s.
- **3D integrated circuit (3D IC)** – Research and development (R&D) on 3D ICs was initiated in 1981 by the Research and Development Association for Future (New) Electron Devices in Japan, and the earliest 3D IC successfully fabricated was by a Fujitsu research team in 1983. 3D ICs allow multiple dies to be stacked vertically, making it possible to extend Moore's law vertically into the third dimension.
- **Multi-gate MOSFET (MuGET) and 3D transistor** – A double-gate MOSFET was first demonstrated by Electrotechnical Laboratory researchers in 1984, leading to the invention of a non-planar double-gate 3D transistor by Hitachi's team in 1989. A popular type of 3D transistor, the FinFET (fin field-effect transistor), was developed by TSMC and UC Berkeley researchers, who achieved 17 nm nanoelectronic fabrication in 1998 and 10 nm fabrication in 2001.
- **Deep UV excimer laser photolithography** – Invented by IBM in 1980 has enabled the smallest features in ICs to shrink from 800 nm in 1990 to as low as 10 nanometers in 2016. Prior to this, excimer lasers had been mainly used as research devices since their development in the 1970s. From a broader scientific perspective, the invention of excimer laser lithography has been highlighted as one of the major milestones in the 50-year history of the LASERS.
- **Interconnect innovations** – Interconnect innovations of the late 1990s, including chemical-mechanical polishing (CMP) or chemical mechanical planarization (CMP), trench isolation, and copper interconnects—although not directly a factor in creating smaller transistors—have enabled improved wafer yield, additional layers of metal wires, closer spacing of devices, and lower electrical resistance.
- **Intel was reported in 2005 as stating that the downsizing of silicon chips with good economics could continue during the following decade, and in 2008 as predicting the trend through 2029.**

Introduction to Microelectronics

Recent Advancements

Challenge - One of the key challenges of engineering future nanoscale transistors is the design of gates. As device dimension shrinks, controlling the current flow in the thin channel becomes more difficult. Compared to FinFETs, which have gate dielectric on three sides of the channel, gate-all-around MOSFET (GAAFET) structure has even better gate control.

- A **gate-all-around MOSFET** was first demonstrated in 1988, by a Toshiba research team, who demonstrated a vertical nanowire GAAFET which he called a "surrounding gate transistor" (SGT).
- In 2006, a team of Korean researchers from the Korea Advanced Institute of Science and Technology (KAIST) and the National Nano Fab Center developed a **3 nm transistor**, the world's smallest nanoelectronic device, based on FinFET technology.
- In 2010, researchers at the Tyndall National Institute in Cork, Ireland announced a **junctionless transistor**. A control gate wrapped around a silicon nanowire can control the passage of electrons without the use of junctions or doping. They claim these may be produced at 10-nanometer scale using existing fabrication techniques.
- In 2011, researchers at the University of Pittsburgh announced the development of a single-electron transistor, 1.5 nanometers in diameter, made out of oxide based materials. Three "wires" converge on a central "island" that can house one or two electrons. Electrons tunnel from one wire to another through the island. Conditions on the third wire result in distinct conductive properties including the ability of the transistor to act as a solid state memory. Nanowire transistors could spur the creation of microscopic computers.
- In 2012, a research team at the University of New South Wales announced the development of the first working **transistor consisting of a single atom placed precisely in a silicon crystal** (not just picked from a large sample of random transistors). Moore's law predicted this milestone to be reached for ICs in the lab by 2020.
- In 2015, **IBM demonstrated 7 nm node chips with silicon-germanium** transistors produced using Extreme UV Lithography. The company believes this transistor density would be four times that of current 14 nm chips.

Introduction to Microelectronics

- In 2018, Samsung Electronics **began commercial production of a 5 nm process**, using FinFET and EUV technology.
- In early 2019, TSMC began commercial production of their own **5 nm process**.
- Samsung and TSMC have presented plans to **manufacture 3 nm GAAFET** nodes by 2021–2022.
- New technology advances may help sustain Moore's law through improved performance, with or without reduced feature size.
- A Toshiba research developed a "**System Block Module**" wafer bonding process for manufacturing 3D integrated circuit (3D IC) packages in 2001. In April 2007, Toshiba introduced an eight-layer 3D IC, the 16 GB THGAM embedded NAND flash memory chip which was manufactured with eight stacked 2 GB NAND flash chips.
- In September 2007, Hynix introduced **24-layer 3D IC**, a 16 GB flash memory chip that was manufactured with 24 stacked NAND flash chips using a wafer bonding process.
- V-NAND, also known as 3D NAND, allows flash memory cells to be stacked vertically using charge trap flash technology, significantly increasing the number of transistors on a flash memory chip. 3D NAND was first announced by Toshiba in 2007. V-NAND was first commercially manufactured by Samsung Electronics in 2013.
- In 2008, researchers at HP Labs announced a **working memristor**, a fourth basic passive circuit element whose existence only had been theorized previously. **The memristor's unique properties permit the creation of smaller and better-performing electronic devices.**
- In 2014, bioengineers at Stanford University developed a circuit modeled on the human brain. Sixteen "Neurocore" chips simulate one million neurons and billions of synaptic connections, claimed to be 9,000 times faster as well as more energy efficient than a typical PC.
- In 2015, Intel and Micron announced 3D XPoint, a non-volatile memory claimed to be significantly faster with similar density compared to NAND. Production scheduled to begin in 2016 was delayed until the second half of 2017.

Introduction to Microelectronics

- In 2017, Samsung combined its V-NAND technology with eUFS 3D IC stacking to produce a 512 GB flash memory chip, with eight stacked 64-layer V-NAND dies.
- In 2019, Samsung produced a 1 TB flash chip with eight stacked 96-layer V-NAND dies, along with quad-level cell (QLC) technology (4-bit per transistor), equivalent to 2 trillion transistors, the highest transistor count of any IC chip.
- While physical limits to transistor scaling such as source-to-drain leakage, limited gate metals, and limited options for channel material have been reached, new avenues for continued scaling are open.
 - The most promising of these approaches rely on using the **spin state of electron i.e., spintronics, tunnel junctions, and advanced confinement of channel materials** via nano-wire geometry.
- A comprehensive list of available device choices shows that a wide range of device options is open for continuing Moore's law into the next few decades.
 - Spin-based logic and memory options are being developed actively in industrial labs, as well as academic labs.

Introduction to Microelectronics

→ Today's microprocessors contain about 100 million transistors in a chip area of approximately $3\text{ cm} \times 3\text{ cm}$. (The chip is a few hundred microns thick.) Suppose integrated circuits were not invented and we attempted to build a processor using 100 million "discrete" transistors. If each device occupies a volume of $3\text{ mm} \times 3\text{ mm} \times 3\text{ mm}$, determine the minimum volume for the processor. What other issues would arise in such an implementation?

→ How much power would such a system consume if each transistor dissipates $10\text{ }\mu\text{W}$?

Answer

Introduction to Microelectronics

- Some examples of Basic Electronic Systems
 - **Mobile/Cellular Telephone**

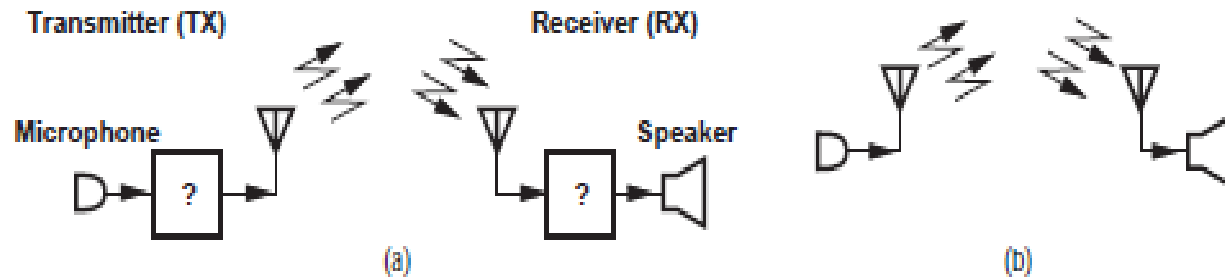


Figure 1.1 (a) Simplified view of a cellphone, (b) further simplification of transmit and receive paths.

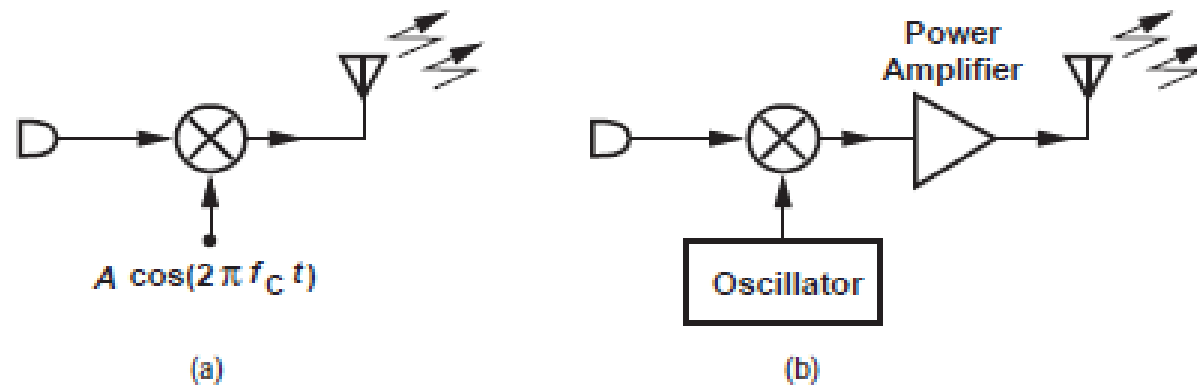


Figure 1.3 (a) Simple transmitter, (b) more complete transmitter.

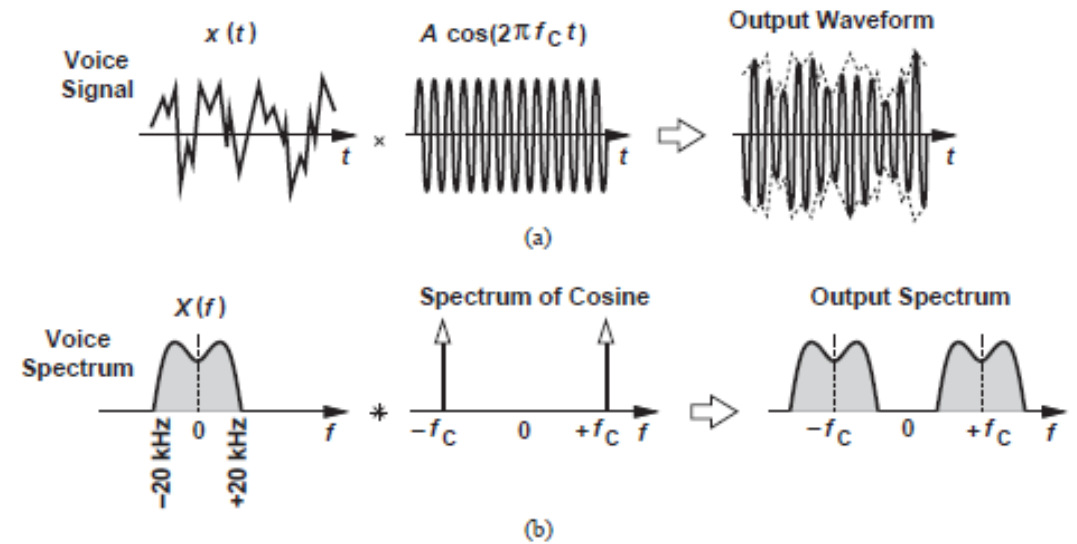


Figure 1.2 (a) Multiplication of a voice signal by a sinusoid, (b) equivalent operation in the frequency domain.

Introduction to Microelectronics

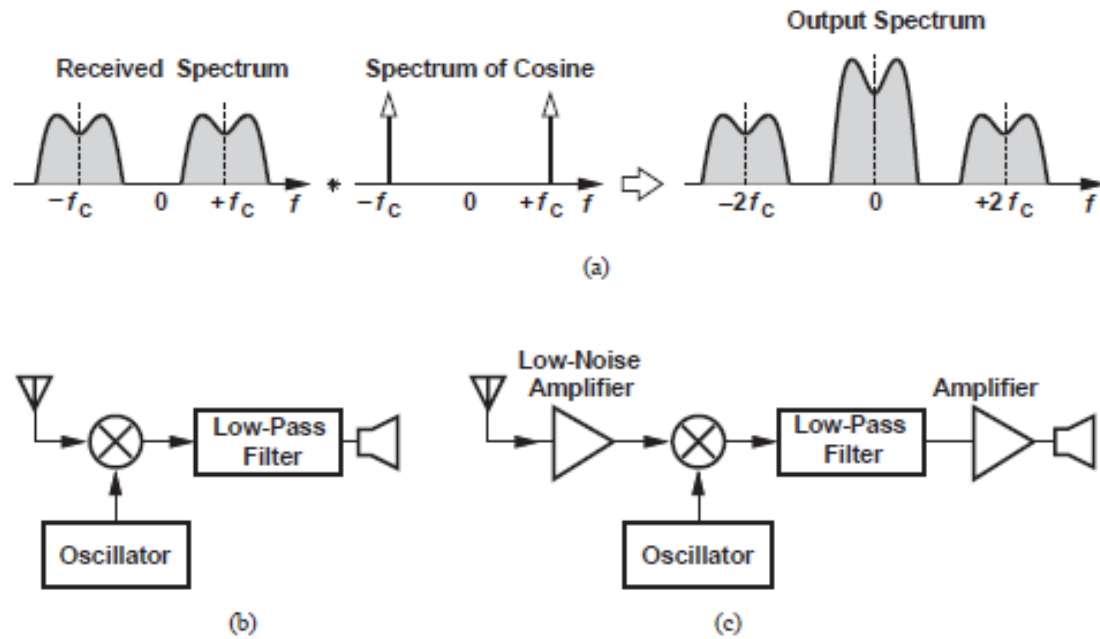
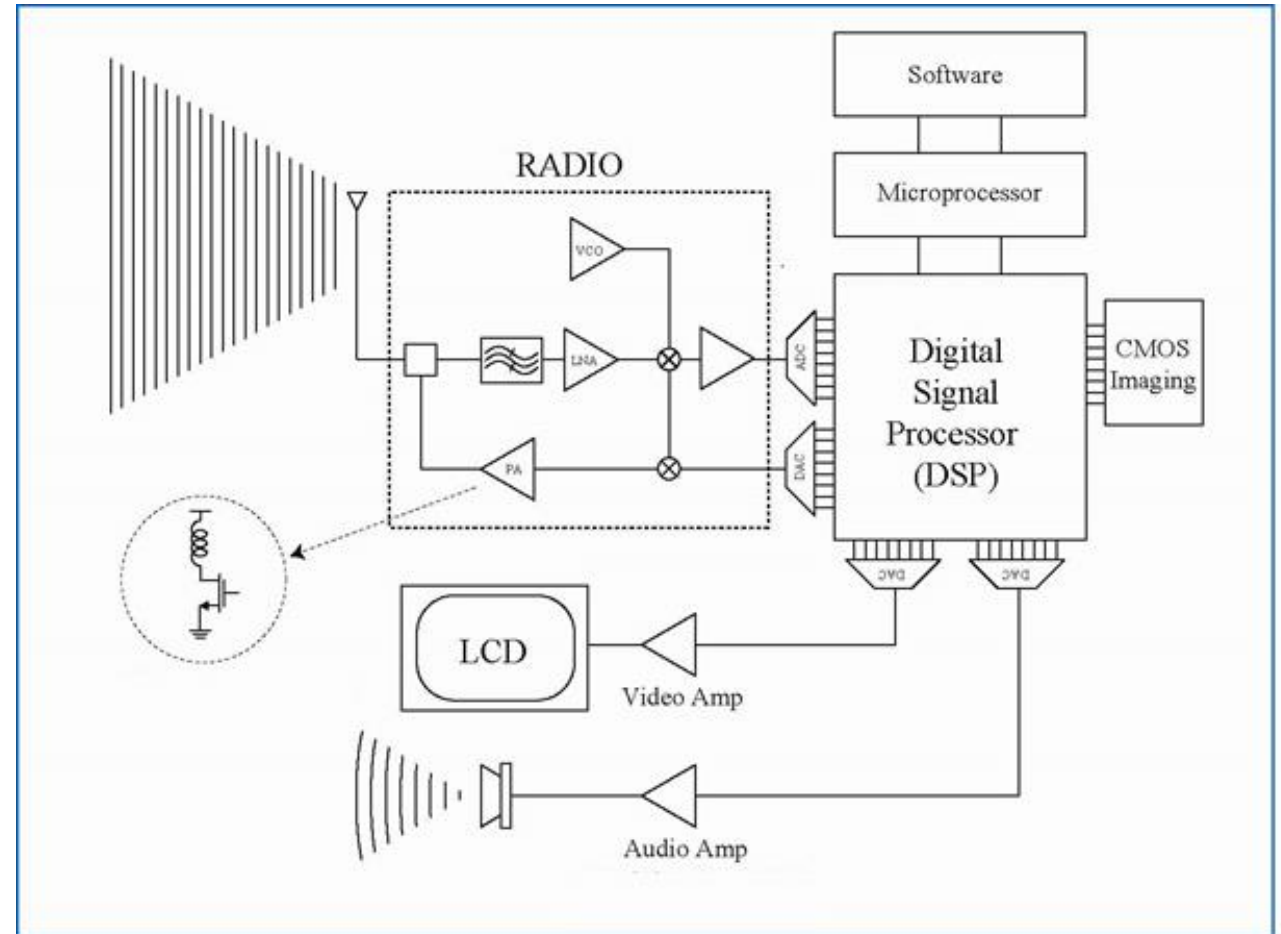


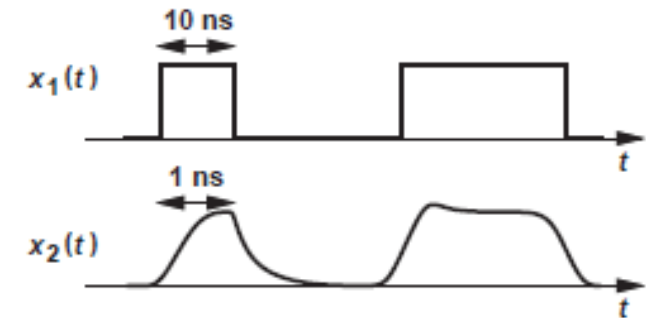
Figure 1.4 (a) Translation of modulated signal to zero center frequency, (b) simple receiver, (c) more complete receiver.

System Level Block Diagram



Introduction to Microelectronics

- Why are amplifiers used ?
- Why are oscillators used ?
- Why are multipliers used ?
- Can you apply a 20uV signal received by an antenna to a digital gate/switch/transistor ?
- **Analog to Digital Conversion**
 - A key component for the modern electronic systems
 - Limited by response (rise and fall) time at high speeds
 - Not every function (signal) can be digitized



Data waveforms at 100 Mb/s and 1 Gb/s.

How your cell phone camera converts light to images ? Think in terms of electronic components involved.

Basic Physics of Semiconductors

□ Semiconductor Materials and Properties

1. Identify charge carriers in solids and formulate their role in current flow
2. Examine means of modifying the density of charge carriers to create desired current flow properties
3. Determine current flow mechanisms

Basic Physics of Semiconductors

□ Charge Carriers in Solids

- Let us go back to our high school Chemistry / Physics course and try to remember about “**Valence Electrons**” or “**Electrons in the outer most shell**” in an atom

Basic Physics of Semiconductors

□ Charge Carriers in Solids

□ Let us go back to our high school Chemistry / Physics course and try to remember about “**Valence Electrons**” or “**Electrons in the outer most shell**” in an atom

□ How many electrons are there in the outer most shell (or valence electrons) of the following atoms

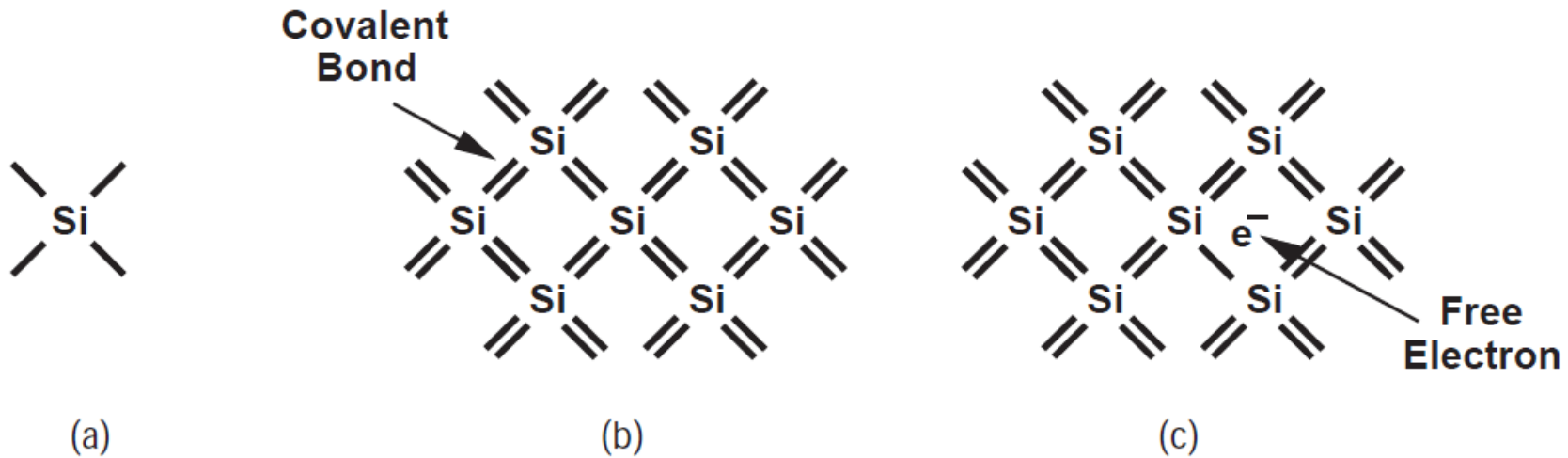
1. Hydrogen
2. Sodium
3. Neon
4. **Silicon**

	III	IV	V	
	Boron (B)	Carbon (C)		
• • •	Aluminum (Al)	Silicon (Si)	Phosphorus (P)	• • •
	Galium (Ga)	Germanium (Ge)	Arsenic (As)	
		• • •		

Basic Physics of Semiconductors

❑ Charge Carriers in Solids

- ❑ Silicon atom has a crystalline structure
- ❑ Silicon undergoes covalent bonding



Basic Physics of Semiconductors

□ Charge Carriers in Solids

Q: What are the valence electrons doing at room temperature ?

Q: What happens to the valence electrons as we decrease the temperature to 0K (absolute zero) ?

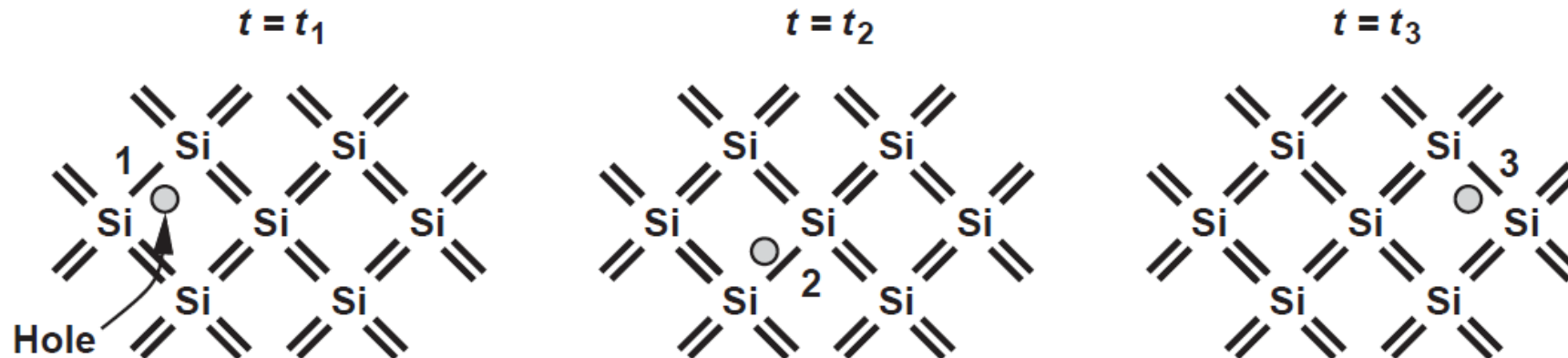
Q: What happens to the valence electrons as we increase the temperature ?

Basic Physics of Semiconductors

□ Charge Carriers in Solids

□ Holes

- When freed from a covalent bond, an electron leaves a “void” behind because the bond is now incomplete
- This void is called a “hole”
- This void can readily absorb a free electron if one becomes available
- An “**electron-hole pair generation**” occurs when an electron is freed from a covalent bond
- An “**electron-hole recombination**” occurs when an electron “falls” into a hole
- Movement of electrons (or holes) in a crystal is shown below



Basic Physics of Semiconductors

❑ Charge Carriers in Solids

❑ Bandgap Energy (E_g)

- ❑ A minimum energy is required to dislodge an electron from a covalent bond
- ❑ This energy is known as the “bandgap energy” and denoted by E_g
- ❑ This minimum energy is a fundamental property of the material

- ❑ The concept of bandgap energy is related to “Quantization”
 - ❑ Remember that the Angular Momentum of an electron is quantized

- ❑ 1eV is the unit of Energy
- ❑ $1\text{eV} = 1.6 \times 10^{-19}$ Joules

$$n_i = 5.2 \times 10^{15} T^{3/2} \exp \frac{-E_g}{2kT} \text{ electrons/cm}^3$$

Example
2.1

Determine the density of electrons in silicon at $T = 300$ K (room temperature) and $T = 600$ K.

Basic Physics of Semiconductors

□ Charge Carriers in Solids

□ Bandgap Energy (E_g)

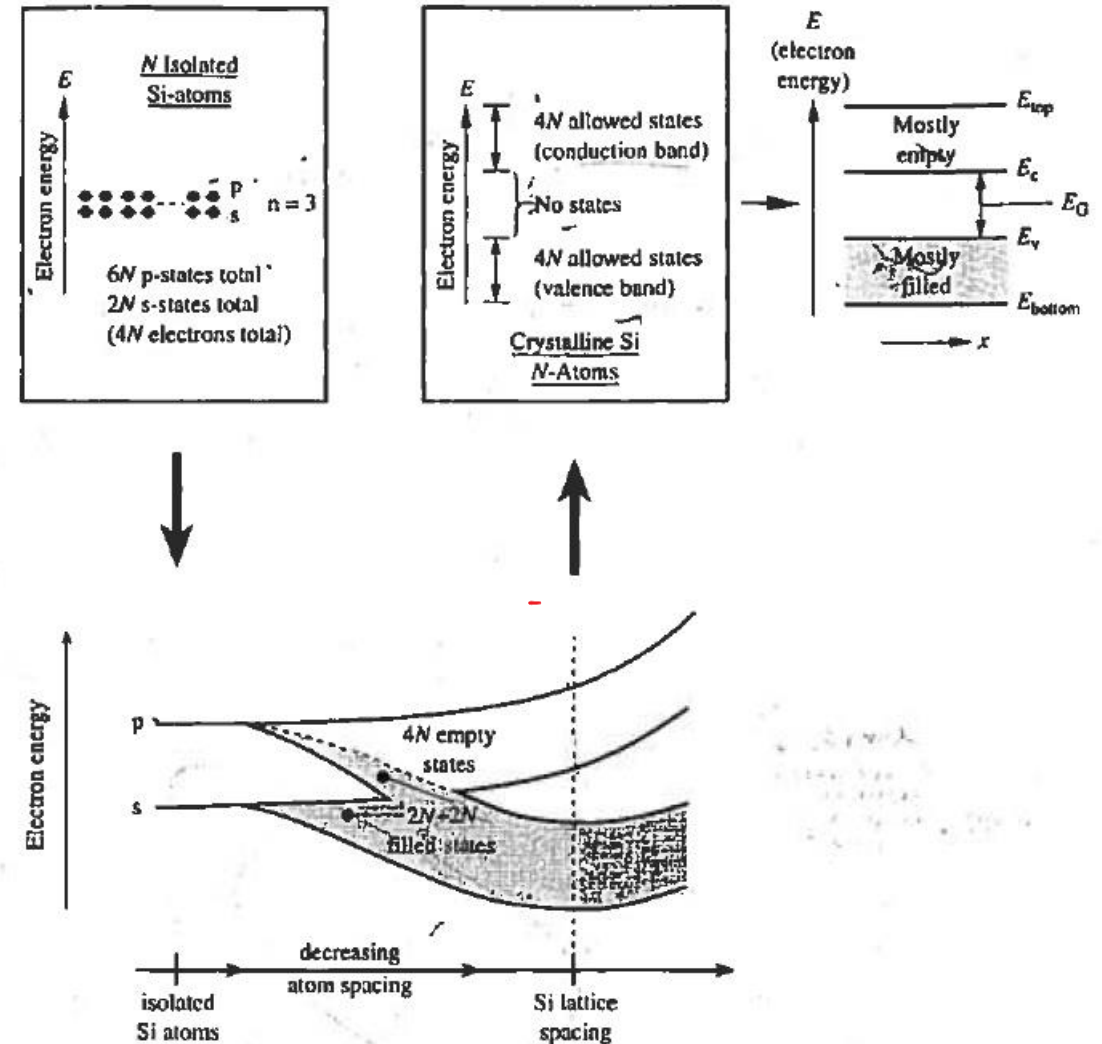


Figure 2.5 Conceptual development of the energy band model starting with N isolated Si atoms on the top left and concluding with a "dressed-up" version of the energy band model on the top right.

- **Reading for this Lecture**

- Chapter # 1 of the Semiconductor Device Fundamentals by Robert F. Pierret

Basic Physics of Semiconductors

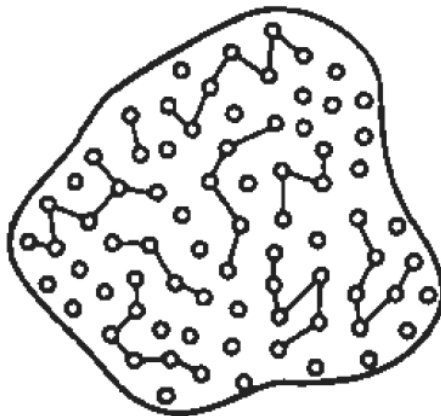
□ Classifications of Solids

□ A general degree of classification of solids is based upon the degree of atomic order. The types of solids as per this classification are

□ Amorphous

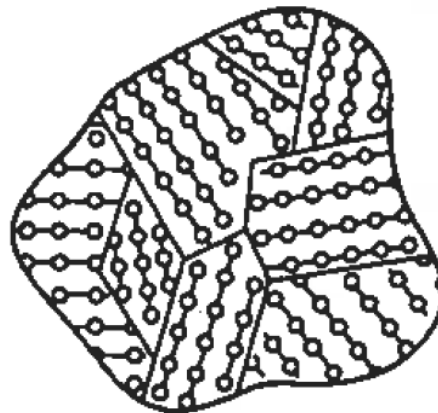
□ Polycrystalline

□ Crystalline



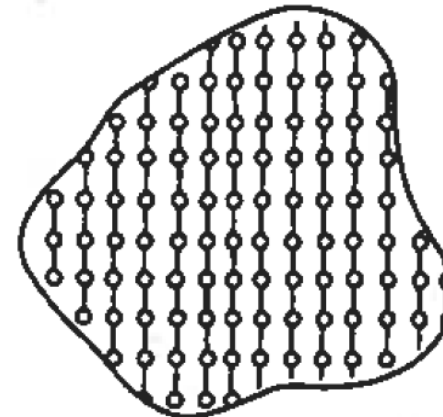
(a) Amorphous

No recognizable
long-range order



(b) Polycrystalline

Completely ordered
in segments



(c) Crystalline

Entire solid is made up of
atoms in an orderly array

Basic Physics of Semiconductors

❑ Classifications of Solids

- ❑ Amorphous Silicon thin film transistors are used in Liquid Crystal Displays (LCDs)
- ❑ Polycrystalline Silicon (Poly-Si) is used for fabricating the “gate region” in Metal Oxide Semiconductor Field Effect Transistors (MOSFETs)
- ❑ A wide variety of electronic devices have their “active region” fabricated in crystalline silicon (or other crystalline materials)

Basic Physics of Semiconductors

☐ Visualizing Crystalline Structures

- ☐ We will examine how one goes about describing the spatial positioning of atoms within crystals
- ☐ We will visualize 3D lattices (atomic arrangements), followed by examining the semiconductor lattices (primarily Silicon)
- ☐ We will talk about Miller Indices
 - ☐ Miller Indices are a widely used notation for identification of specific planes and directions within crystals

Basic Physics of Semiconductors

□ Unit Cell

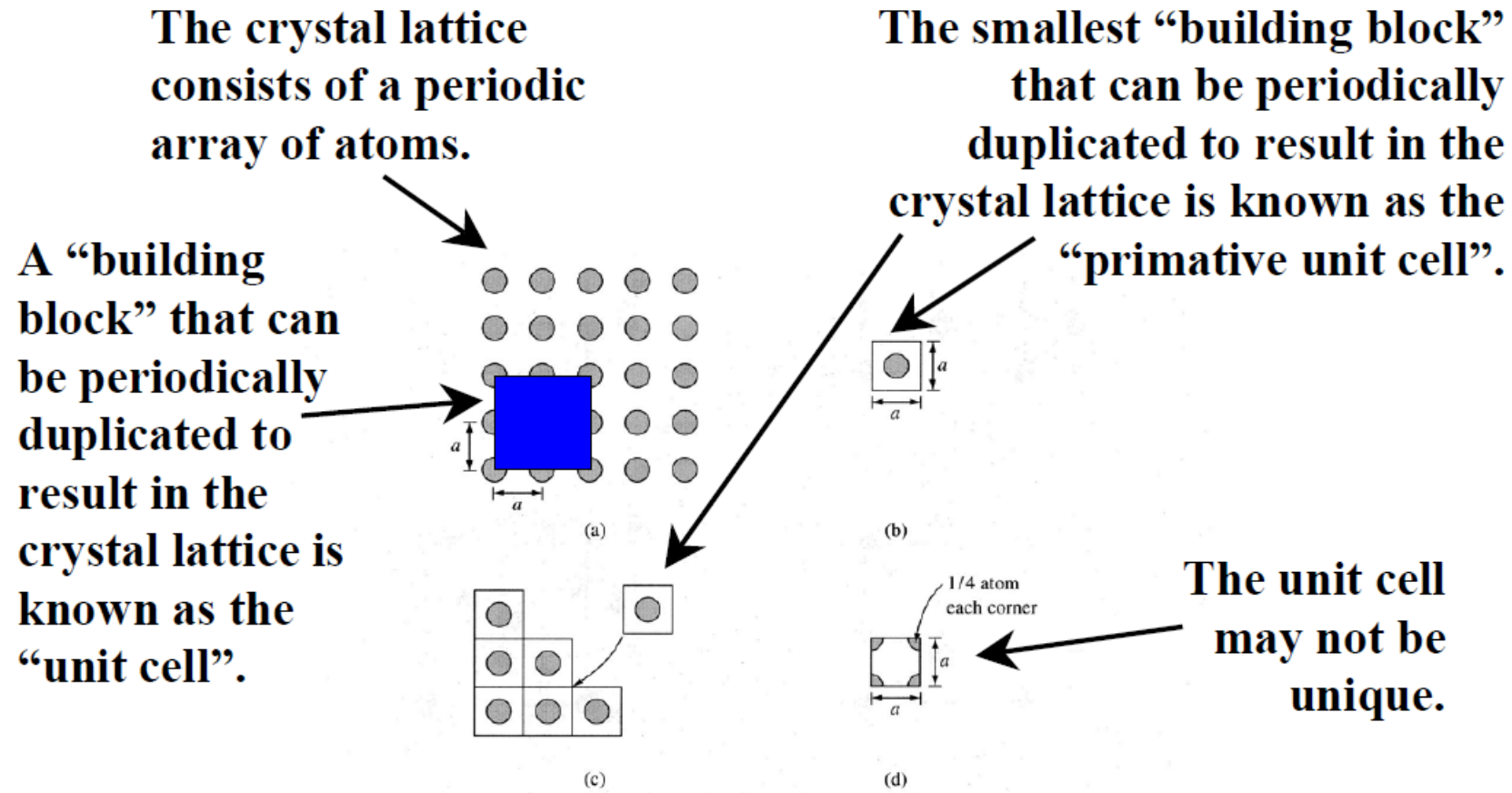
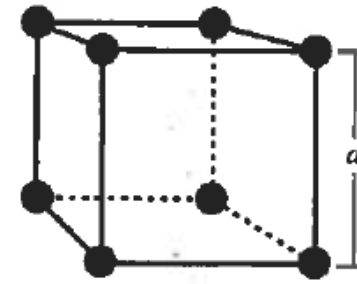


Figure 1.2 Introduction to the unit cell method of describing atomic arrangements within crystals. (a) Sample two-dimensional lattice. (b) Unit cell corresponding to the part (a) lattice. (c) Reproduction of the original lattice. (d) An alternative unit cell.

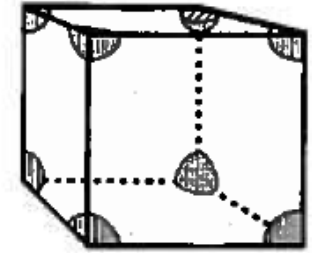
Basic Physics of Semiconductors

□ Simple 3D Unit Cells

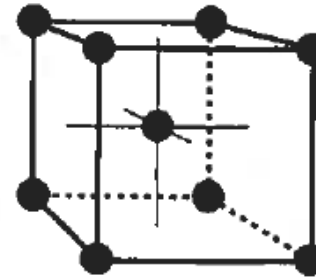
- Semiconductor crystals are 3D structures and therefore defined using 3D unit cells
- Figure 1.3 shows some of the 3D units cells used to define semiconductor materials.
- Figure 1.3 (a) shows the simplest of 3D unit cells i.e., “Cubic unit cell”
- The simple cubic cell is an equal-sided cube with an atom positioned at each corner of the cube
- The simple cubic lattice associated with the thin cell is constructed in a manner paralleling the two-dimensional case



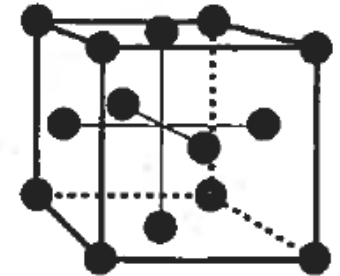
(a) Simple cubic



(b) Pedantically correct simple cubic



(c) bcc



(d) fcc

Figure 1.3 Simple three-dimensional unit cells. (a) Simple cubic unit cell. (b) Pedantically correct simple cubic unit cell including only the fractional portion ($1/8$) of each corner atom actually within the cell cube. (c) Body centered cubic unit cell. (d) Face centered cubic unit cell.

Basic Physics of Semiconductors

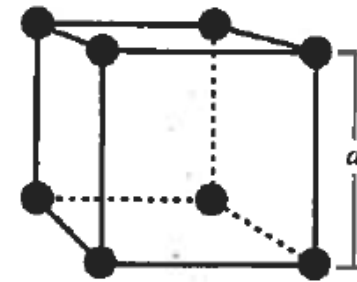
□ Simple 3D Unit Cells

□ Planes of atoms parallel to the base-plane are separated from one another by a unit cell length or “lattice constant, a ”

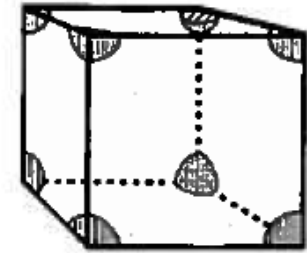
□ When viewed normal to the base plane, the atoms in a given place are positioned directly over the atoms in the underlying plane

*Question:

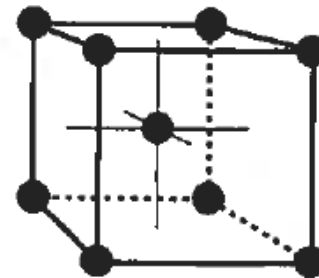
How many atoms are there across each face in a simple cubic unit cell, fcc unit cell, and a bcc unit cell ?



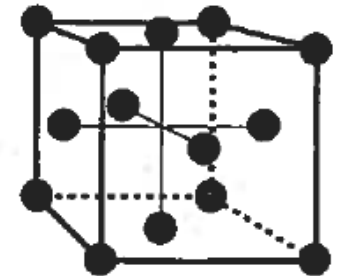
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Basic Physics of Semiconductors

□ Lattice Constant

Lattice Constant: A length that describes the unit cell. It is normally given in Å, angstroms = $1\text{e-}10$ meters.

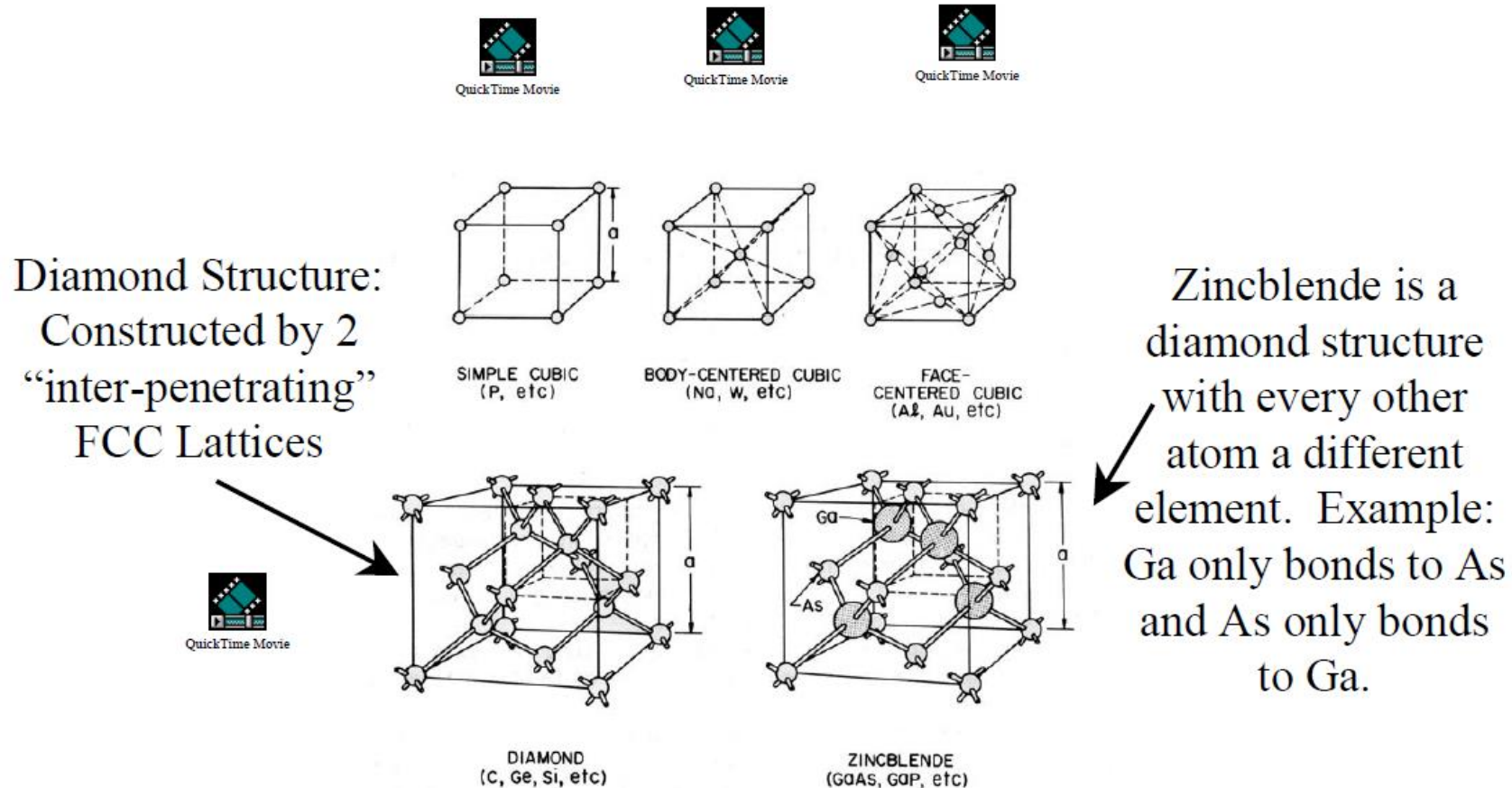


Fig. 1 Some important unit cells (direct lattices) and their representative elements or compounds; a is the lattice constant.

Basic Physics of Semiconductors

□ Unit Cell Concept

□ Silicon at room temperature has a unit cell length (lattice constant, a) of $5.43 \text{ \AA}$

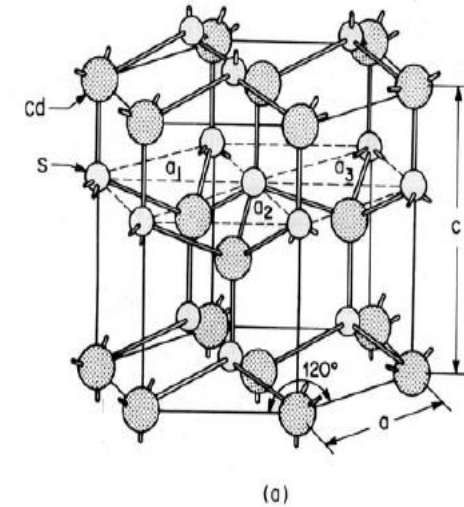
□ There are 8 atoms per unit volume of a cubic cell with the unit cell volume of $a^3 \text{ cm}^3$

□ The number of Silicon atoms per unit volume are _____

Remember

- $\text{\AA} = 10^{-10} \text{ m (or } 10^{-8} \text{ cm)}$

Some unit cells have hexagonal symmetry.



Rocksalt unit cells are one of the simplest practical unit cells.

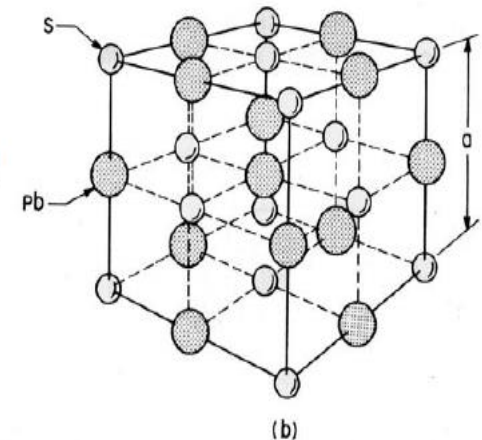
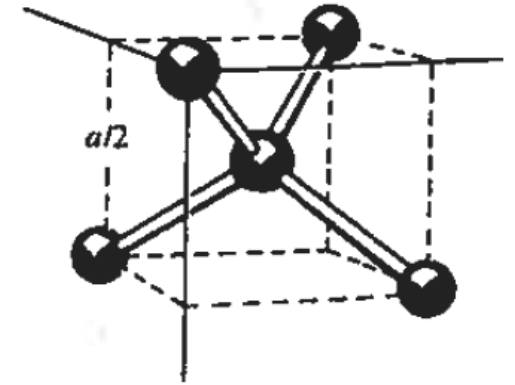


Fig. 2 Two unit cells of compound semiconductors. (a) Wurtzite lattice (CdS , ZnS , etc.). (b) Rock-salt lattice (PbS , PbTe , etc.).

Basic Physics of Semiconductors

□ Unit Cell Concept

- Silicon at room temperature has a unit cell length (lattice constant, a) of $5.43 \text{ \AA}$. The unit cell structure is shown in the figure.
- What is the distance (d) between the center of the Silicon atom to the center of its nearest neighbor?



- The distance (d) is _____

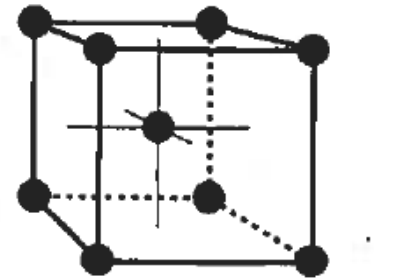
Remember

- $\text{\AA} = 10^{-10} \text{ m (or } 10^{-8} \text{ cm)}$

Basic Physics of Semiconductors

□ Unit Cell Concept

- Silicon at room temperature has a unit cell length (lattice constant, a) of $5.43 \text{ \AA}$. The unit cell structure is shown in the figure.
- What is the atomic density for Silicon atoms if arranged in a BCC lattice structure ?



- The atomic density is _____ atoms/cm³

Remember

- $\text{\AA} = 10^{-10} \text{ m (or } 10^{-8} \text{ cm)}$

Basic Physics of Semiconductors

□ Miller Indices

- The specification of crystallographic planes is of practical importance
 - Crystalline surface of silicon wafers plays a key role in device fabrication process
 - The surface of wafers are specifically oriented to lie across the desired crystalline plane
- Miller indices are the conventional means of identifying the planes and directions within a crystal

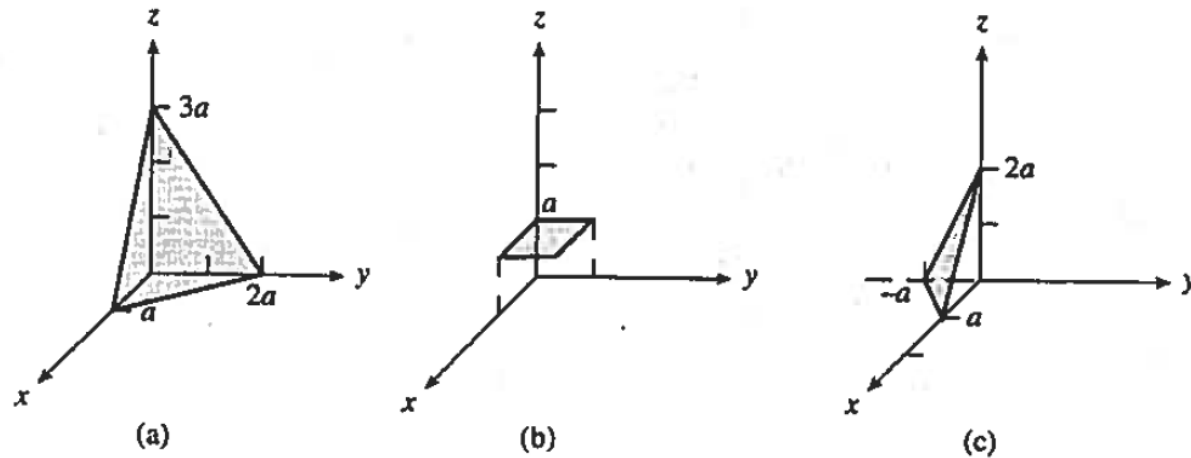


Figure 1.6 Sample cubic crystal planes. (a) The (632) plane used in explaining the Miller indexing procedure. (b) The (001) plane. (c) The $(\bar{2}21)$ plane.

Basic Physics of Semiconductors

□ Miller Indices

- The Miller indices for any given plane of atoms within a crystal are obtained using the below mentioned procedure

Indexing Procedure	Sample Implementation
After setting up coordinate axes along the edges of the unit cell, note where the plane to be indexed intercepts the axes. Divide each intercept value by the unit cell length along the respective coordinate axis. Record the resulting normalized (pure-number) intercept set in the order x, y, z .	1, 2, 3
Invert the intercept values; that is, form $[1/\text{intercept}]$ s.	1, $\frac{1}{2}$, $\frac{1}{3}$
Using an appropriate multiplier, convert the $1/\text{intercept}$ set to the smallest possible set of whole numbers.	6, 3, 2
Enclose the whole-number set in curvilinear brackets.	(632)

Basic Physics of Semiconductors

□ Miller Indices

To complete the description of the plane-indexing procedure, the user should also be aware of the following special facts:

- (i) If the plane to be indexed is parallel to a coordinate axis, the “intercept” along the axis is taken to be at infinity. Thus, for example, the plane shown in Fig. 1.6(b) intercepts the coordinate axes at ∞ , ∞ , 1, and is therefore a (001) plane.
- (ii) If the plane to be indexed has an intercept along the negative portion of a coordinate axis, a minus sign is placed *over* the corresponding index number. Thus the Fig. 1.6(c) plane is designated a ($\bar{2}\bar{2}$ 1) plane.
- (iii) Referring to the diamond lattice of Fig. 1.4(a), note that the six planes passing through the cube faces contain identical atom arrangements; that is, because of crystal symmetry, it is impossible to distinguish between the “equivalent” (100), (010), (001), ($\bar{1}$ 00), (0 $\bar{1}$ 0), and (00 $\bar{1}$) planes; or, it is impossible to distinguish between [100] planes. A group of equivalent planes is concisely referenced in the Miller notation through the use of { } braces.
- (iv) Miller indices cannot be established for a plane passing through the origin of coordinates. The origin of coordinates must be moved to a lattice point not on the plane to be indexed. This procedure is acceptable because of the equivalent nature of parallel planes.

Basic Physics of Semiconductors

□ Miller Indices

□ Usually encountered planes are (111), (110), [001]

□ For cubic crystals, a plane and the direction normal to plane have precisely the same indices

□ Example – [110] direction is normal to the (110) plane

Table 1.3 Miller Convention Summary.

<i>Convention</i>	<i>Interpretation</i>
(hkl)	Crystal plane
$\{hkl\}$	Equivalent planes
$[hkl]$	Crystal direction
$\langle hkl \rangle$	Equivalent directions

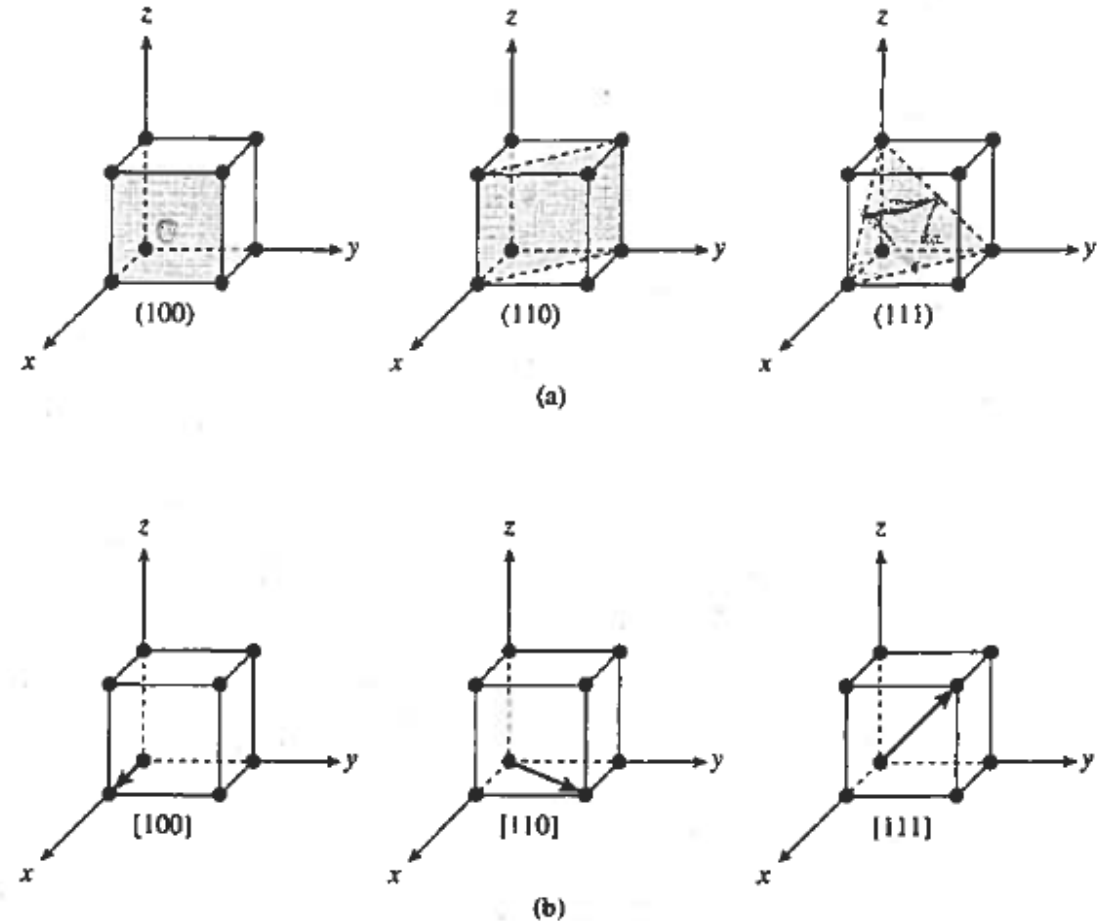
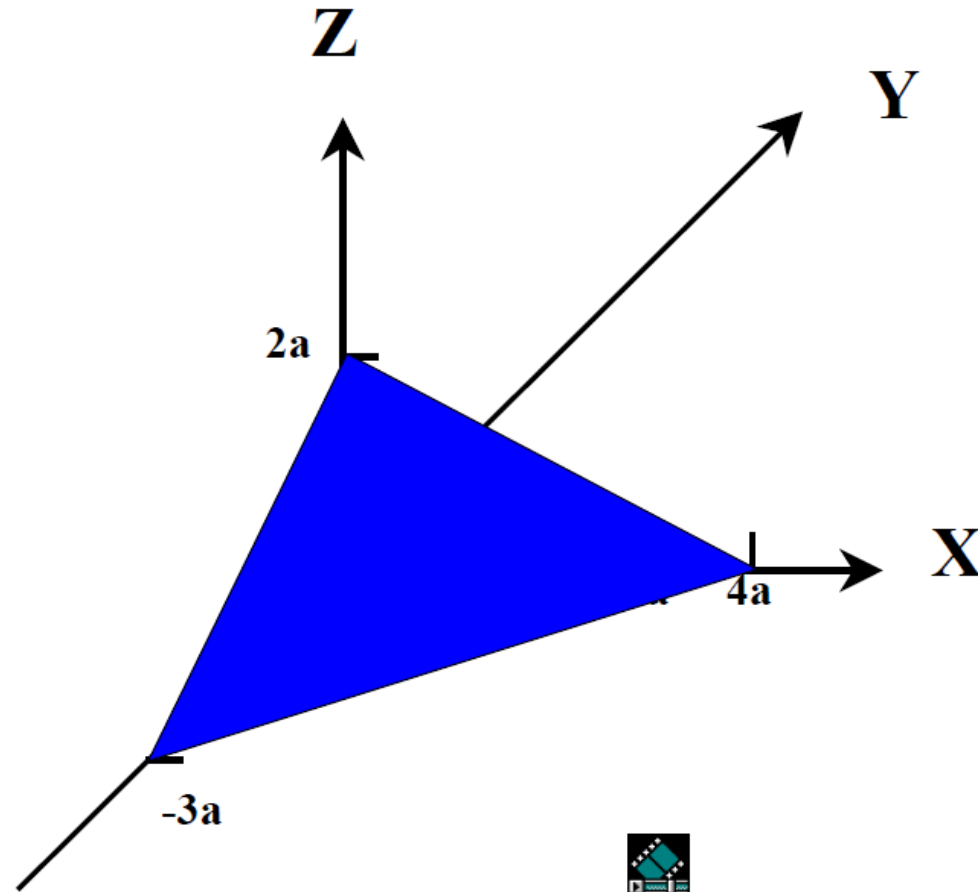


Figure 1.7 Visualization and Miller indices of commonly encountered (a) crystalline planes and (b) direction vectors.

Basic Physics of Semiconductors

□ Miller Indices



Identify Intercepts in x,y,z
order = $4a, -3a, 2a$

Divide by unit cell length in each
direction x,y,z order = $4, -3, 2$

Invert the values = $1/4, -1/3, 1/2$

Multiply by a number (12 in this
example) to give smallest whole
number set = $3, -4, 6$

Place any minus signs over their
index and place set in parenthesis
= $(\bar{3}46)$

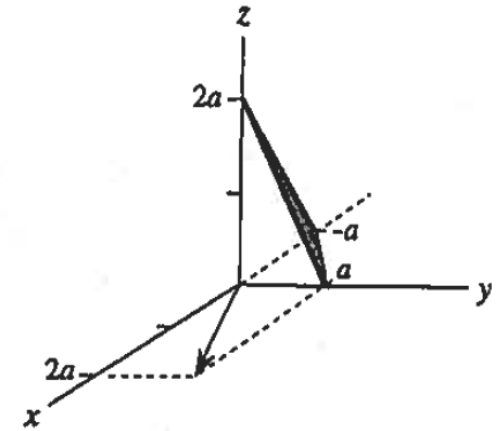
Basic Physics of Semiconductors

□ Miller Indices

Question

For cubic crystal lattice,

- (a) Determine the indices for the plane and direction vector shown in the figure
- (b) Sketch the plane and direction vector characterized by (011) and [011], respectively



- **Reading for this Lecture**

- Chapter # 2 of the Semiconductor Device Fundamentals by Robert F. Pierret
- Chapter # 2 of the Introduction to Microelectronics by Behzad Razavi

Charge Carriers

□ Hydrogen atom

$$E_H = -\frac{m_0 q^4}{2(4\pi\epsilon_0 \hbar n)^2} = -\frac{13.6}{n^2} \text{ eV}, \quad n = 1, 2, 3, \dots$$

E_H = Binding Energy of electrons within Hydrogen atom

m_0 = mass of free electron

q = charge on electrons

$\hbar$ = planks constant

n = energy quantum number (or orbit identifier)

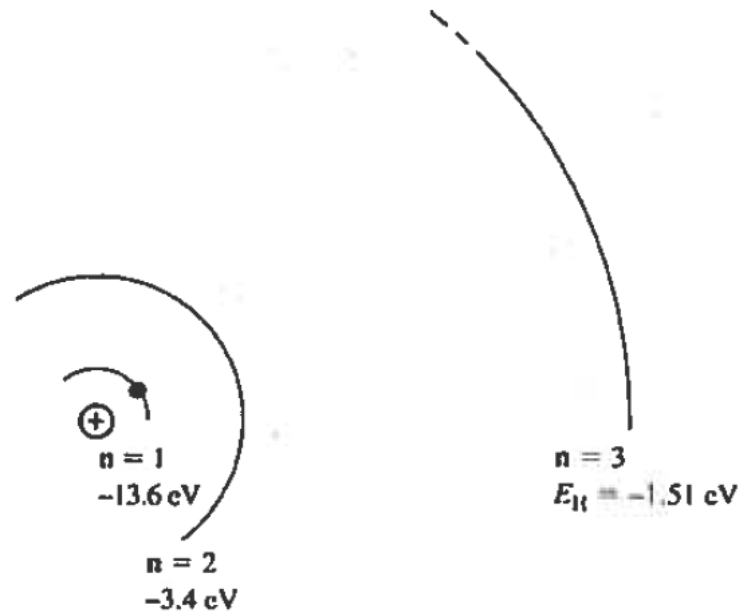


Figure 2.1 The hydrogen atom—idealized representation showing the first three allowed electron orbits and the associated energy quantization.

Charge Carriers

□ Silicon atom

- The binding of 10 electrons closest to the nucleus is so strong that they remain unperturbed during chemical reactions
- The remaining 4 electrons are weakly bound to the nucleus and readily participate in chemical reactions
- These four electrons are known as valence electrons

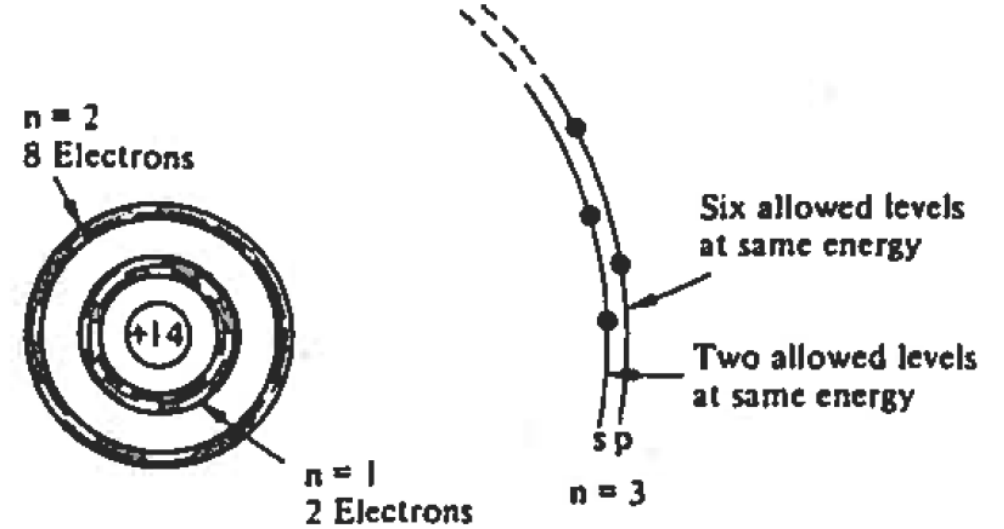


Figure 2.2 Schematic representation of an isolated Si atom.

Charge Carriers

□ Energy Band Model

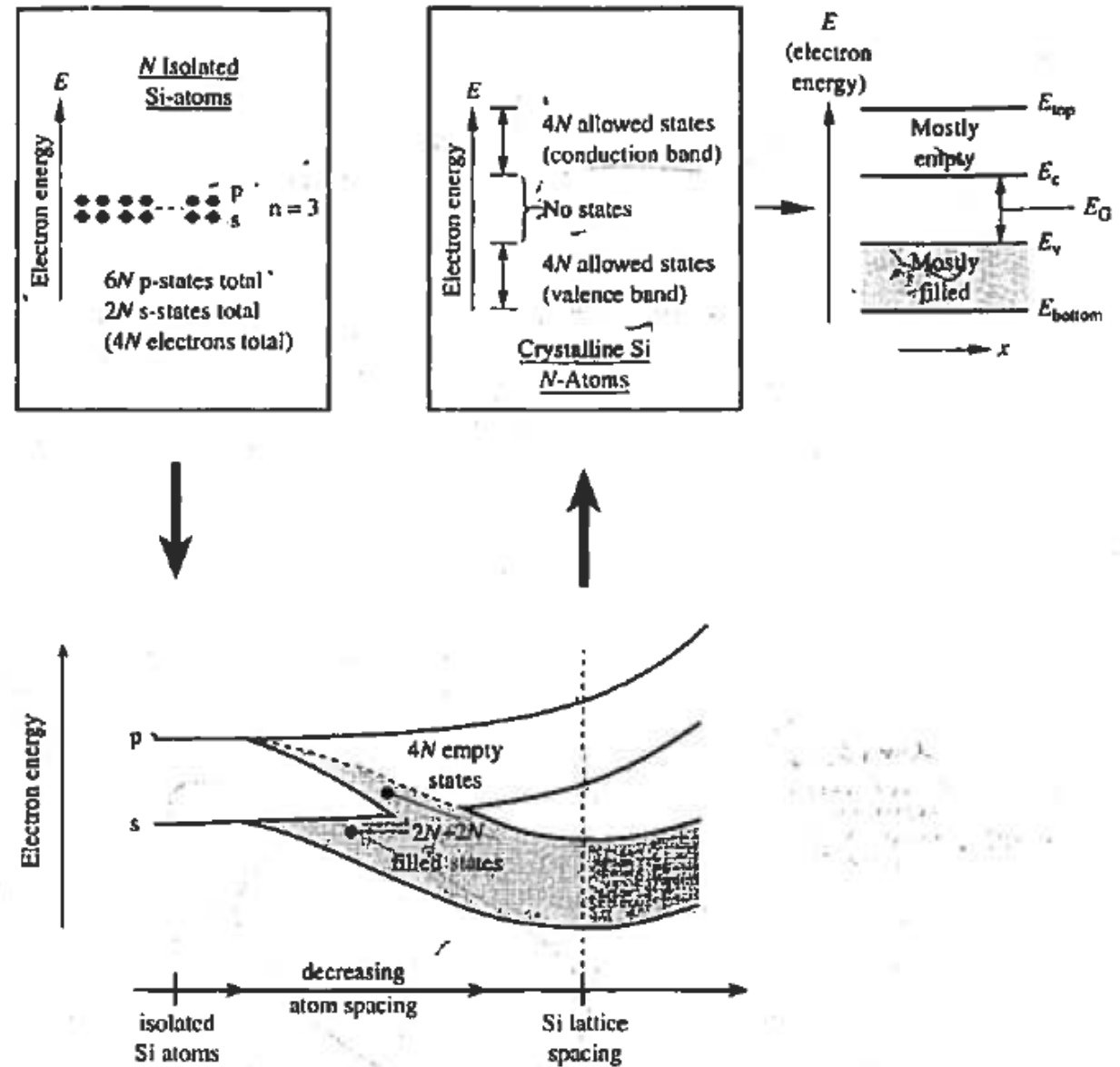
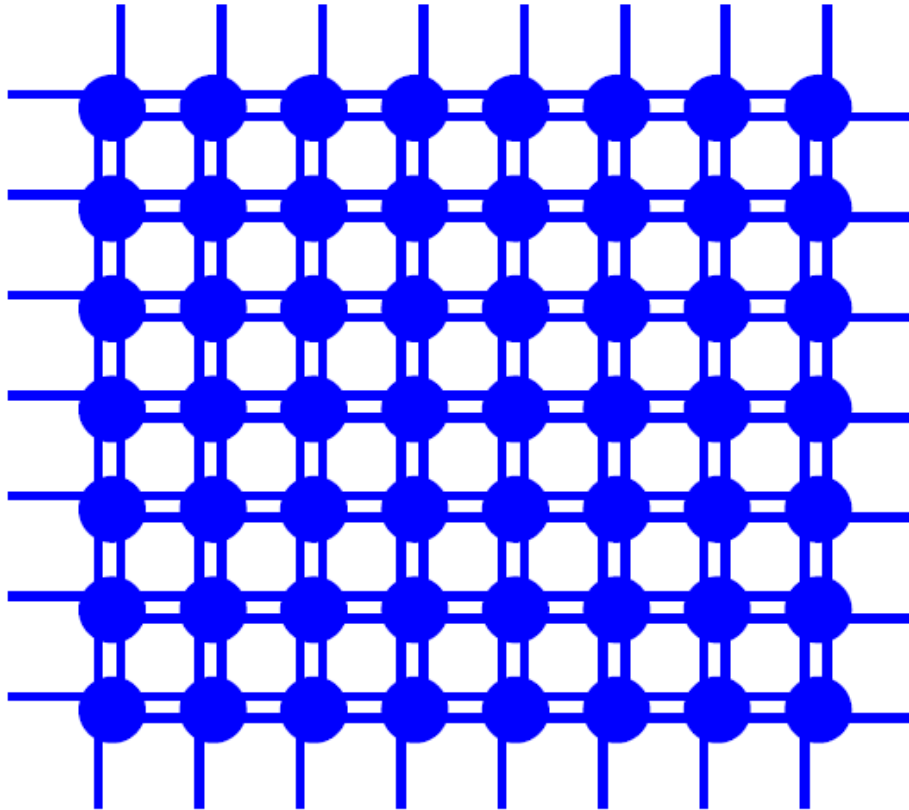


Figure 2.5 Conceptual development of the energy band model starting with N isolated Si atoms on the top left and concluding with a “dressed-up” version of the energy band model on the top right.

Charge Carriers

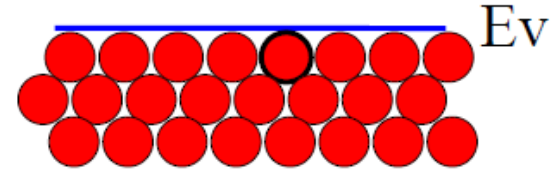
□ Band Occupation at Low Temperature ($T = 0\text{K}$)

For ($E_{\text{thermal}} = kT$) = 0



No electrons in
conduction band means
no electron conduction
is possible

————— E_c

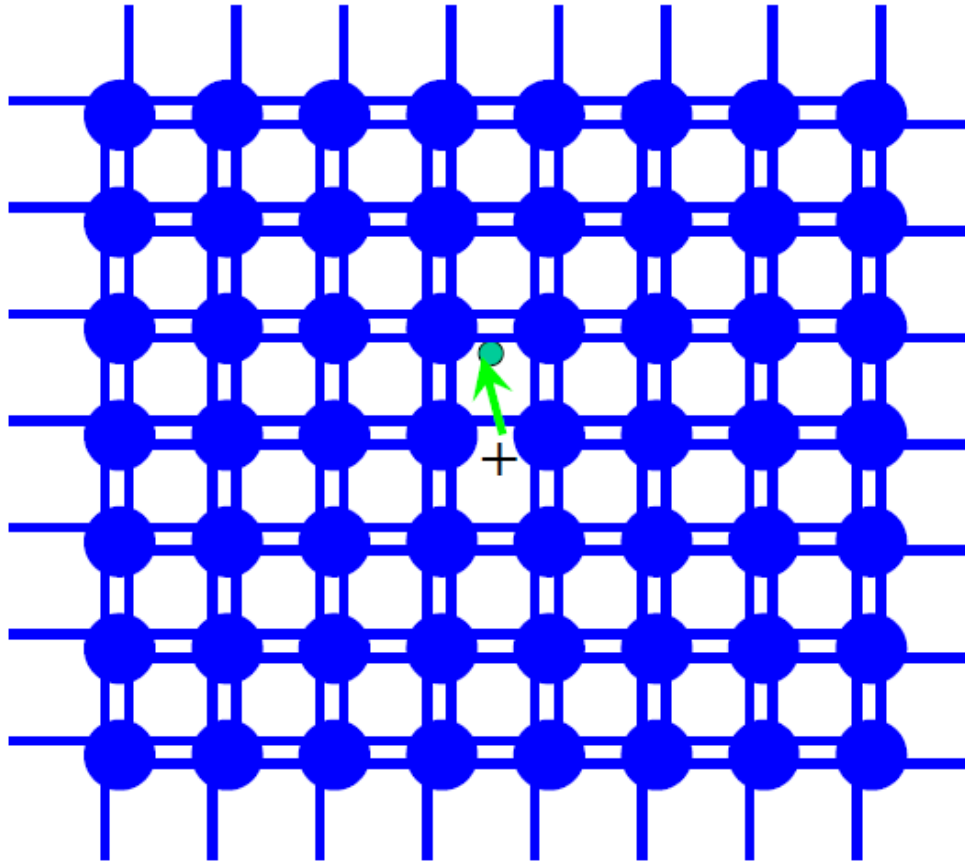


No “Holes” valence
band means no “hole”
conduction is possible

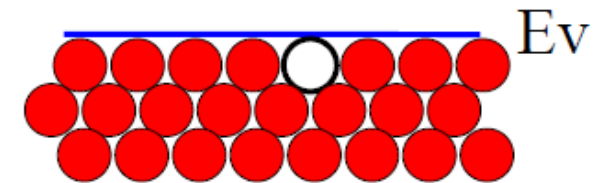
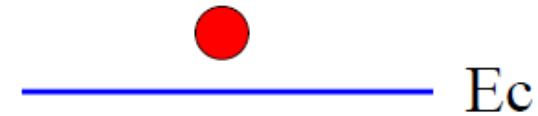
Charge Carriers

□ Band Occupation at High Temperature ($T > 0\text{K}$)

For ($E_{\text{thermal}} = kT$) > 0



Electron free to move
in conduction band

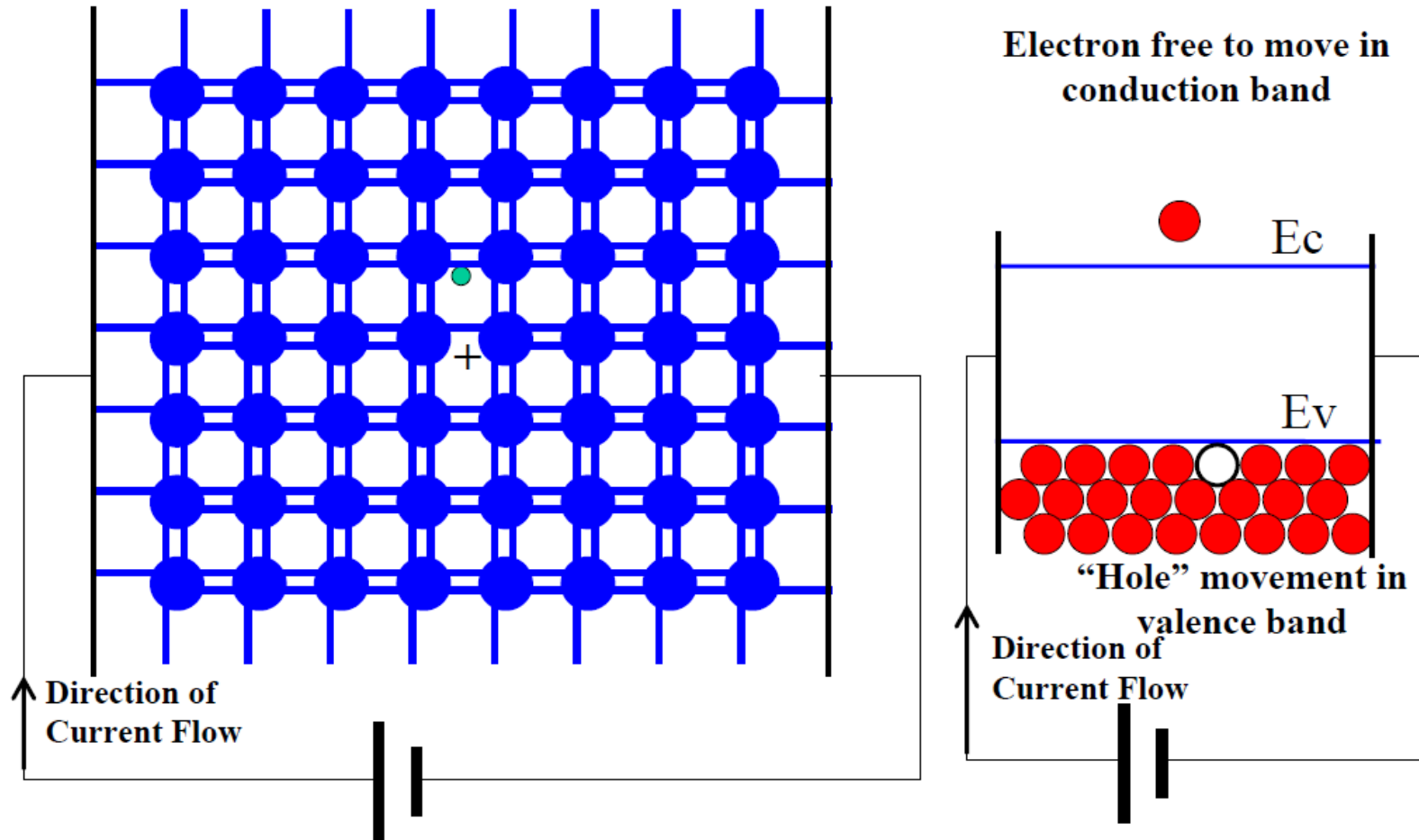


“Hole” free to move in
valence band

Charge Carriers

□ Band Occupation at Applied Bias

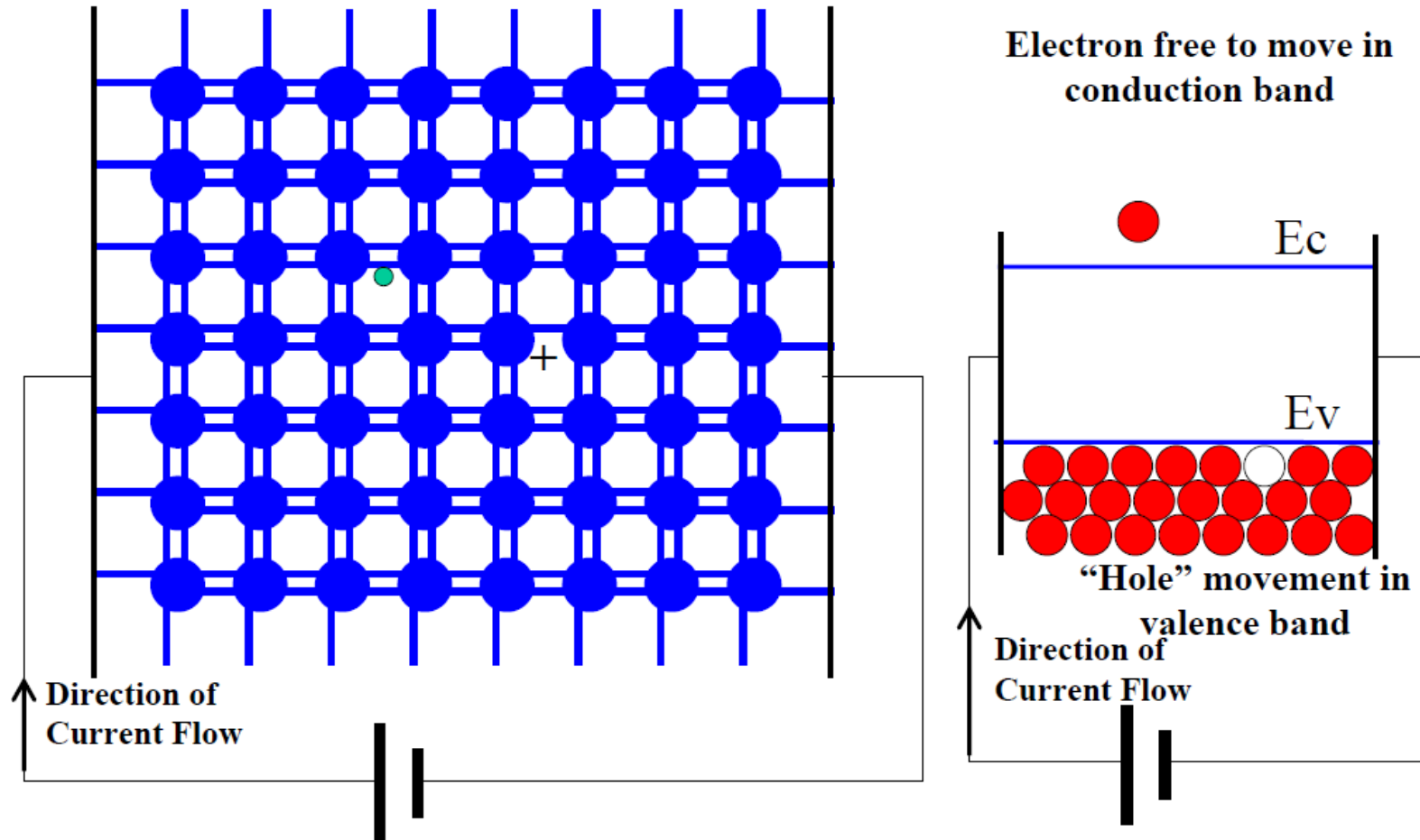
For $(E_{\text{thermal}} = kT) > 0$



Charge Carriers

□ Band Occupation at Applied Bias

For $(E_{\text{thermal}} = kT) > 0$



Charge Carriers

□ Bandgap and Material Classifications

Ease of achieving thermal population of conduction band determines whether a material is an insulator, semiconductor, or metal

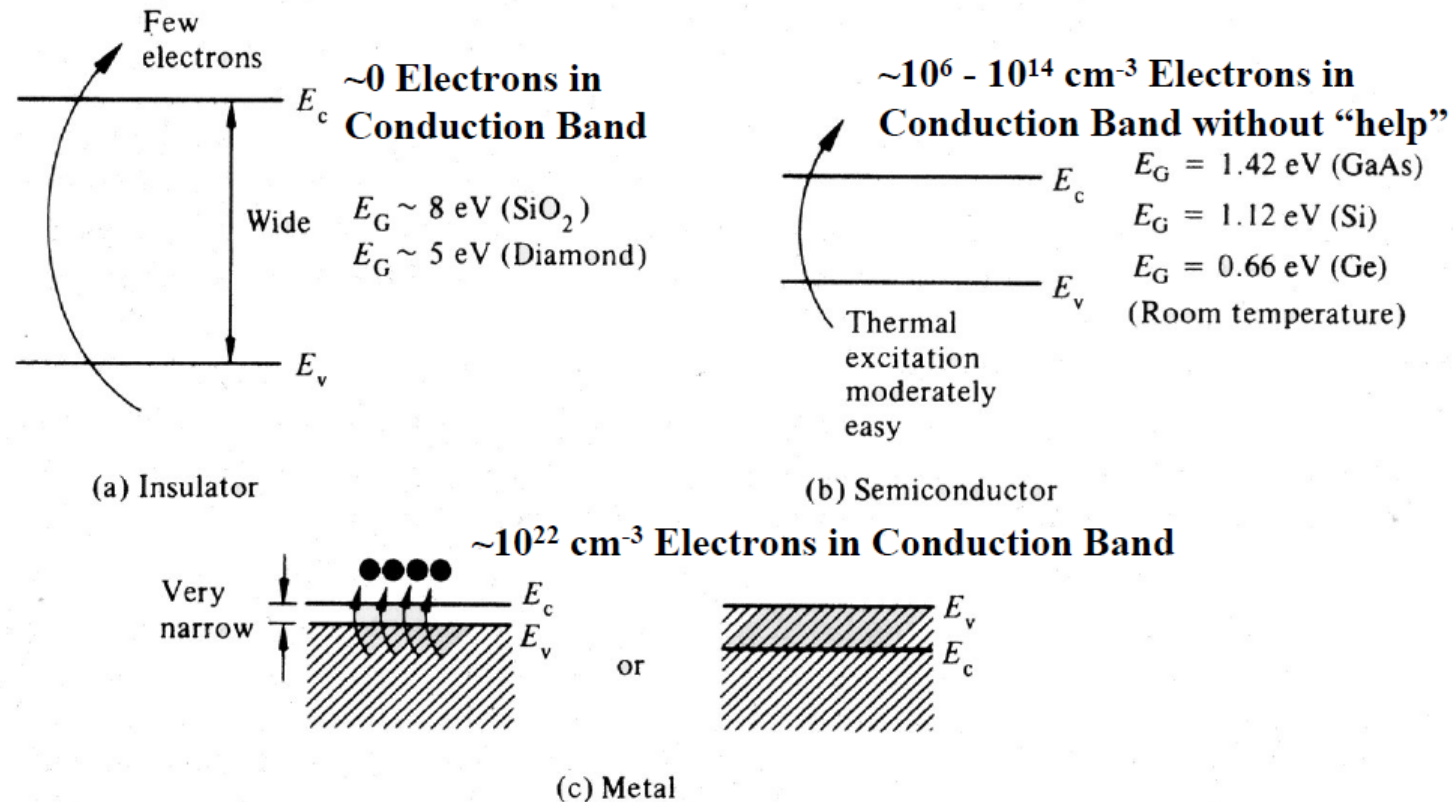


Figure 2.8 Explanation of the distinction between (a) insulators, (b) semiconductors, and (c) metals using the energy band model.

Charge Carriers

□ Effective Mass of Charge Carriers

We account for this complex interaction with the crystal (each crystal behaving differently) by using an “effective mass”

$$F = -qE = m_n^* \frac{dv}{dt}$$

$F \equiv$ force, $v \equiv$ velocity, $t \equiv$ time,

$q \equiv$ electronic charge e ,

$m_n^* \equiv$ electron effective mass

$$F = qE = m_p^* \frac{dv}{dt}$$

$F \equiv$ force, $v \equiv$ velocity, $t \equiv$ time,

$q \equiv$ electronic charge e ,

$m_p^* \equiv$ hole effective mass

Table 2.1 Density of States Effective Masses at 300 K.

Material	m_n^*/m_0	m_p^*/m_0
Si	1.18	0.81
Ge	0.55	0.36
GaAs	0.066	0.52

Ge and GaAs have “lighter electrons” than Si
which results in faster devices

Charge Carriers

□ Intrinsic Charge Carrier Concentrations

- For each electron promoted to the conduction band, one hole is left in the valence band. Thus, the number of electrons in the conduction band is equal to the number of holes in the valence band unless there is “help” to change the relative populations in each band.
- Intrinsic carrier concentration is the number of electron (=holes) per cubic centimeter populating the conduction band (or valence band) is called the intrinsic carrier concentration, n_i
- $n_i = f(T)$ that increases with increasing T (more thermal energy)

$$np = n_i^2.$$

n = concentration of electrons

p = concentration of holes

n_i = intrinsic concentration of charge carriers

At Room Temperature ($T=300$ K)

$n_i \sim 2 \times 10^6 \text{ cm}^{-3}$ for GaAs with $E_g = 1.42 \text{ eV}$,

$n_i \sim 1 \times 10^{10} \text{ cm}^{-3}$ for Si with $E_g = 1.1 \text{ eV}$,

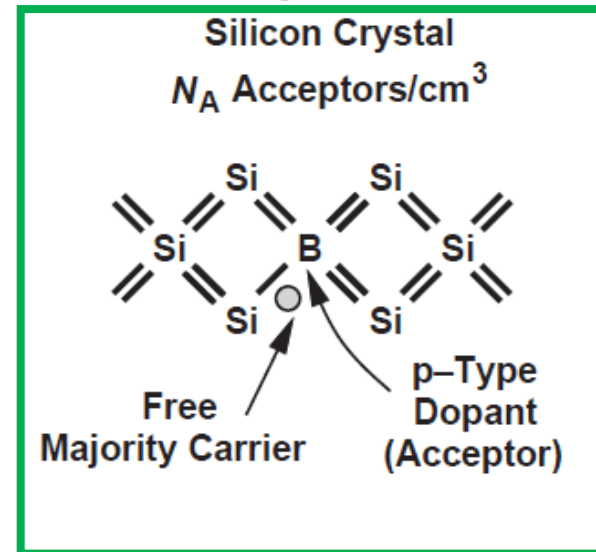
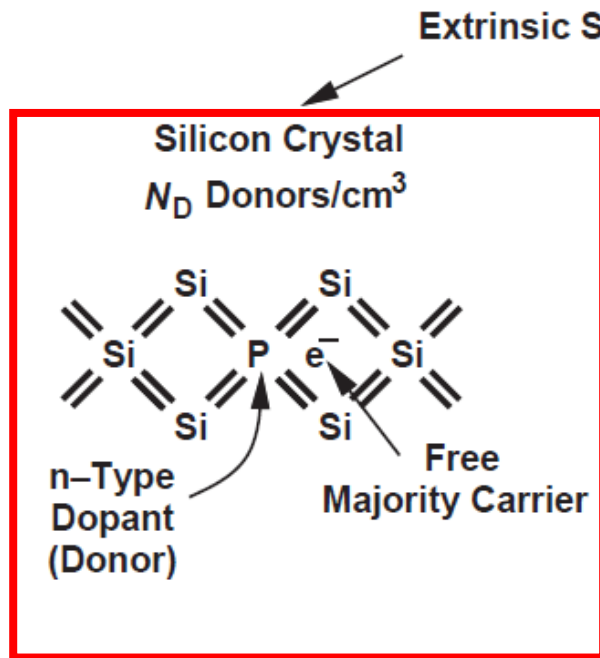
$n_i \sim 2 \times 10^{13} \text{ cm}^{-3}$ for Ge with $E_g = 0.66 \text{ eV}$,

$n_i \sim 1 \times 10^{-14} \text{ cm}^{-3}$ for GaN with $E_g = 3.4 \text{ eV}$

Charge Carriers

□ Extrinsic Semiconductor

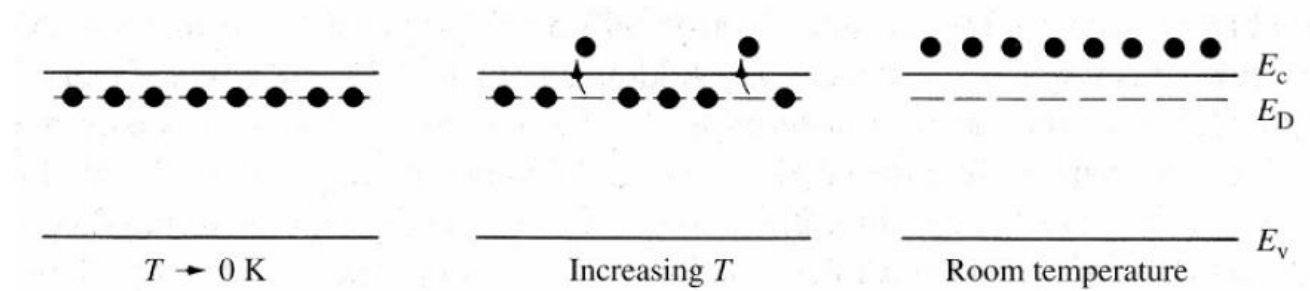
Majority Carriers: $n \approx N_D$
Minority Carriers: $p \approx \frac{n_i^2}{N_D}$



Majority Carriers: $p \approx N_A$
Minority Carriers: $n \approx \frac{n_i^2}{N_A}$

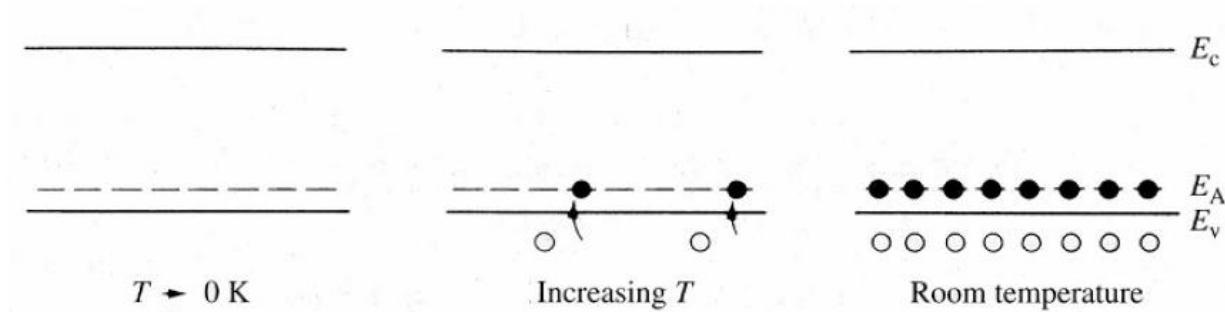
Charge Carriers

□ Band Diagram - Donor “adding extra” electrons



Charge Carriers

□ Band Diagram - Acceptor “adding extra” holes



Charge Carriers

Example 2.3

A piece of crystalline silicon is doped uniformly with phosphorus atoms. The doping density is 10^{16} atoms/cm³. Determine the electron and hole densities in this material at the room temperature.

Charge Carriers – Summary of Important terms

Bandgap Energy: Energy required to remove a valence electron and allow it to freely conduct.

Intrinsic Semiconductor: A “native semiconductor” with no dopants. Electrons in the conduction band equal holes in the valence band. The concentration of electrons (=holes) is the intrinsic concentration, n_i .

Extrinsic Semiconductor: A doped semiconductor. Many electrical properties controlled by the dopants, not the intrinsic semiconductor.

Donor: An impurity added to a semiconductor that adds an additional electron not found in the native semiconductor.

Acceptor: An impurity added to a semiconductor that adds an additional hole not found in the native semiconductor.

Dopant: Either an acceptor or donor.

N-type material: When electron concentrations (n =number of electrons/cm³) exceed the hole concentration (normally through doping with donors).

P-type material: When hole concentrations (p =number of holes/cm³) exceed the electron concentration (normally through doping with acceptors).

Majority carrier: The carrier that exists in higher population (ie n if $n > p$, p if $p > n$)

Minority carrier: The carrier that exists in lower population (ie n if $n < p$, p if $p < n$)

Other important terms (among others): Insulator, semiconductor, metal, amorphous, polycrystalline, crystalline (or single crystal), lattice, unit cell, primitive unit cell, zincblende, lattice constant, elemental semiconductor, compound semiconductor, binary, ternary, quaternary, atomic density, Miller indices, various notations, etc...

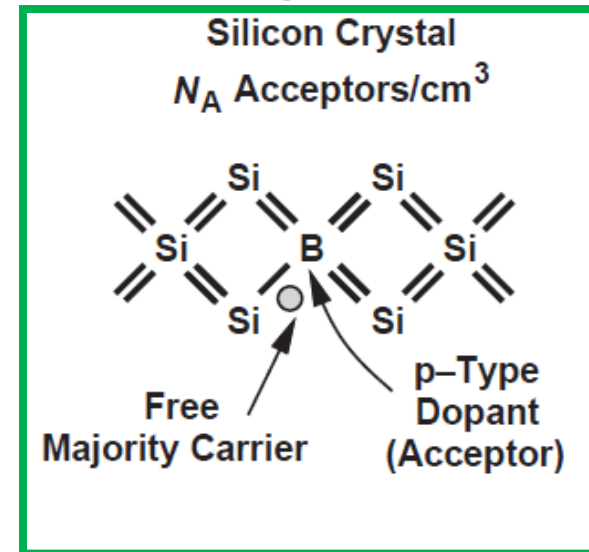
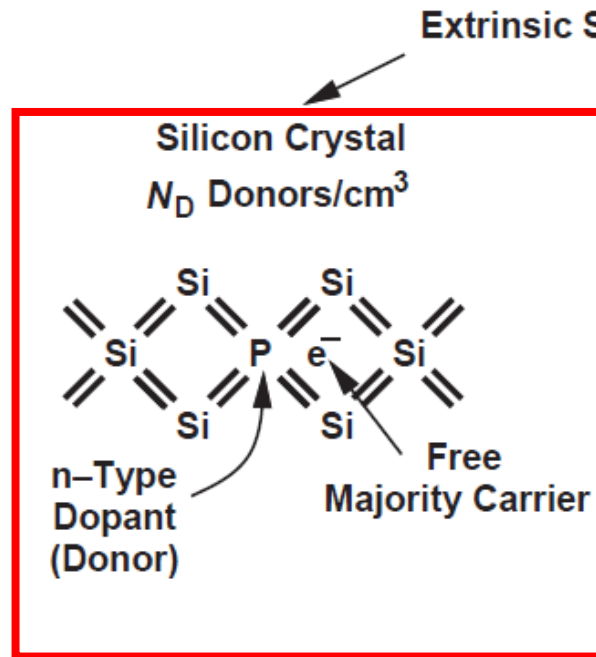
- **Reading for this Lecture**

- Chapter # 2 of the Introduction to Microelectronics by Behzad Razavi

Charge Carriers

□ Extrinsic Semiconductor

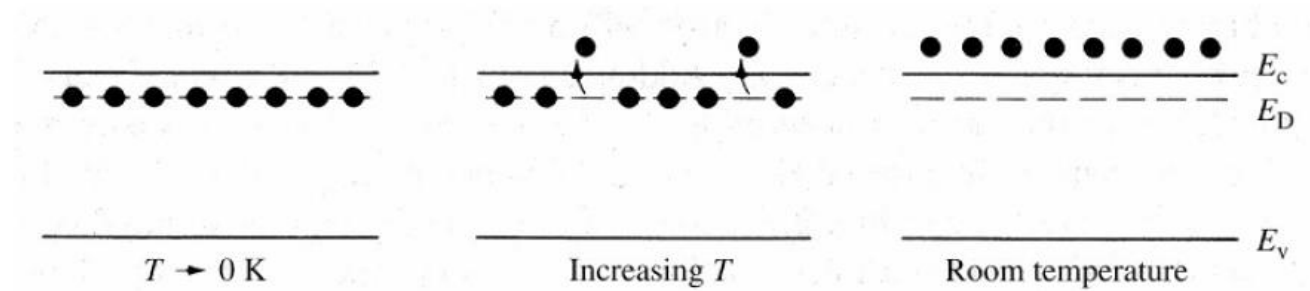
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Minority Carriers: $p \approx \frac{n_i^2}{N_D}$



Majority Carriers: $p \approx N_A$
Minority Carriers: $n \approx \frac{n_i^2}{N_A}$

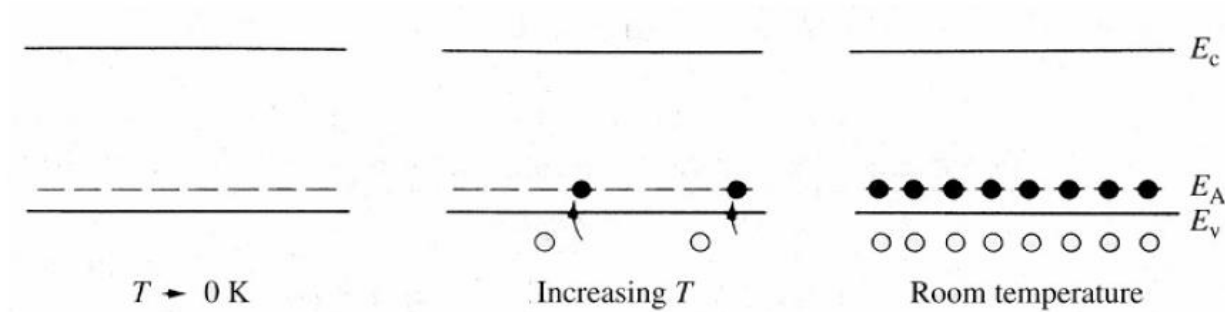
Charge Carriers

□ Band Diagram - Donor “adding extra” electrons



Charge Carriers

□ Band Diagram - Acceptor “adding extra” holes



Charge Carriers

Example 2.3

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Other important terms (among others): Insulator, semiconductor, metal, amorphous, polycrystalline, crystalline (or single crystal), lattice, unit cell, primitive unit cell, zincblende, lattice constant, elemental semiconductor, compound semiconductor, binary, ternary, quaternary, atomic density, Miller indices, various notations, etc...

Charge Carriers

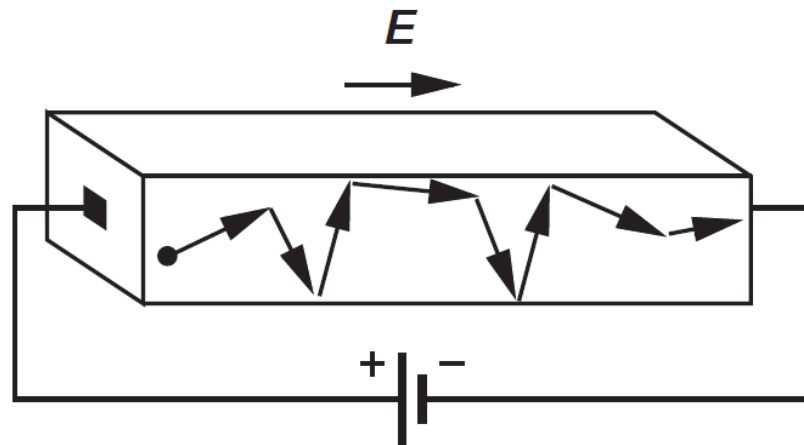
□ Transport of Charge Carriers

- Movement of charge carriers in semiconductor materials is known as “Transport”
- Charge carriers are holes and electrons
- Two types of transport mechanisms
 1. Drift of carriers
 2. Diffusion of carriers

Charge Carriers

□ Drift of Charge Carriers

- A material can conduct current in response to a potential difference and hence an electric field (Ohm's Law)
- The field accelerates the charge carriers in the material, forcing some to flow from one end to the other
- Movement of charge carriers due to an electric field is called “Drift”



Charge Carriers

□ Drift of Charge Carriers

- The charge carriers are accelerated by the field and accidentally collide with the atoms in the crystal, eventually reaching the other end and flowing into the battery
- The acceleration due to the field and the collision with the crystal counteract, leading to a constant velocity for the carriers
- The velocity is proportional to the applied electric field

$$v \propto E$$

$$v = \mu E$$

where μ is called the “mobility” and usually expressed in $\text{cm}^2/(\text{V} \cdot \text{s})$

Charge Carriers

□ Drift of Charge Carriers

- The velocity is proportional to the applied electric field

$$v = \mu E$$

where μ is called the “mobility” and usually expressed in $\text{cm}^2/(\text{V} \cdot \text{s})$

$$\vec{v}_e = -\mu_n \vec{E}.$$

$$\vec{v}_h = \mu_p \vec{E}.$$

- In **Silicon**,

the mobility of electrons, $\mu_n = 1350 \text{ cm}^2/(\text{V} \cdot \text{s})$, and that of holes, $\mu_p = 480 \text{ cm}^2/(\text{V} \cdot \text{s})$

Charge Carriers

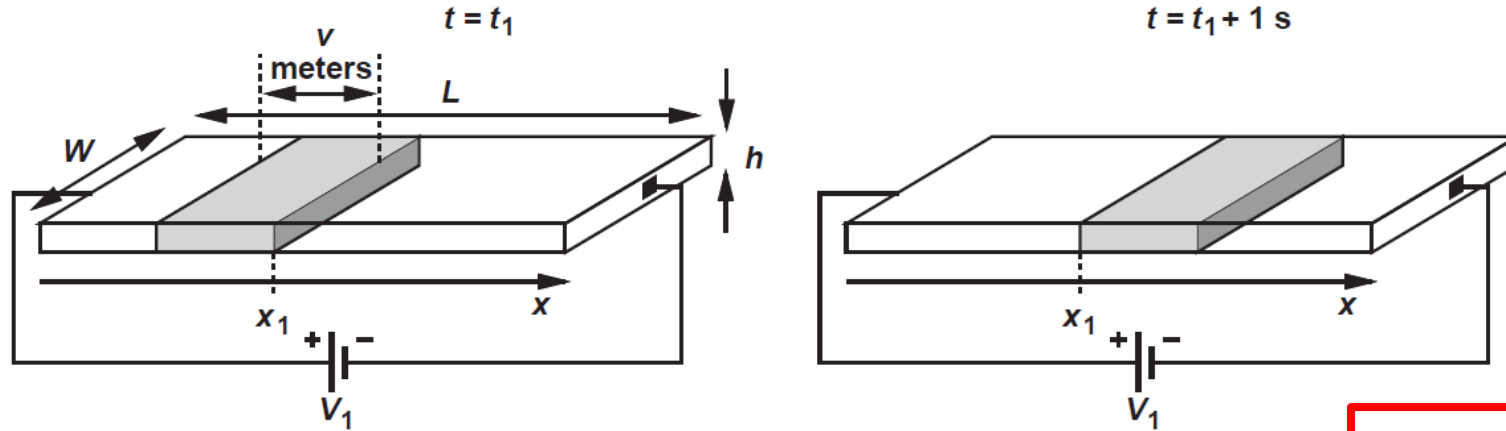
□ Drift of Charge Carriers

Example 2.5

A uniform piece of n -type of silicon that is $1\text{ }\mu\text{m}$ long senses a voltage of 1 V . Determine the velocity of the electrons.

Charge Carriers

□ Current flow in terms of charge density

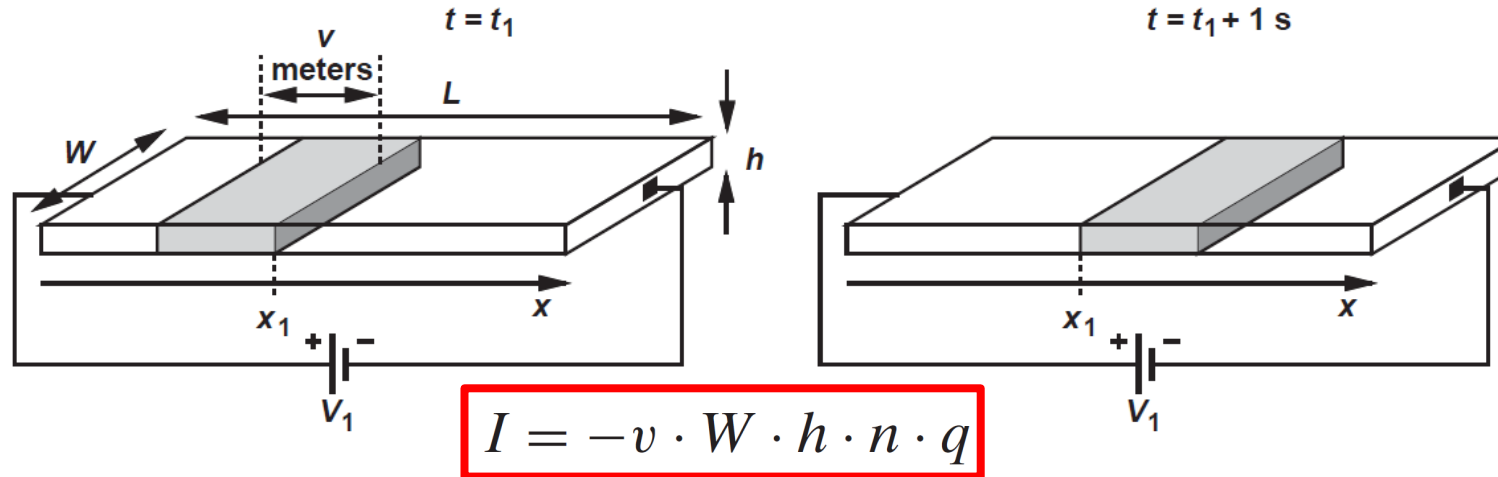


$$I = -v \cdot W \cdot h \cdot n \cdot q$$

- Suppose
 - Uniform voltage V_1 is applied across a semiconductor bar
 - Width of the semiconductor slab is “ W ”
 - Free electron density is “ n ” and charge on one electron “ q ” is $1.6 \times 10^{-19} \text{ C}$
 - Electron velocity is “ v ” (m/s)
 - Distance travelled by electrons in 1 second is “ v ” meters
- Current is the “total charge enclosed per unit length (area or volume)”
 - Remember that the current flows in 3D, so its good to see it as charge flowing per unit volume

Charge Carriers

□ Current flow in terms of charge density



- Current density is denoted by “J” (A/m²)
 - Current density of electrons is denoted by $J_n = \mu_n E \cdot n \cdot q$
 - Current density of holes is denoted by $J_p = \mu_p E \cdot p \cdot q$
 - Total drift current density in a semiconductor material is given as $J_{tot} = J_n + J_p$

$$J_{tot} = \mu_n E \cdot n \cdot q + \mu_p E \cdot p \cdot q$$
$$= q(\mu_n n + \mu_p p) E.$$

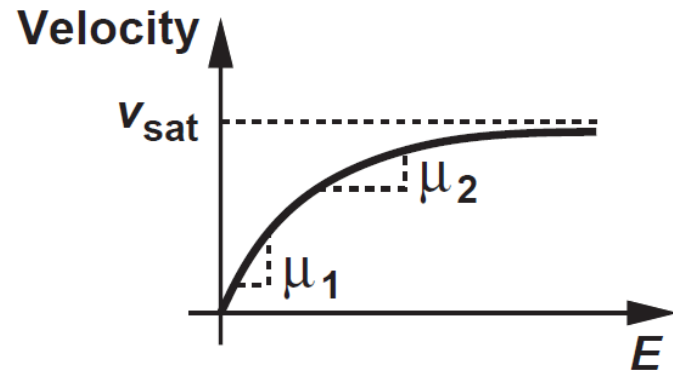
Charge Carriers

□ Velocity Saturation

- We have thus far assumed that the mobility of carriers in semiconductors is independent of the electric field and the velocity rises linearly with E according to $v = \mu E$
- In reality, if the electric field approaches sufficiently high levels, “ v ” no longer follows “ E ” linearly
- This is because the carriers collide with the lattice so frequently and the time between the collisions is so short that they cannot accelerate much
- As a result, “ v ” varies “sub-linearly” at high electric fields, eventually reaching a saturated level “ v_{sat} ”. This is called “velocity saturation”
- This effect manifests itself in some modern transistors, limiting the performance of circuits

Charge Carriers

□ Velocity Saturation



$$\mu = \frac{\mu_0}{1 + bE}$$

$$v_{sat} = \frac{\mu_0}{b}$$

$$v = \mu E$$

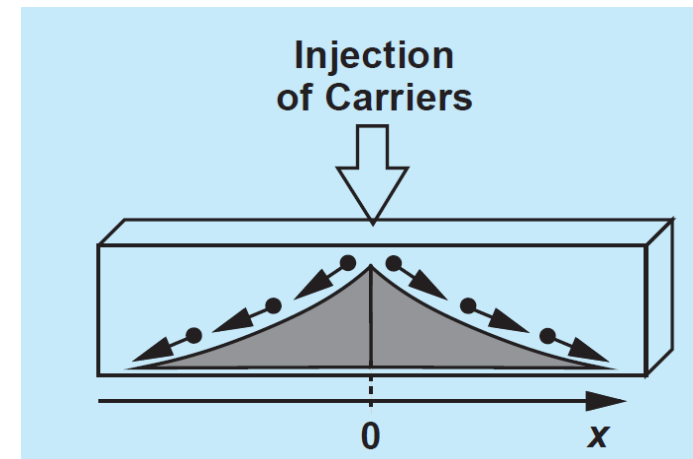
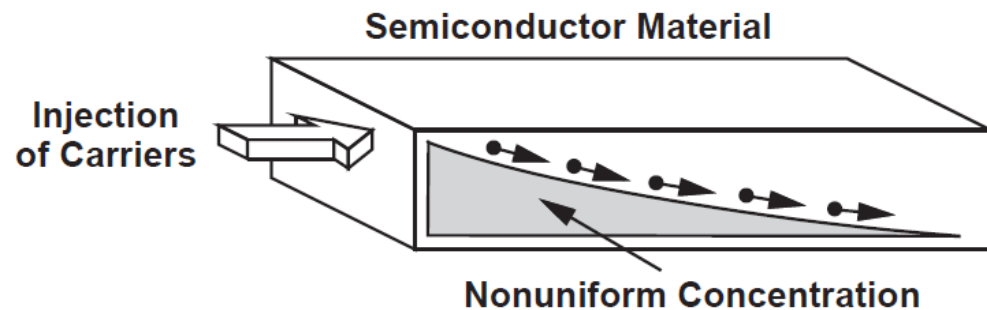
$$v = \frac{\mu_0}{1 + bE} E$$

$$v = \frac{\mu_0}{1 + \frac{\mu_0 E}{v_{sat}}} E.$$

Charge Carriers

□ Diffusion

- When charge carrier are injected into the semiconductor material with the help of doping process, they tend to move the towards the regions of low concentration from a region of higher concentration, in the absence of electric field
- This movement of charge carriers continues till an equilibrium is reached
- This creates a “non-uniform” density of charge carriers in doped semi-conductor materials



Charge Carriers

□ Diffusion

- The diffusion current “I” (due to electron diffusion) is given as

$$I = AqD_n \frac{dn}{dx}$$

where

A = cross sectional area

q = charge of an electron

D_n = diffusion constant for electrons (cm²/s)

dn/dx = concentration gradient (with respect to length of the material)

- Intrinsic Silicon has a diffusion constant for electrons as $D_n = 34$ cm²/s and a diffusion constant for hole as $D_p = 12$ cm²/s

Charge Carriers

□ Diffusion current density

- The diffusion current density because of the electrons is given as

$$J_n = qD_n \frac{dn}{dx}$$

- The diffusion current density because of the holes is given as

$$J_p = -qD_p \frac{dp}{dx}$$

- The total current density because of diffusion is given as

$$J_{tot} = q \left(D_n \frac{dn}{dx} - D_p \frac{dp}{dx} \right)$$

Charge Carriers

□ Einstein Relationship

- The mobility of electrons (or holes) and the diffusion constant of electrons (or holes) are related and the relationship is known as “Einstein Relationship”

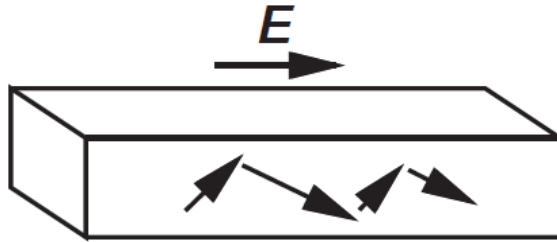
$$\frac{D}{\mu} = \frac{kT}{q}$$

- Note that $kT/q = 26\text{mV}$ for $T = 300\text{K}$ (where k = Boltzmann constant)

Charge Carriers

□ Summary of Charge Transport

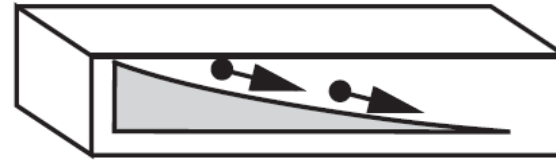
Drift Current



$$J_n = q n \mu_n E$$

$$J_p = q p \mu_p E$$

Diffusion Current

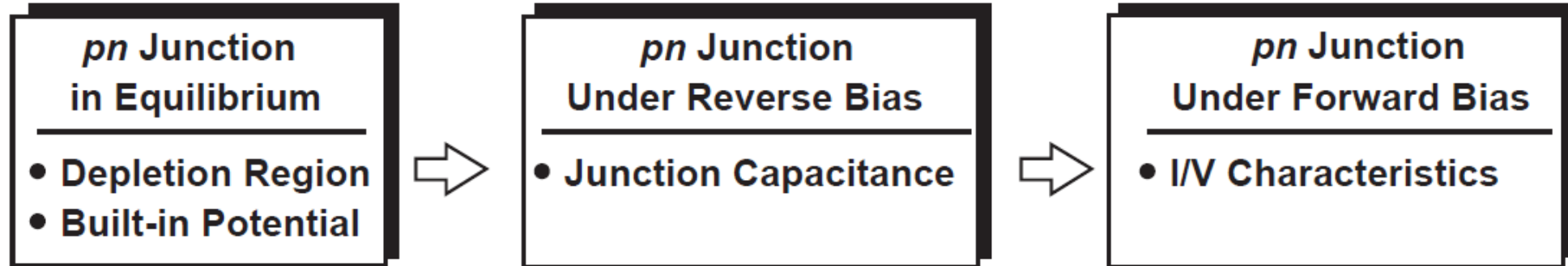


$$J_n = q D_n \frac{dn}{dx}$$

$$J_p = -q D_p \frac{dp}{dx}$$

pn-Junction

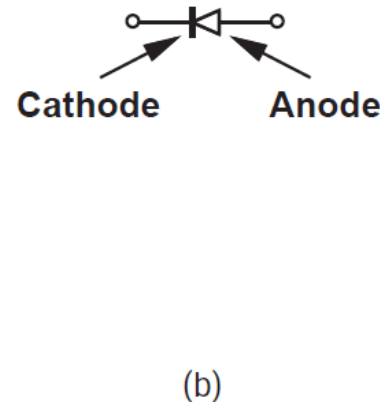
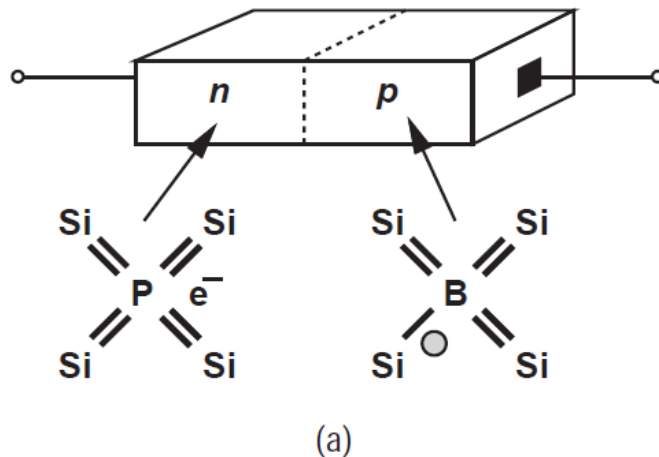
□ What we shall be discussing in this topic



pn-Junction

□ Basics of a pn-Junction

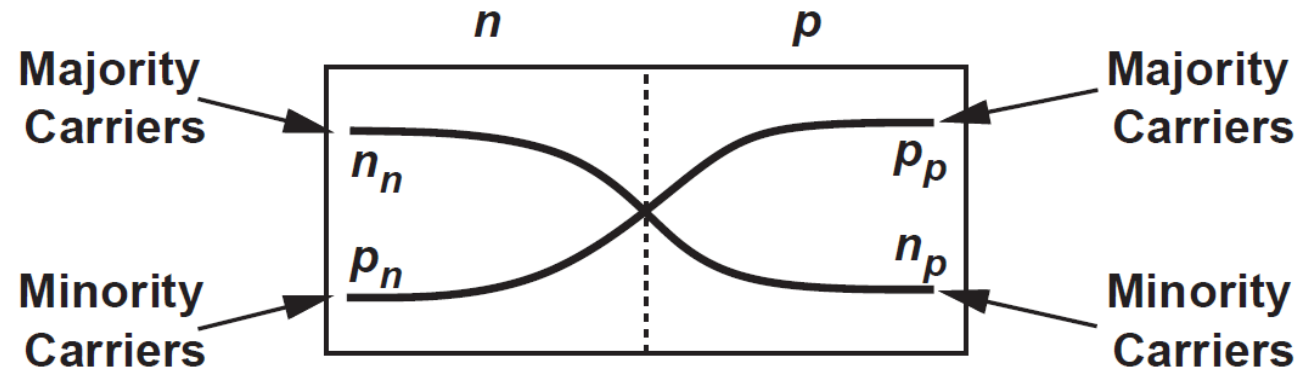
- The basics of semiconductor devices starts with the details of a pn-junction
- A pn-junction is simplest of semiconductor devices and forms the basis of modern transistors-based semiconductor devices
- A pn-junction is also known as a “Diode”



pn-Junction

□pn-Junction at Equilibrium

- The concentration gradients for charge carriers are shown



n_n : Concentration of electrons on n side

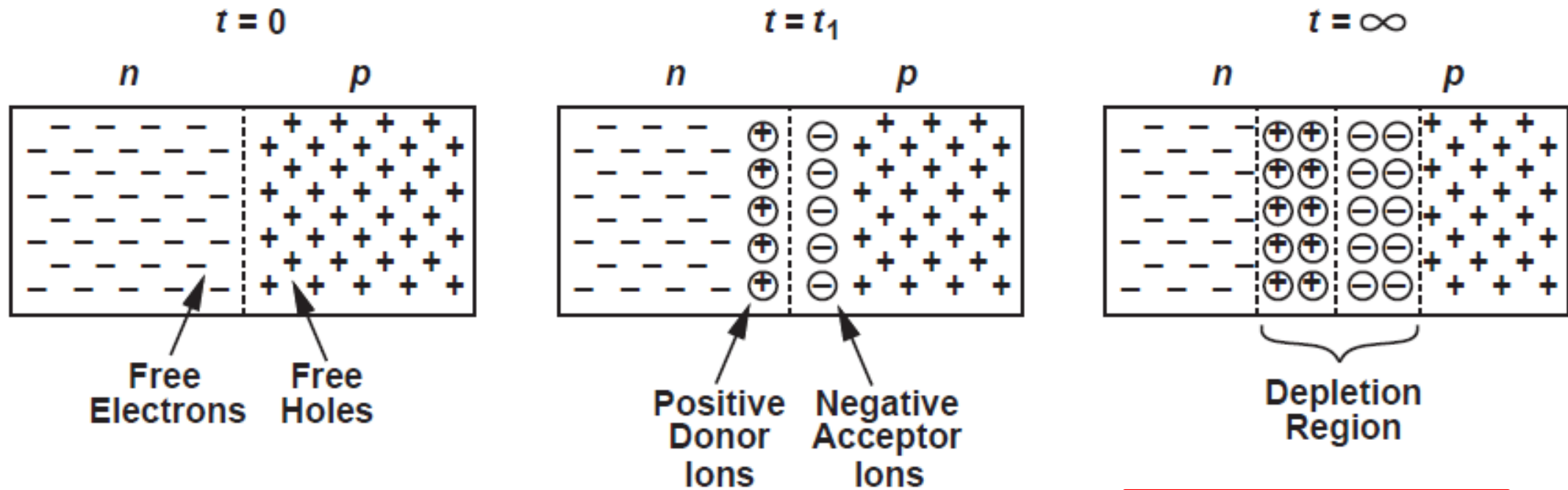
p_n : Concentration of holes on n side

p_p : Concentration of holes on p side

n_p : Concentration of electrons on p side

pn-Junction

□ Evolution of charge concentrations in a pn-Junction



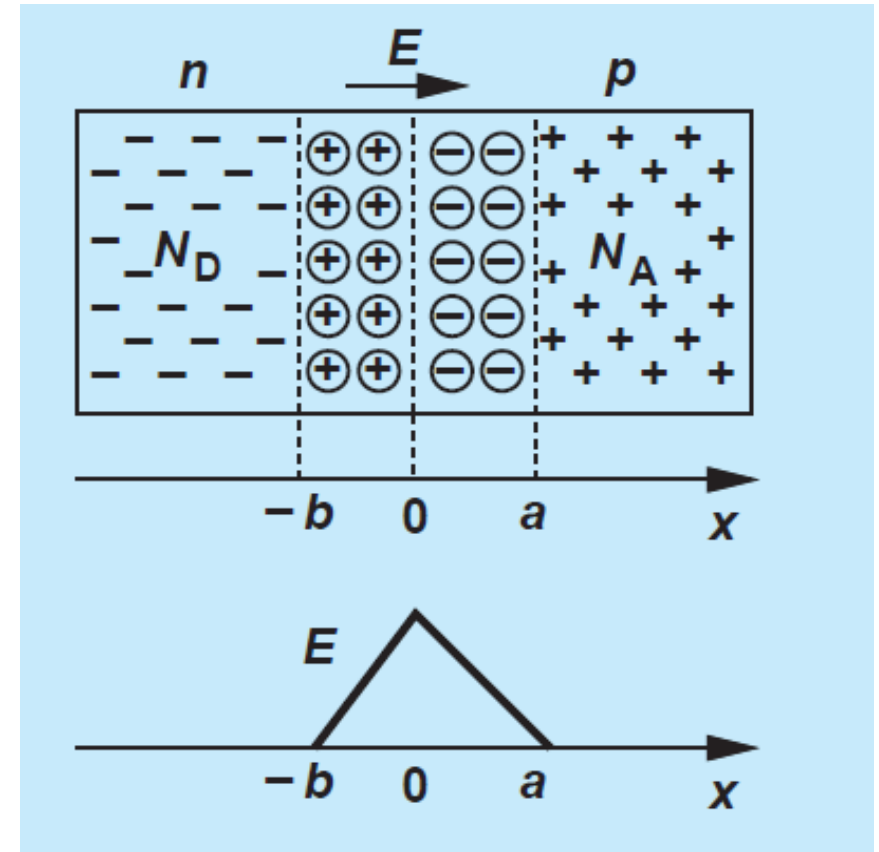
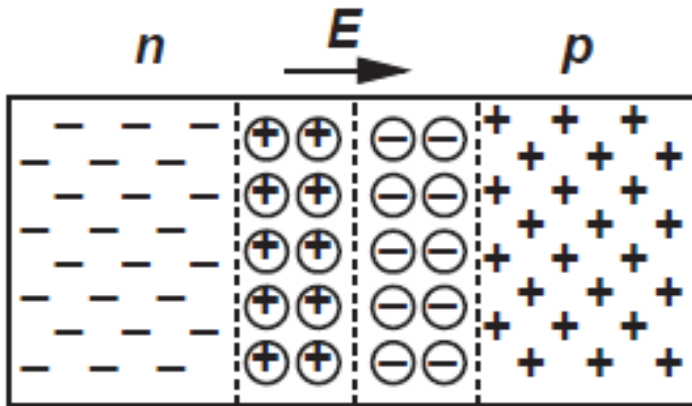
$$|I_{\text{drift},p} + I_{\text{drift},n}| = |I_{\text{diff},p} + I_{\text{diff},n}|$$

$$|I_{\text{drift},p}| = |I_{\text{diff},p}|$$

$$|I_{\text{drift},n}| = |I_{\text{diff},n}|$$

pn-Junction

□ Electric Field in a pn-Junction



pn-Junction

□pn-Junction at Equilibrium

Example 2.11

A *pn* junction employs the following doping levels: $N_A = 10^{16} \text{ cm}^{-3}$ and $N_D = 5 \times 10^{15} \text{ cm}^{-3}$. Determine the hole and electron concentrations on the two sides.

(Assume $n_i = 1.08 \times 10^{10} \text{ cm}^{-3}$ for Silicon at $T = 300\text{K}$)

pn-Junction

□ Built-in Potential

- The existence of an electric field within the depletion region suggests that the junction may exhibit a “built-in potential”

$$V_0 = \frac{kT}{q} \ln \frac{N_A N_D}{n_i^2}$$

Example 2.13

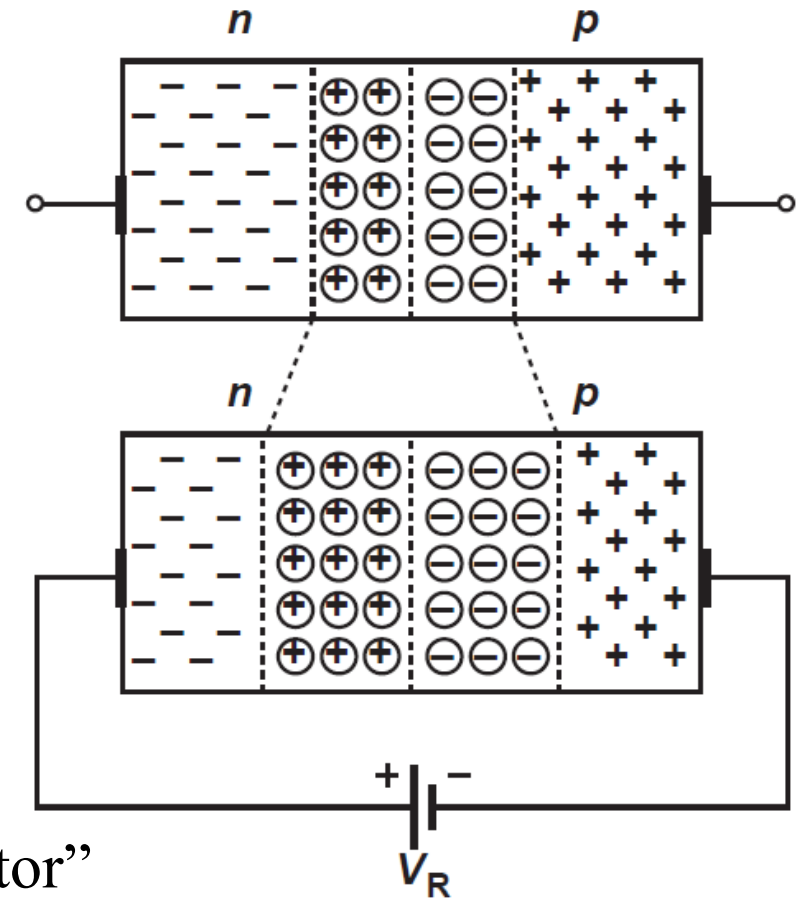
A silicon *pn* junction employs $N_A = 2 \times 10^{16} \text{ cm}^{-3}$ and $N_D = 4 \times 10^{16} \text{ cm}^{-3}$. Determine the built-in potential at room temperature ($T = 300 \text{ K}$).

(Assume $n_i = 1.08 \times 10^{10} \text{ cm}^{-3}$ for Silicon at $T = 300 \text{ K}$)

pn-Junction

□pn-Junction under Reverse Bias

- The E-field is enhanced under reverse bias
- Widening of depletion region
- No junction current flows during reverse bias
- The depletion region can be considered as a “Capacitor”
- The depletion region displays **“voltage-dependent Capacitance”**
- The capacitance is non-linear in nature as it does not follow $Q = CV$ relationship



pn-Junction

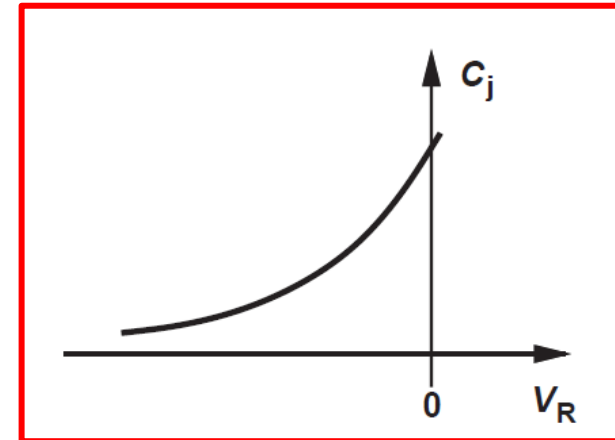
□pn-Junction as a “Voltage-Dependent Capacitor”

- The capacitance of the junction per unit area is given as

$$C_j = \frac{C_{j0}}{\sqrt{1 - \frac{V_R}{V_0}}}$$

where

$$C_{j0} = \sqrt{\frac{\epsilon_{si} q}{2} \frac{N_A N_D}{N_A + N_D} \frac{1}{V_0}}$$



Note: ϵ_{Si} is the dielectric constant of Silicon and is given as 8.85×10^{-14} F/cm

pn-Junction

□pn-Junction as a “Voltage-Dependent Capacitor”

Example 2.15

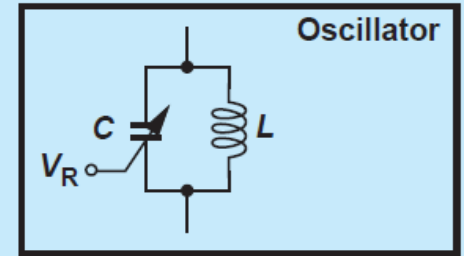
A *pn* junction is doped with $N_A = 2 \times 10^{16} \text{ cm}^{-3}$ and $N_D = 9 \times 10^{15} \text{ cm}^{-3}$. Determine the capacitance of the device with (a) $V_R = 0$ and $V_R = 1 \text{ V}$.

pn-Junction

□ Practical Example – A pn-Junction under Reverse Bias

Example 2.16

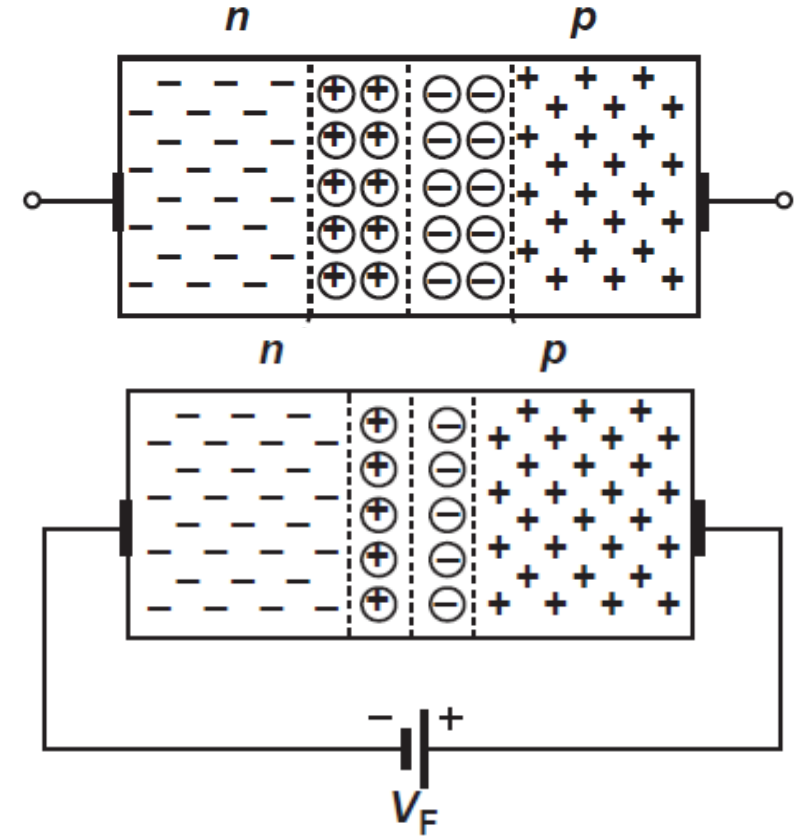
A cellphone incorporates a 2-GHz oscillator whose frequency is defined by the resonance frequency of an LC tank (Fig. 2.26). If the tank capacitance is realized as the pn junction of Example 2.15, calculate the change in the oscillation frequency while the reverse voltage goes from 0 to 2 V. Assume the circuit operates at 2 GHz at a reverse voltage of 0 V, and the junction area is $2000 \mu\text{m}^2$.



pn-Junction

□pn-Junction under Forward Bias

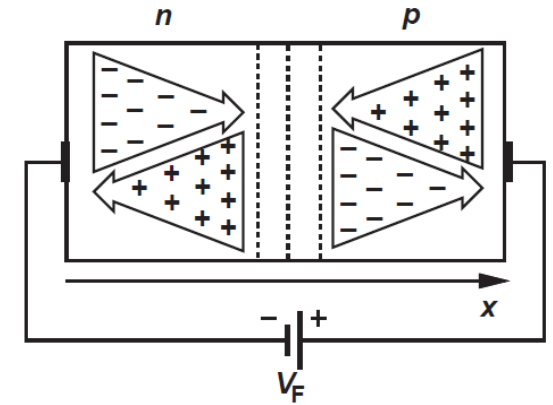
- The built-in E-field is reduced under forward bias
- Reduction of the depletion region
- Junction current flows during forward bias



pn-Junction

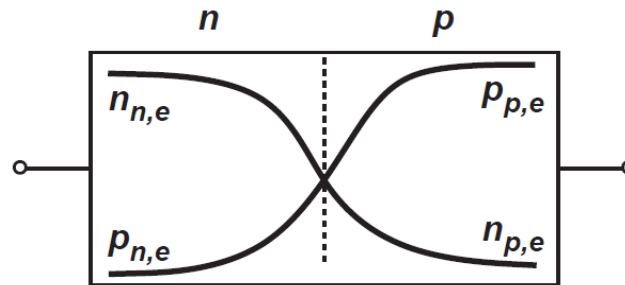
□pn-Junction under Forward Bias – Carrier Profiles

- The minority carrier concentration on the p side rises rapidly
- The majority carrier concentration remains relatively constant
- The same happens with the carriers at the n side of the device

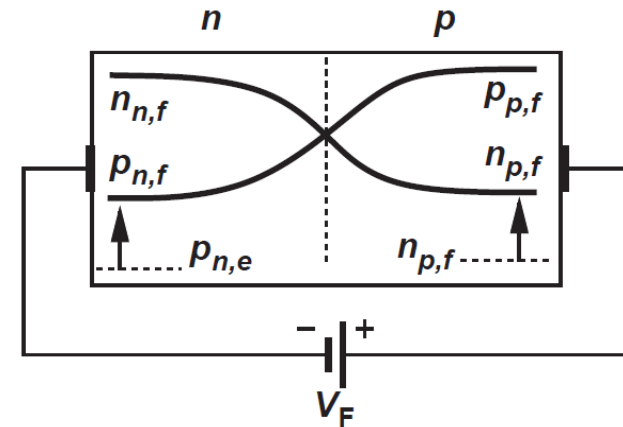


Minority and majority carrier currents.

Why ?



(a)



(b)

pn-Junction

□pn-Junction under Forward Bias – The I-V characteristics

- The current through a forward bias pn-junction is given as

$$I_D = I_S \left(\exp \frac{V_D}{V_T} - 1 \right)$$

$$I_{tot} = I_S \left(\exp \frac{V_F}{V_T} - 1 \right)$$

where “ I_S ” is the “reverse saturation current”

$$I_S = Aqn_i^2 \left(\frac{D_n}{N_A L_n} + \frac{D_p}{N_D L_p} \right)$$

where

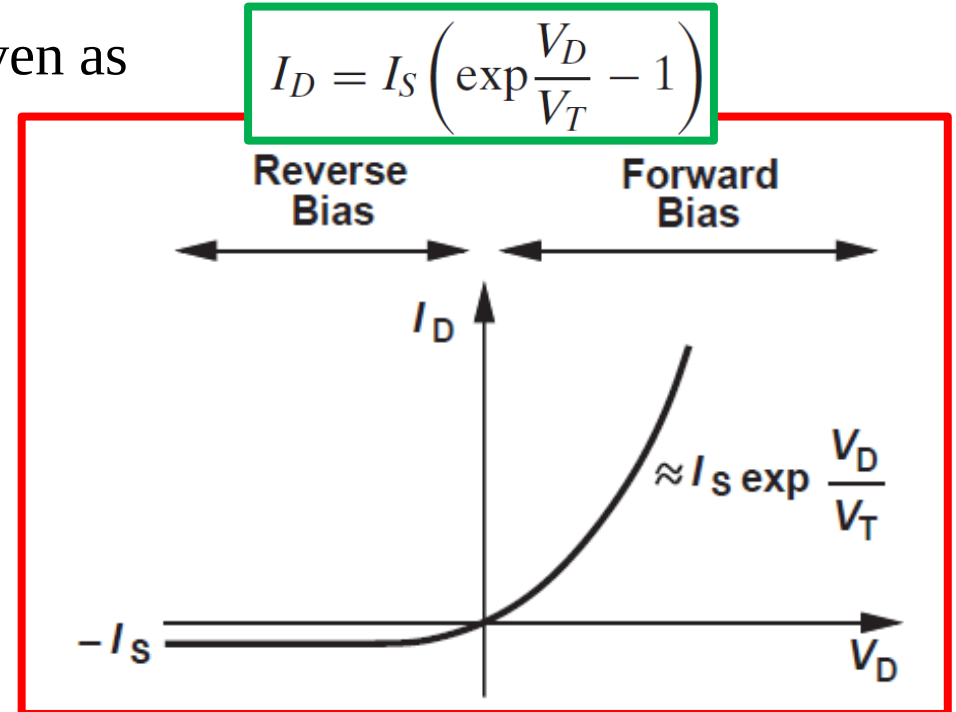
A = Cross section area

L_n = Diffusion length of the electrons

L_p = Diffusion length of the holes

D_n = Diffusion constant for electrons

D_p = Diffusion constant for holes



- **Reading for this Lecture**

- Chapter # 3 of the Introduction to Microelectronics by Behzad Razavi
- Chapter # 3 of Electronic Devices and Circuits by Manzar Saeed (Vol.1)

Diode Models and Circuits

☐ Ideal Diode Model

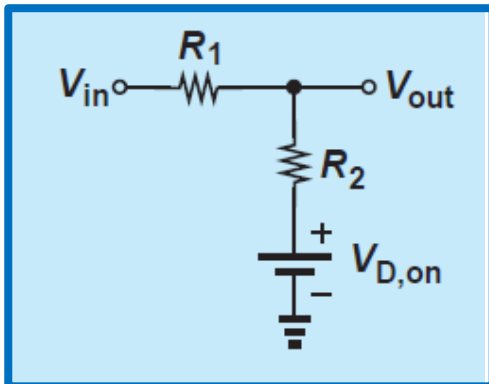
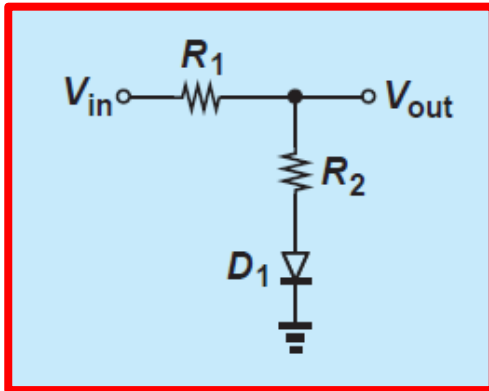
☐ Constant-Voltage Model

Diode Models and Circuits

□ Input-Output Characteristics

Example 3.12

Plot the input/output characteristic of the circuit shown in Fig. 3.14(a) using (a) the ideal model and (b) the constant-voltage model.

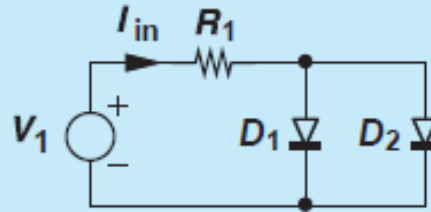


Diode Models and Circuits

□ Input-Output Characteristics

Example 3.13

In the circuit of Fig. 3.15, D_1 and D_2 have different cross section areas but are otherwise identical. Determine the current flowing through each diode.

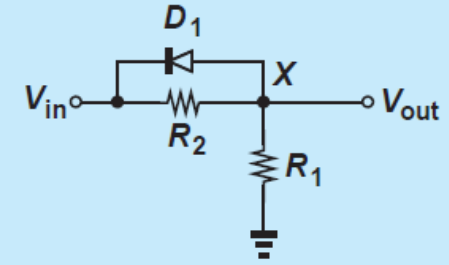


Diode Models and Circuits

□ Input-Output Characteristics

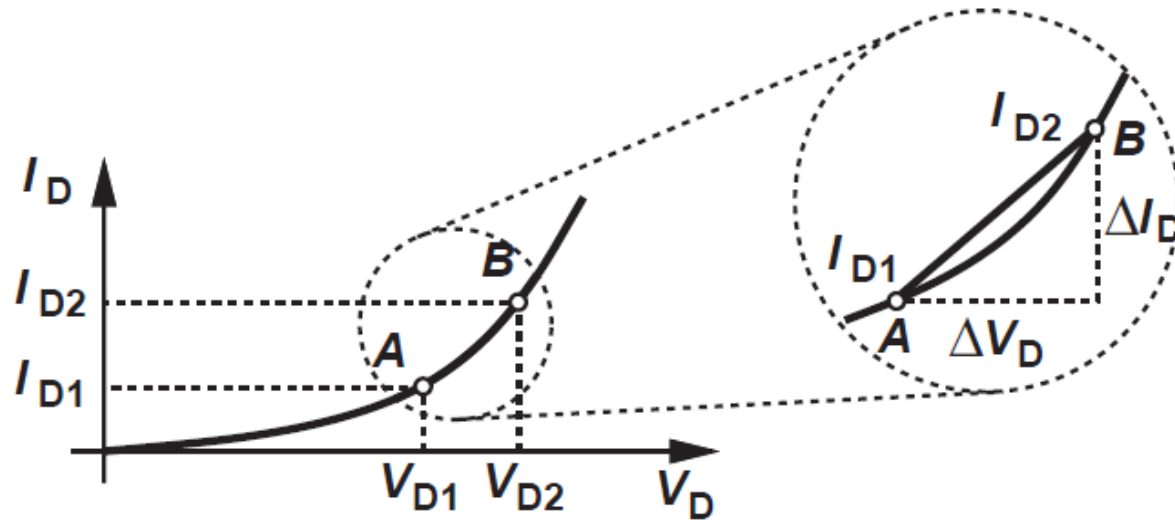
Example 3.15

Plot the input/output characteristic for the circuit shown in Fig. 3.17(a). Assume a constant-voltage model for the diode.



Diode Models and Circuits

□ Large-Signal and Small-Signal Operation



$$\Delta I_D = \frac{\Delta V}{V_T} I_{D1}$$

$$r_d = \frac{V_T}{I_D}$$

$$\begin{aligned} \frac{\Delta I_D}{\Delta V_D} &= \left. \frac{dI_D}{dV_D} \right|_{V_D=V_{D1}} \\ &= \frac{I_S}{V_T} \exp \frac{V_{D1}}{V_T} \\ &= \frac{I_{D1}}{V_T}, \end{aligned}$$

Diode Models and Circuits

□ Large-Signal and Small-Signal Operation

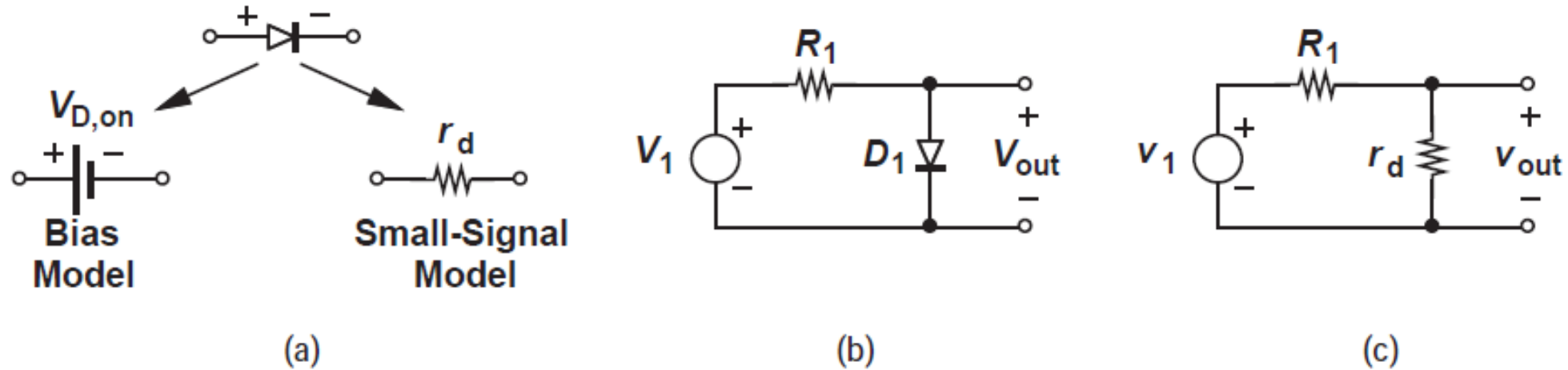


Figure 3.22 Summary of diode models for bias and signal calculations, (b) circuit example, (c) small-signal model.

$$\Delta I_D = \frac{\Delta V}{V_T} I_{D1}$$

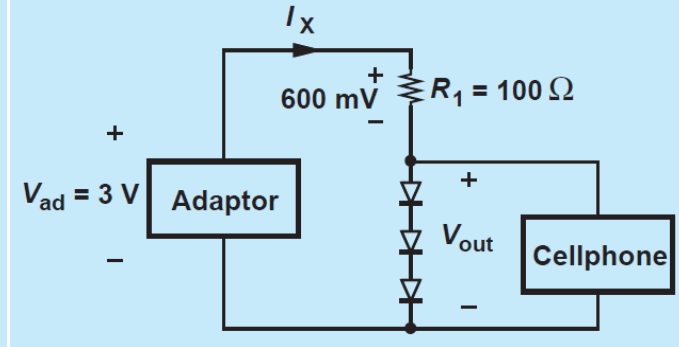
$$r_d = \frac{V_T}{I_D}$$

Diode Models and Circuits

□ Large-Signal and Small-Signal Operation

Example 3.17

Having lost his 2.4-V cellphone charger, an electrical engineering student tries several stores but does not find adaptors with outputs less than 3 V. He then decides to put his knowledge of electronics to work and constructs the circuit shown in Fig. 3.19, where three identical diodes in forward bias produce a total voltage of $V_{out} = 3V_D \approx 2.4$ V and resistor R_1 sustains the remaining 600 mV. Neglect the current drawn by the cellphone.⁶ (a) Determine the reverse saturation current, I_{S1} so that $V_{out} = 2.4$ V. (b) Compute V_{out} if the adaptor voltage is in fact 3.1 V.



Diode Models and Circuits

□ Large-Signal and Small-Signal Operation

Example 3.18

A diode is biased at a current of 1 mA. (a) Determine the current change if V_D changes by 1 mV. (b) Determine the voltage change if I_D changes by 10%.

Clippers and Clampers Overview

Clipper Circuits

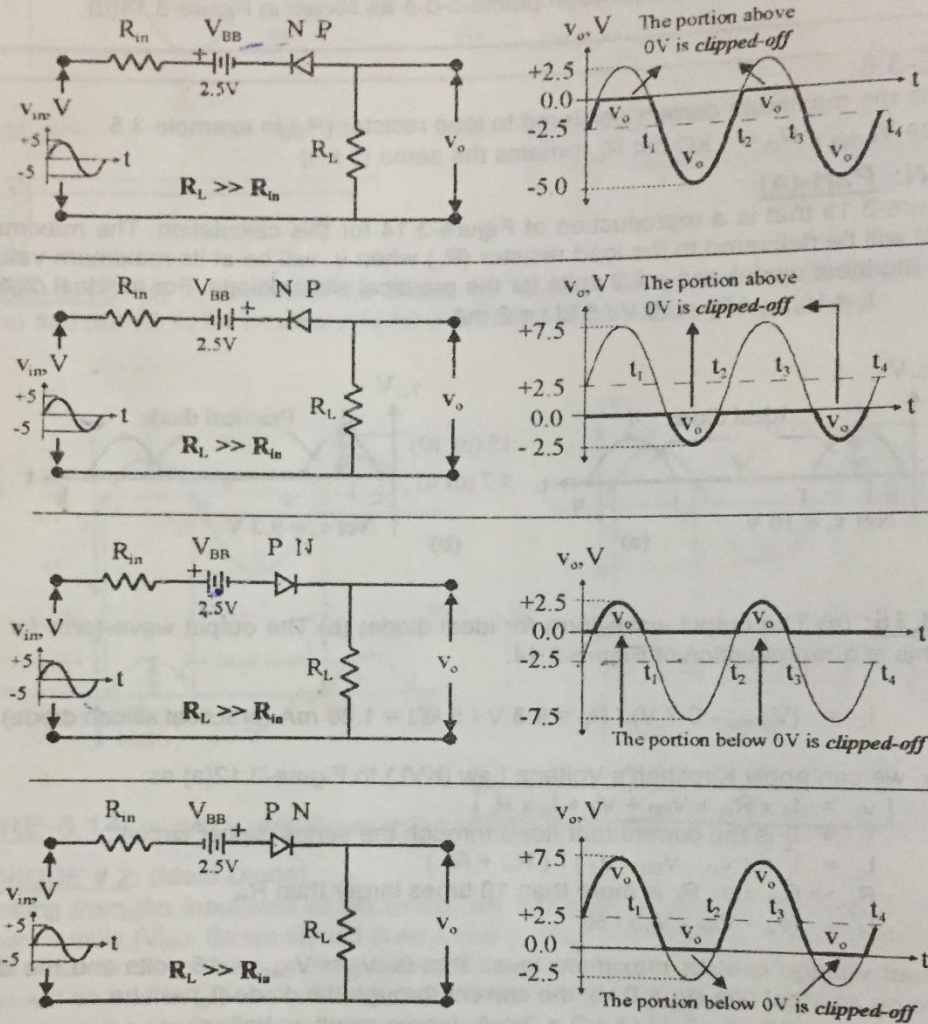


FIGURE-3.16: Summary of the series clipping circuits. The darker areas correspond to v_o that follows v_{in} when the diode is forward-biased. It stays at appropriate (such as $+2.5V$ etc.,) level when the diode is reverse-biased.

Design of Clipper Circuits

DESIGN EXAMPLE-3.1:

Design a clamper circuit that should be able to clamp the positive peak of a square wave input voltage to negative 5 volts. The peak to peak amplitude of the square wave is 20 volts at a frequency of 1 kHz. Assume that an ideal diode is used.

SOLUTION: Frequency of the square wave is 1 kHz, so $T = 1/f = 1/1 \text{ kHz} = 1 \text{ ms}$. The C is directly chosen as 10 nF or 0.01 μF (see equation-3.6). If we like we may choose C as 100 nF or any other available value keeping in view equation-3.6. From equation-3.6(a) R_L can be designed as under:

$$R_L = \frac{1.5 T}{C} = \frac{1.5 \times 1 \times 10^{-3}}{0.01 \times 10^{-6}} = 15 \text{ k}\Omega$$

From equation-3.6(b) R_{in} can, now, be designed as under:

$$R_{in} = \frac{T}{150 C} = \frac{1 \times 10^{-3}}{150 \times 0.01 \times 10^{-6}} = 66.67 \Omega$$

$$T = 1 \text{ ms}, C = 0.01 \mu\text{F}, R_L = 15 \text{ k}\Omega, \text{ and } R_{in} = 66.67 \Omega$$

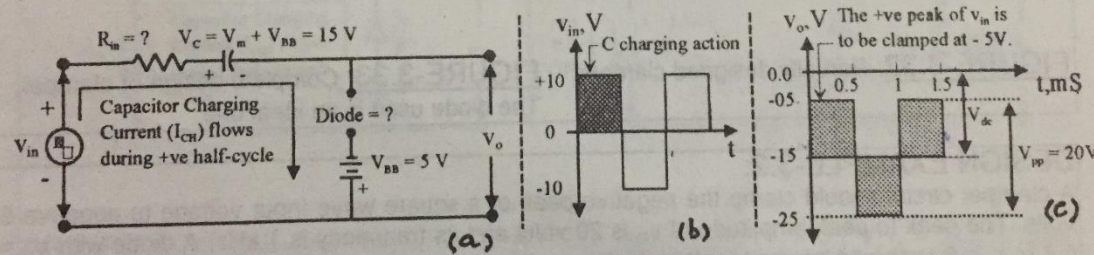


FIGURE-3.31: (a) Shows charging of C. Watch direction of I_{CH} ; (b) Input voltage wave form and (c) Output voltage wave form drawn from statement of the problem.

THE DIRECTION OF DIODE:

From design statement the positive peak of v_{in} should be clamped at negative 5 volts. For this to happen the positive peak of v_{in} must move from +10 V to -5 V, so that V_{dc} introduced in v_{in} will be -15 V. The wave forms of given v_{in} and would be v_o are sketched in Figure-3.31(b and c). To introduce V_{dc} of -15 V in output voltage means that after the charging action is complete, the net voltage on capacitor must be -15 volts. That is, $V_C = -15$ volts --- Looking from the output side as shown in Figure-3.31(a). If the capacitor is to charge with a negative voltage on its right hand plate (RHP), the charging current must flow into its left hand plate (LHP). That is, it must flow in the clock wise direction as clearly marked in the same figure. Now with this information we can figure out that the diode must be connected with its P-side up and N-side down. Finally to decide about the direction and magnitude of V_{BB} we must proceed as follows.

THE MAGNITUDE AND POLARITY OF V_{BB} :

For the capacitor to have 15 V with its LHP positive with respect to its RHP, we should have a net voltage of 15 V in the circuit. Since $v_{in, \max}$ can only be +10 V, so the LHP of the capacitor can be ideally charged to +10 V. From where we will get the extra 5 V. Of course, the remaining 5 V will be supplied by properly connecting V_{BB} in the circuit. This discussion reveals that magnitude of V_{BB} must be 5 V. Now its direction should be such as to charge the RHP of the capacitor to -5 V, so its must be connected with its negative terminal up as depicted in Figures-3.32 and 3.33. With this the design of the clamper is complete and is shown in Figure-3.33 with the designed values:

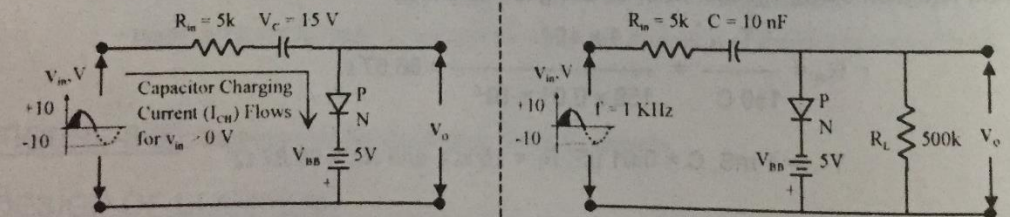


FIGURE-3.32: Partially designed clamper.

FIGURE-3.33: Complete design of clamper. The diode used is an ideal one.

Design of Clipper Circuits

DESIGN EXAMPLE-3.2:

A clamper circuit should clamp the negative peak of a square wave input voltage to negative 5 volts. The peak to peak amplitude of v_{in} is 20 volts and its frequency is 1 kHz. A diode with $V_d = 0.7$ V, $r_d = 0 \Omega$ should be used for this design.

SOLUTION:

The calculations of C , R_L and R_{in} will be the same as for design example-3.1 and are not repeated here. We should only try to ascertain the magnitude and polarity of V_{BB} and the direction in which P-side and N-sides of diode should be connected in the clamper circuit.

THE DIRECTION OF DIODE:

Since the negative peak is to be clamped at - 5 volts, so, v_{in} must move up by the same amount. Therefore, v_o must be as shown in Figure-3.34(b). Observe from this figure that $V_{dc} = + 5$ volts. This means the RHP of the capacitor must be at + 5 V. For this to happen the capacitor charging current (I_{CH}) must flow in the counter clock wise direction as indicated in Figure-3.34 (c). Hence the diode must be connected with its P-side pointing up as shown in Figure-3.35(a).

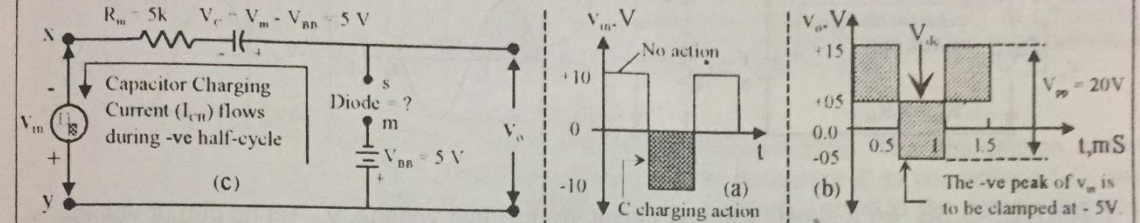


FIGURE-3.34: (a) v_{in} applied to the circuit; (b) The resulting v_o ; (c) Equivalent circuit valid during negative half cycle $T/2 < t < T$ only. It shows how C should be charged.

THE MAGNITUDE AND POLARITY OF V_{BB} :

Since we want to have only 5 V on capacitor, so $v_{in,max}$ and V_{BB} must be connected in opposite direction. This means $v_{in,max} - V_{BB}$ must be 5 V. Since $v_{in,max}$ is 10 V, so V_{BB} must be 5 V. The direction of V_{BB} should be as indicated in Figure-3.35(a) for ideal diode and Figure-3.35(b) for practical diode. Finally, it should be noted that V_{BB} must be 4.3 volts instead of 5 volts so as to allow a drop of 0.7 V for a practical silicon diode.

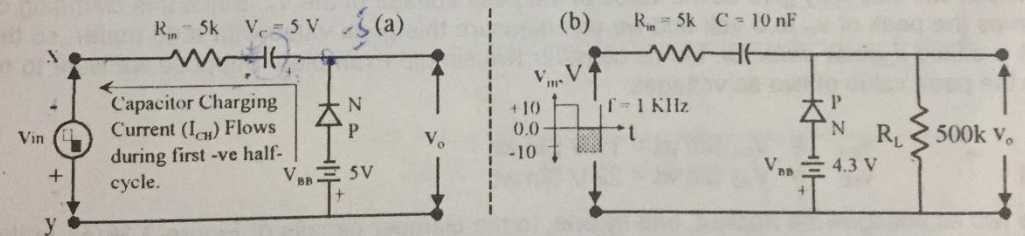


FIGURE-3.35: Complete design of clamper. The silicon diode ($V_d = 0.7$ V, $r_d = 0 \Omega$) is used.

Design of Clamper Circuits

DESIGN OF CLAMPERS

Design procedure of clamping circuits is the reverse of analysis procedure (thoroughly discussed above). A step by step design is explained below:

First the time period (T) should be determined from the frequency of the input voltage signal, $v_{in} = V_{max} \sin \omega t$, where: $\omega = 2 \pi f$ and $T = 1/f$. A suitable value of C is selected from a range of 1 nF to 100 μF . Usually a very small or a very large value of C is not suitable from practical point of view. In this textbook C will be chosen between 1 nF to 100 μF .

$$1 \text{ nF} < C < 100 \text{ } \mu F \quad 3.6$$

R_L can be chosen from equation-3.6(a) as explained below. First let us ponder over equation-3.3.

$$5 R_L C \gg T/2 \quad 3.3$$

From a practical point of view $5 R_L C$ should be 10 to 20 times larger than $T/2$. We, however, may choose a mean value of 15. $5 R_L C = (T/2) 15 = 7.5 T$.

$$R_L = 7.5 T / 5 C \quad 3.6(a)$$

R_{in} can be chosen from equation-3.6(b) as explained below. Consider equation-3.3(a) first as:

$$5 R_{in} C \ll T/2 \quad \dots\dots\dots 3.3(a)$$

The time constant $5 R_{in} C$ should be 10 to 20 times smaller than $T/2$. We may, again, choose a mean value of 15, therefore: $5 C R_{in} = (T/2) (1/15) = T/30$.

$$R_{in} = T / 150 C \quad 3.6(b)$$

The magnitude of dc power supply (battery, V_{BB}), is directly obtained from the information available from the design statement of the problem. It will be soon explained with the help of a few examples. Similarly direction of the diode (up or down) and the polarity of V_{BB} can be ascertained from the design statement and will be soon explained.

Voltage Multipliers – Voltage Doubler

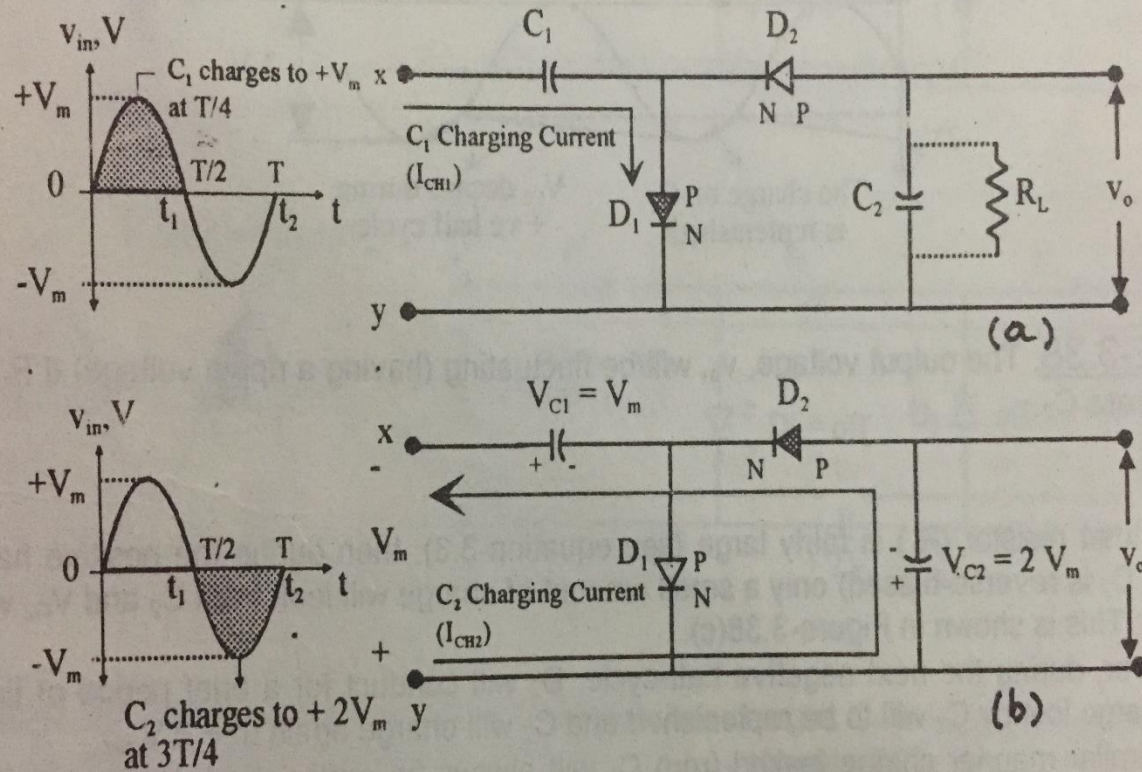


FIGURE-3.37: (a) C_1 charges during the positive half-cycle to V_{max} at $t = T/4$; (b) During the negative half-cycle D_1 remains OFF, V_{C1} and v_{in} are series additive. D_2 turns ON and C_2 charges to $2V_{max}$ with its polarity marked.

Voltage Multipliers – Voltage Tripler

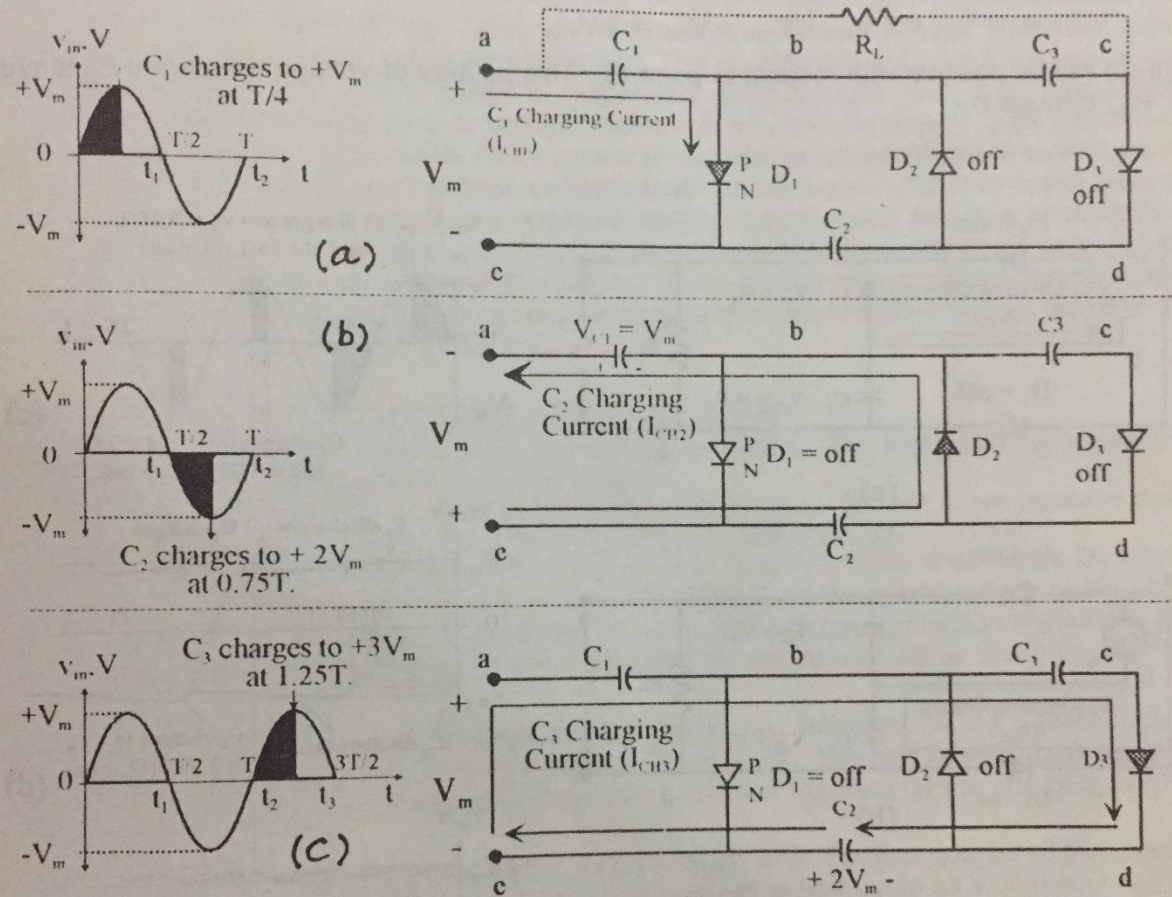


FIGURE-3.39: (a) The basic voltage-tripler circuit. I_{CH1} charges C_1 to V_{max} ; (b) I_{CH2} charges C_2 to $2 V_{max}$. D_1 and D_3 remain off, only D_2 conducts; (c) I_{CH3} charges C_3 to $2 V_{max}$ through D_3 . Diodes D_1 and D_2 remain off.

Rectifier Circuits Overview and Design

Rectifier Circuits

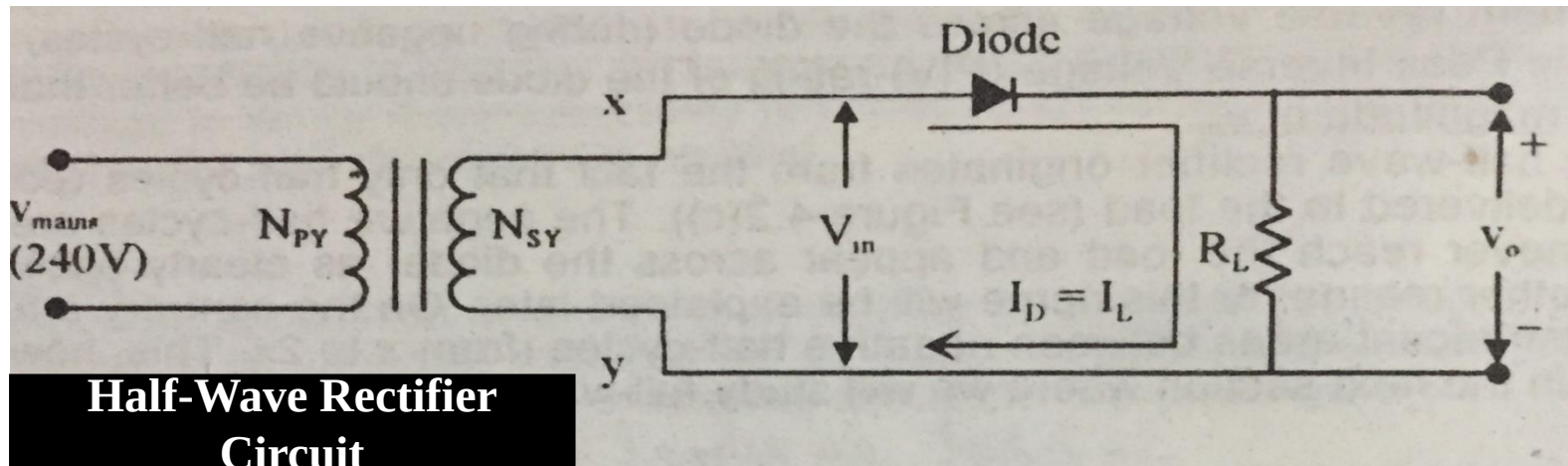
□Types of Rectifier Circuits

1. Half Wave (HW) Rectifiers
2. Full Wave (FW) Rectifiers
 - a) Center Tapped Rectifiers
 - b) Bridge Rectifiers

Half Wave Rectifiers

- Assuming ideal diode

- $V_{dc} = V_m / \pi = 0.318 V_m$
- $I_{dc} = V_{dc} / R_L = 0.318 V_m / R_L$
- $V_{rms} = V_m / 2$
- Efficiency = $P_{o(dc)} / P_{in(AC)} = 40.52\%$
- $V_{R,rms} = 0.385 V_m$
- Ripple Frequency = f_{in}
- Ripple Factor = Ripple RMS / Average Value = $V_{R,rms} / V_{dc} = 1.21$
- Peak Inverse Voltage Rating = PIV $> V_m$



Half Wave Rectifiers

□ Ripple Factor

- Ripple content is defined as the amount of AC content present in the output DC. If the ripple factor is less, the rectifier performance will be more.

$$\text{Ripple Factor} = \sqrt{\left(\frac{I_{RMS}}{I_{DC}}\right)^2 - 1}$$

Half Wave Rectifiers

□ How to calculate V_{DC} (or V_{AVG}) of the output voltage ?

Half Wave Rectifiers

□ How to calculate V_{RMS} of the output voltage ?

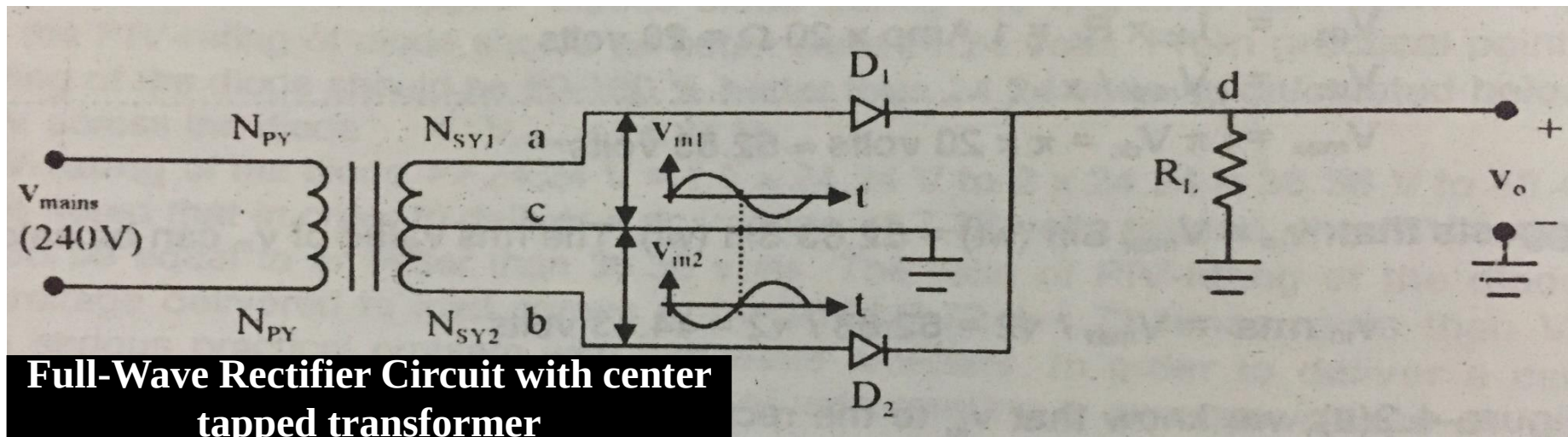
Half Wave Rectifiers

□ How to calculate efficiency of the rectifier ?

Full Wave Rectifier with Center Tapped Transformer

- Assuming ideal diode

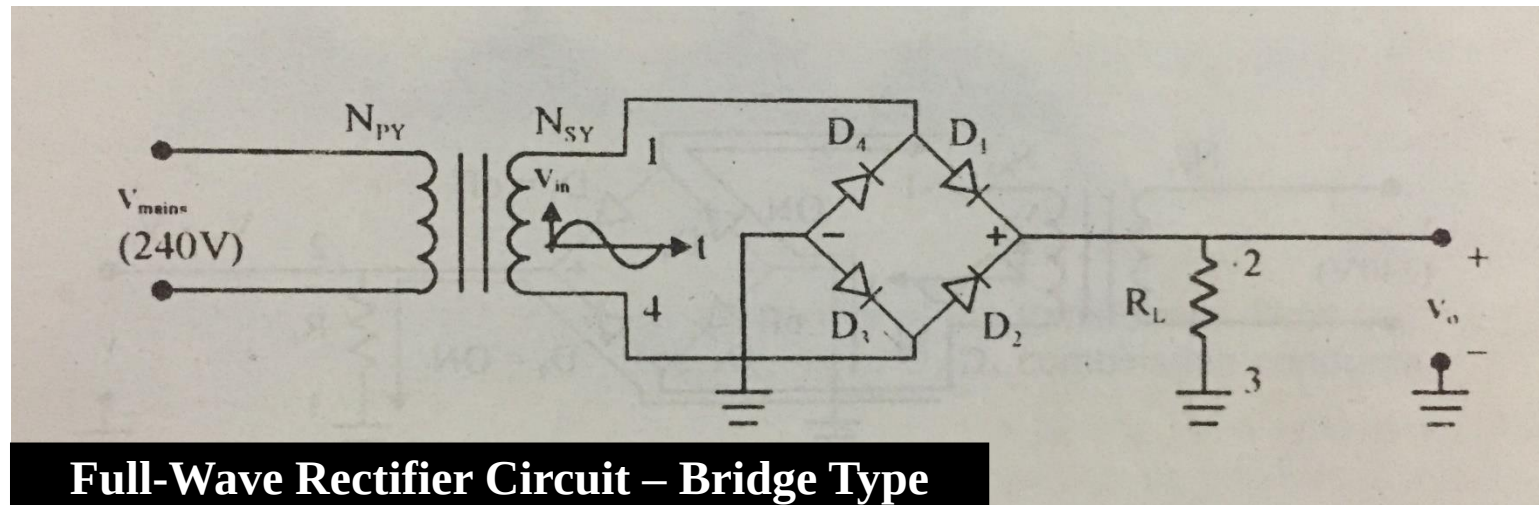
- $V_{dc} = 2V_m / \pi = 0.636 V_m$
- $I_{dc} = V_{dc} / R_L = 0.636 V_m / R_L$
- $V_{rms} = V_m / \sqrt{2} = 0.707 V_m$
- Efficiency = $P_{o(dc)} / P_{in(AC)} = 81.06\%$
- $V_{R,rms} = 0.308 V_m$
- Ripple Frequency = $2f_{in}$
- Ripple Factor = Ripple RMS / Average Value = $V_{R,rms} / V_{dc} = 0.484$
- Peak Inverse Voltage Rating = PIV $> 2V_m$



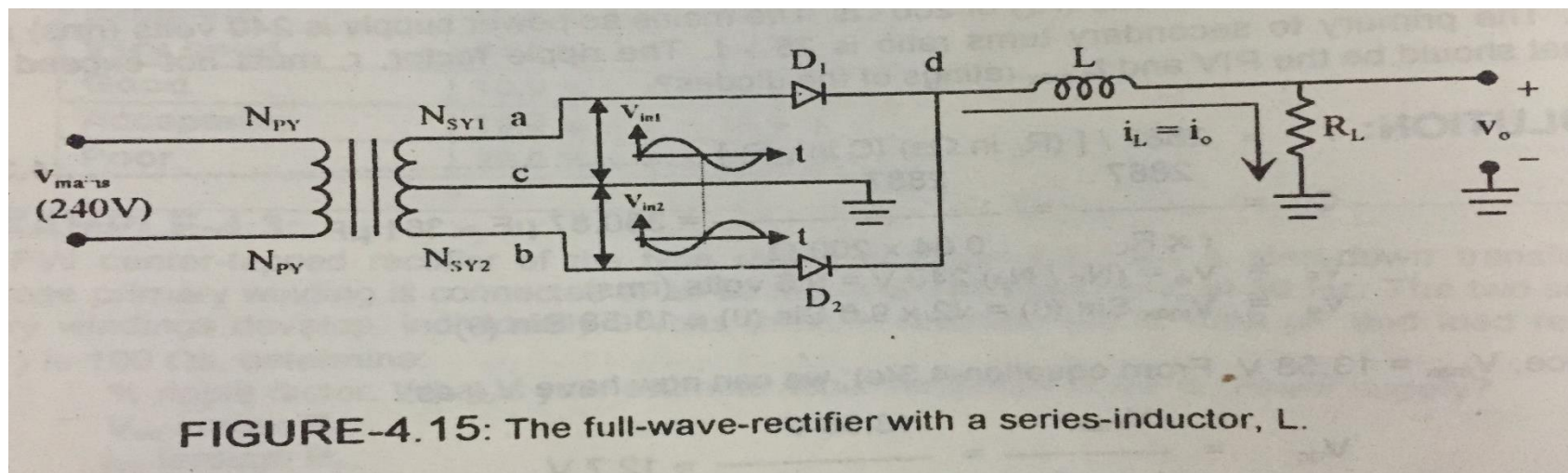
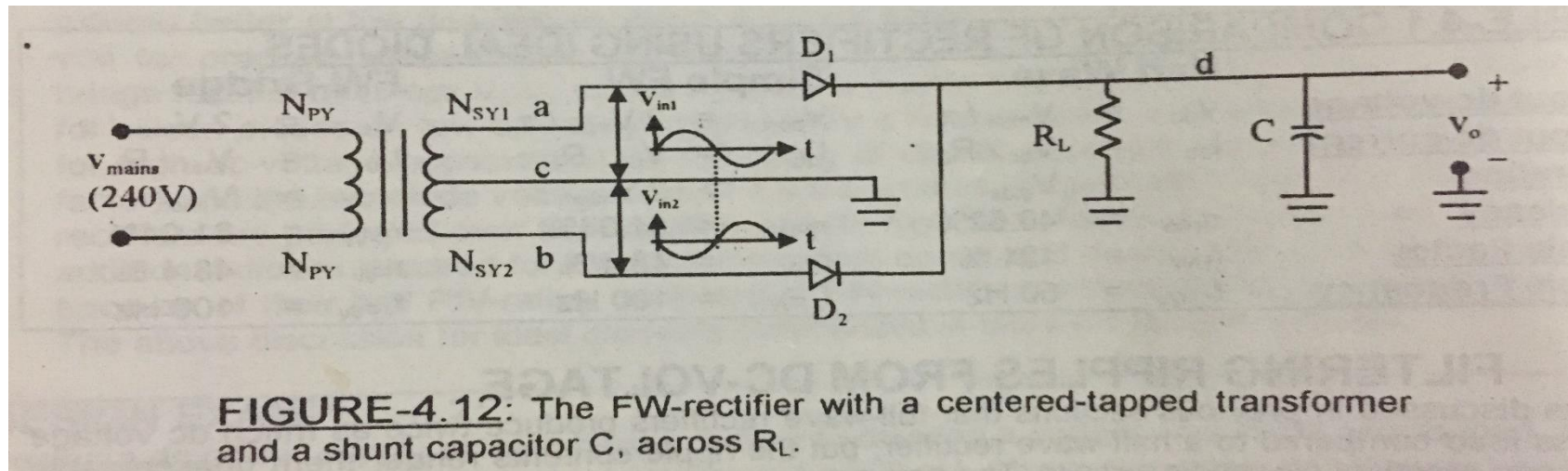
Full Wave Rectifier – Bridge Type

- Assuming ideal diode

- $V_{dc} = 2V_m / \pi = 0.636 V_m$
- $I_{dc} = V_{dc} / R_L = 0.636 V_m / R_L$
- $V_{rms} = V_m / \sqrt{2} = 0.707 V_m$
- Efficiency = $P_{o(dc)} / P_{in(AC)} = 81.06\%$
- $V_{R,rms} = 0.308 V_m$
- Ripple Frequency = $2f_{in}$
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- Peak Inverse Voltage Rating = PIV $> V_m$



Practical Full Wave Rectifiers with out Capacitors and Inductors



Design Example – Half Wave Rectifier

DESIGN EXAMPLE-4.1:

Design a half-wave rectifier circuit that should be capable of delivering 1 ampere of dc current to a load of $20\ \Omega$. Assume an ideal diode so that $V_d = 0\text{ V}$ and r_d is negligible. Specify the turns ratio for the step down transformer, the PIV-rating and the $I_{d,\max}$ rating of the silicon diode used. The mains ac voltage is 240 volts (rms) at 50 Hz.

SOLUTION:

First we can determine the dc voltage across the load resistor of $20\ \Omega$ when a current of 1 Ampere will flow through it. Obviously:

$$\begin{aligned} V_{dc} &= I_{dc} \times R_L = 1\text{ Amp} \times 20\ \Omega = 20\text{ volts} \\ V_{dc} &= V_{\max} / \pi \dots\dots\dots (4.1) \\ \therefore V_{\max} &= \pi V_{dc} = \pi \times 20\text{ volts} \approx 62.83\text{ volts} \end{aligned}$$

This suggests that: $v_{in} = V_{\max} \sin(\omega t) = 62.83 \sin(\omega t)$. The rms value of v_{in} can be calculated as:

$$v_{in,rms} = V_{\max} / \sqrt{2} = 62.83 / \sqrt{2} = 44.43\text{ volts}$$

From Figure-4.2(a), we know that v_{in} to the rectifier circuit is the same as the v_s (or secondary voltage of the transformer). Therefore, $v_{s,rms} = v_{in,rms} = 44.43\text{ volts}$.

Now we also know that: $N_p / N_s = v_p / v_s$ [where: v_p is given as 240 volts (rms)]. Hence, $N_p / N_s = 240\text{ V} / 44.43\text{ V} = 5.4 / 1$. The primary to secondary turns ratio of the transformer should be 5.4:1. The PIV-rating of the diode should be better than $V_{\max}$ (i.e., 62.83 volts). From practical point of view it should be 50 to 100 % more. Therefore, PIV-rating of the diode = $62.85 \times 1.5 = 94.25\text{ V}$ to $62.83 \times 2\text{ V} = 125.66\text{ V}$.

The maximum current, $I_{d,\max}$, through the diode will be: $I_{d,\max} = V_{\max} / R_L = 62.83\text{ V} / 20\ \Omega = 3.14\text{ amps}$. This means that the current carrying ability of the diode should be better than 3.14 amps. From practical point of view it should be from 50 to 100 % better. Therefore, $I_{d,\max}$ rating of the diode under discussion should be at least = $3.14 \times 1.5 = 4.71\text{ A}$ to $3.14 \times 2 = 6.28\text{ A}$.

EXERCISE 4.3:

Design a half-wave rectifier circuit that should be capable of delivering 500 mA of dc current to a load of $100\ \Omega$ s. The primary side of the transformer is connected to 240 volts (rms) at 50 Hz. Assume an ideal. Specify turns ratio of the transformer. Also specify the PIV-rating and the $I_{d,\max}$ ratings of the diode.

ANSWER: $N_p:N_s = 2.16:1$, $I_{d,\max}$ rating = 2.36 Amp minimum (3.14 Amp is better)

PIV-rating = 235.62 V minimum (314.16 V is better).

Design Example – Full Wave Rectifier with centered tapped Transformer

DESIGN EXAMPLE-4.3:

In a FW-rectifier with a center-tapped transformer, determine value of the shunt capacitor connected across a resistive load (R_L) of $200\ \Omega$ s. The mains ac power supply is 240 volts (rms) at 50 Hz. The primary to secondary turns ratio is 25 : 1. The ripple factor, r , must not exceed 4%. What should be the PIV and $I_{d,max}$ ratings of the diodes?

SOLUTION: $r \approx \frac{2887}{2887} / \left[(R_L \text{ in } \Omega\text{s}) (C \text{ in } \mu\text{F}) \right] \dots\dots\dots 4.3(c)$

$$C = \frac{2887}{r \times R_L} = \frac{2887}{0.04 \times 200\ \Omega} = 360.87\ \mu\text{F} \approx 361\ \mu\text{F}$$

$$V_s = V_{in} = (N_s / N_p) 240\ \text{V} = 9.6\ \text{volts (rms)}$$

$$V_o = V_{max} \sin(\theta) = \sqrt{2} \times 9.6 \sin(\theta) \approx 13.58 \sin(\theta)$$

Hence, $V_{max} = 13.58\ \text{V}$. From equation-4.3(e); we can now have V_{dc} as:

$$V_{dc} = \frac{V_{max}}{1 + r\sqrt{3}} = \frac{13.58\ \text{V}}{1 + 0.04\sqrt{3}} \approx 12.7\ \text{V}$$

DIODE RATINGS:

$$I_{dc,max} = V_{max} / R_L = 13.58\ \text{V} / 0.2\ \text{k}\Omega = 67.9\ \text{mA}$$

Rating $I_{dc,max} = (1.5 \text{ to } 2) 67.9\ \text{mA} = \text{at least } 101.85\ \text{mA or better } 135.8\ \text{mA}$

Rating $PIV = (1.5 \text{ to } 2) 2 V_{max} = (1.5 \text{ to } 2) 27.16\ \text{V} = 40.74\ \text{V or better } 54.32\ \text{V}$

Design Example – Full Wave Bridge Rectifier

DESIGN EXAMPLE-4.2:

Design a dc power supply using a bridge-rectifier to develop a certain dc-voltage across load, R_L of $15\ \Omega$. The dc-current through the load should be 500 mA. Use a step-down transformer whose primary side is connected to 240 volts (rms) at a mains frequency of 50 Hz. Specify PIV-rating, $I_{d,max}$ rating and the power-rating of the diodes. Consider a practical silicon diode

SOLUTION:

The first step is to find the primary to secondary turns ratio ($N_p : N_s$) of the step down transformer as explained below. From equation-4.2(a), we know that:

$$I_{dc} = 2 I_{o,max} / \pi = (2 V_{o,max} / \pi R_L) \dots\dots\dots 4.2(a)$$

$$\therefore I_{o,max} = I_{dc} (\pi / 2) = 500\text{ mA} (\pi / 2) \approx 785.4\text{ mA}$$

$$V_{o,max} = I_{o,max} R_L = 785.4\text{ mA} \times 15\ \Omega = 11.78\text{ V}$$

Maximum value of $v_s = V_{o,max} + V_d = 11.78 + 0.7\text{ V} = 12.48\text{ volts}$. Note that maximum amplitude of the voltage across the secondary of the transformer ($v_s = 12.48\text{ volts}$) differs from the maximum amplitude of the output voltage ($V_{o,max}$) by one diode drops (0.7 volts). Now the maximum amplitude of the ac mains voltage across the primary side of the transformer can be determined as:

$$V_{p,max} = \sqrt{2} \times 240\text{ volts} = 339.41\text{ volts}$$

$$\therefore N_p / N_s = V_{p,max} / V_{s,max} \\ = 339.41\text{ V} / 12.48\text{ V} \approx 27.2 : 1$$

RATINGS: From table-4.1 (taking $V_d = 0.7\text{ V}$ into account):

$$\text{PIV-rating} > V_{o,max} = 1.5 \text{ to } 2 (11.78\text{ V}) = 17.67\text{ V to } 23.56\text{ V}$$

$$I_{d,max}\text{-rating} > 785.4\text{ mA} > 1.5 \times \text{to } 2 (785.4\text{ A}) = 1.178\text{ A to } 1.57\text{ A}$$

$$P_{d,max}\text{-rating} = I_{d,max} \times V_d = 785.4\text{ mA} \times 0.7\text{ V} = 549.78\text{ mW} \approx 550\text{ mW}$$

Therefore, the $P_{d,max}$ -rating of the diode should be better than 550 mW. In the experience of the author the rating should be at least 50% better. So $P_{d,max}\text{-rating} = 1.5 \times 550 = 825\text{ mW}$. For better safety it should be 100% more than 550 mW, that is, 1.1 W.

Design Example – Full Wave Bridge Rectifier

DESIGN EXAMPLE-4.4:

A FW-rectifier shown in Figure-4.15 employs a series-inductor filter to provide 25 volts dc at 1 Ampere to the load resistor (R_L) with a ripple factor of 5%. The mains ac voltage is 240 V (rms) at a frequency of 50 Hz. Design the complete power supply. Assume ideal diodes and neglect the resistance of the inductor and the secondary winding of the transformer.

SOLUTION: FINDING TRANSFORMER TURNS RATIO:

A complete design would require to find R_L , L , PIV-rating, $I_{d,max}$ -rating of diodes and the transformer primary to secondary turns ratio N_p / N_s . From the given information we know that: $V_{dc} = 25$ volts, $I_{dc} = 1$ amp and $r = 5\% = 0.05$. Re-arranging equation-4.4(c), we can have the maximum voltage of the input V_{max} . After this we can determine the rms value of the input voltage ($V_{in,rms}$). This would be a prelude to find the turns ratio of the transformer. Let us do it now as under.

$$V_{max} = \pi V_{dc} / 2 = \pi 25 \text{ V} / 2 \approx 39.27 \text{ volts}$$

Therefore, the maximum amplitude of the input voltage (v_{in}) across one of the secondary windings will be 39.27 volts. This means that:

$$\begin{aligned} V_{in,rms} &= V_{max} / \sqrt{2} = 39.27 / \sqrt{2} = 27.77 \text{ V} \\ N_p / N_s &= 240 \text{ V} / 27.77 \text{ V} = 8.64 / 1 \end{aligned}$$

FINDING R_L AND L :: As $V_{dc} = I_{dc} \times R_L$

$$\therefore R_L = V_{dc} / I_{dc} = 25 \text{ V} / 1 \text{ A} = 25 \Omega$$

From equation-4.4(m) value of the inductor can be found as:

$$\begin{aligned} r &= R_L / (1333 L) \dots\dots\dots 4.4(m) \\ \text{or} \quad L &= R_L / (1333 r) = 25 \Omega / (1333 \times 0.05) = 0.375 \text{ H} = 375 \text{ mH} \end{aligned}$$

DIODE RATINGS: Diode ratings can be found from table-4.1 as:

$$\begin{aligned} \text{PIV across diodes} &= 2 V_{max} > 2 \times 39.27 \text{ V} = 78.54 \text{ V} \\ \text{or} \quad \text{PIV-rating of diodes} &\geq 1.5 \text{ to } 2 (78.54 \text{ V}) = 117.8 \text{ V (min) or } 157.08 \text{ V (better)} \\ I_{d,max} \text{ through diodes} &= V_{max} / R_L > 39.27 / 25 \Omega > 1.57 \text{ A} \\ I_{d,max} \text{ rating of diodes} &\geq 1.5 \text{ to } 2 (1.57 \text{ A}) = 2.36 \text{ A (min) or } 3.14 \text{ A (better)} \end{aligned}$$

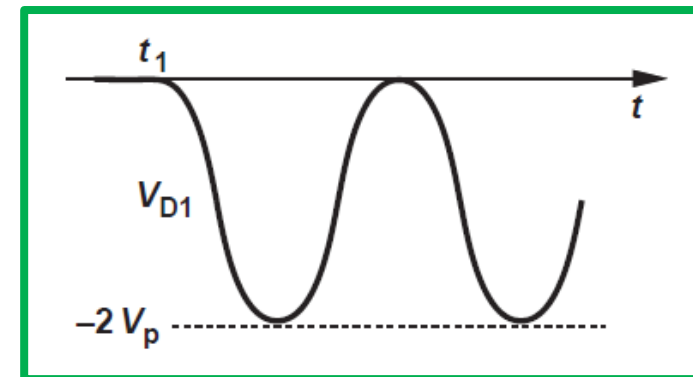
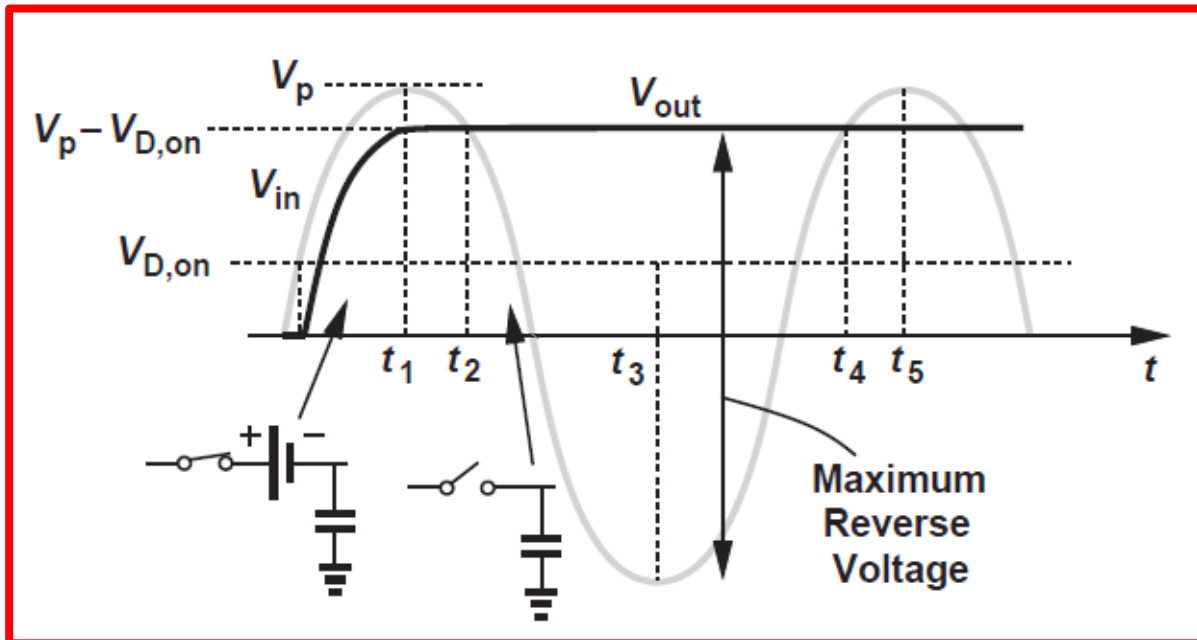
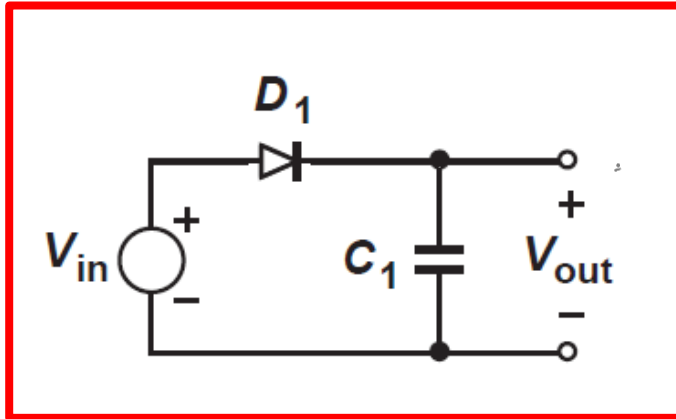
STUDENT EXERCISE-4.6:

Design a FW-rectifier (bridge-type) with a series inductor filter to produce a 12 volts dc to a certain load. The dc-current is 0.5 A with a ripple factor of 10 %. Neglect resistance of the diode, the secondary winding of the transformer and the inductor. Take v_{py} as 240 V_{rms} and V_d as 0 volt.

ANSWER: $N_p : N_s = 18 : 1$, $L = 180 \text{ mH}$, PIV-rating = 28.28 V (min) and 37.7 V (better), $I_{d,max}$ rating = 1.178 A (min) and 1.57 A (better).

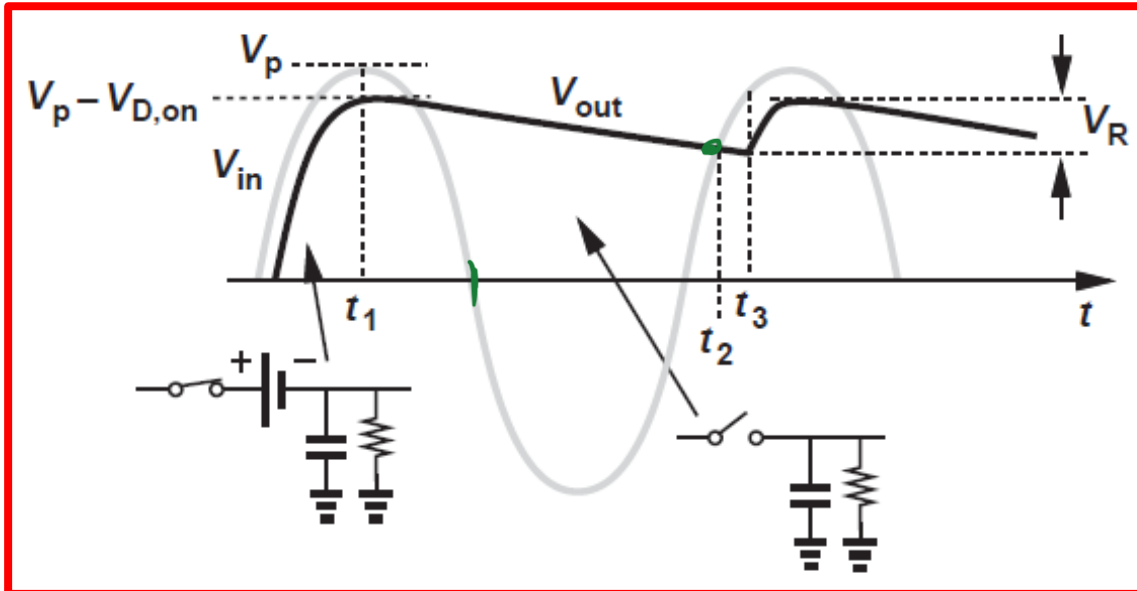
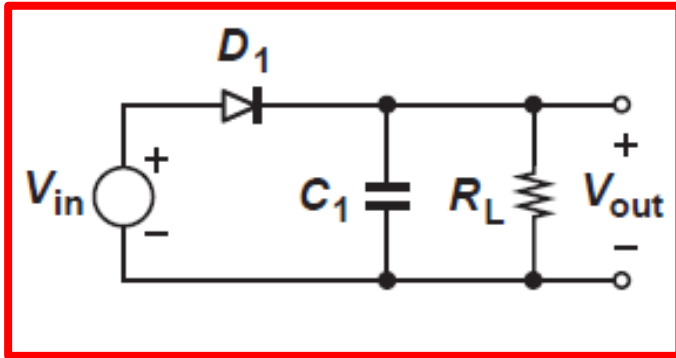
Rectifier Circuits

□ Diode Capacitor Circuit



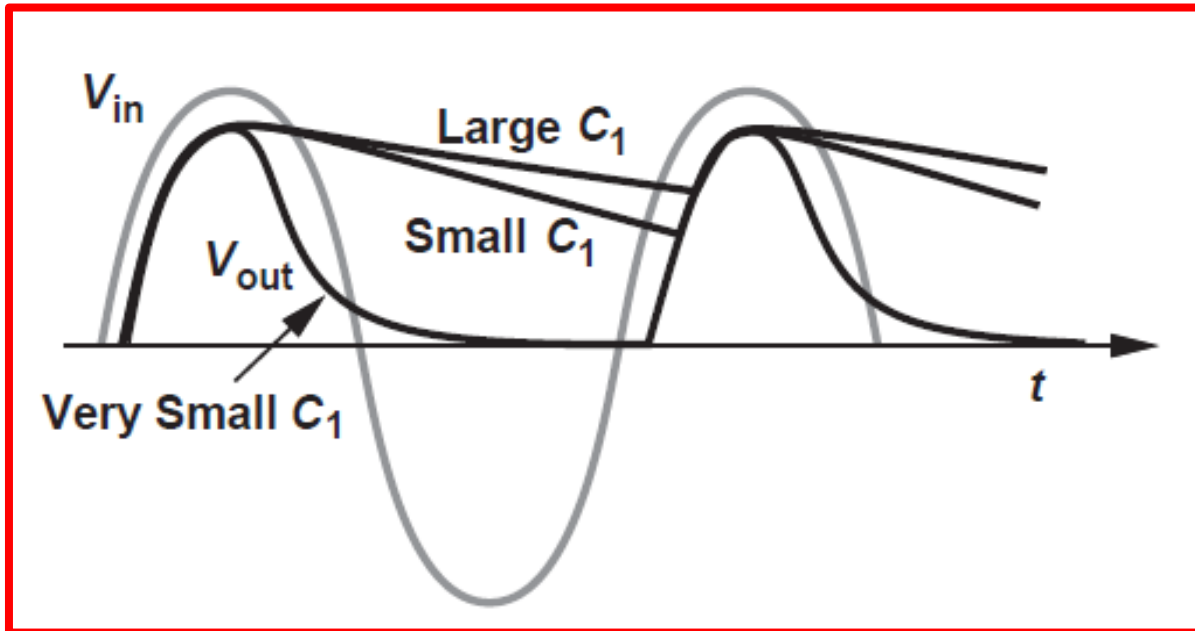
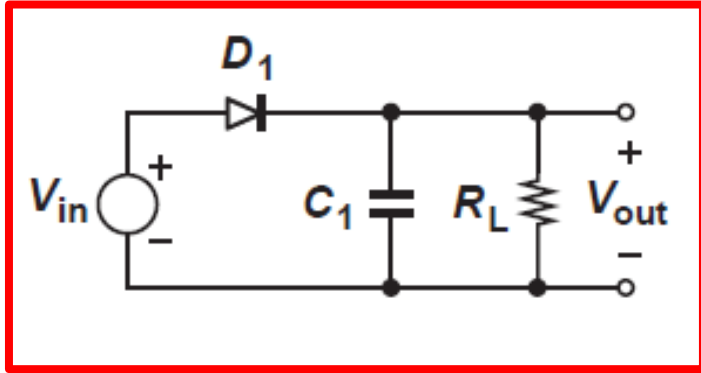
Rectifier Circuits

□ Diode Capacitor Circuit (Rectifier) driving a Load



Rectifier Circuits

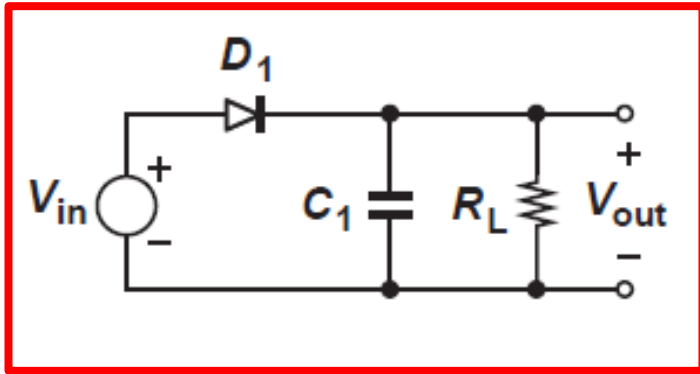
□ Output waveform of rectifier for different values of Capacitor



Rectifier Circuits

□ Ripple Amplitude

- In typical applications, the (peak-to-peak) amplitude of the ripple (V_R) must remain below 5 to 10% of the input peak voltage
- If the maximum current drawn by the load is known, the value of C_1 is chosen large enough to yield an acceptable ripple



$$V_R \approx \frac{V_p - V_{D,on}}{R_L} \cdot \frac{T_{in}}{C_1}$$
$$\approx \frac{V_p - V_{D,on}}{R_L C_1 f_{in}},$$

In many cases, the current drawn by the load is known. Repeating the above analysis with the load represented by a constant current source or simply viewing $(V_p - V_{D,on})/R_L$ in Eq. (3.80) as the load current, I_L , we can write

$$V_R = \frac{I_L}{C_1 f_{in}}. \quad (3.83)$$

Rectifier Circuits

□ Ripple Amplitude

Example 3.27

A transformer converts the 110-V, 60-Hz line voltage to a peak-to-peak swing of 9 V. A half-wave rectifier follows the transformer to supply the power to the laptop computer of Example 3.25. Determine the minimum value of the filter capacitor that maintains the ripple below 0.1 V. Assume $V_{D,on} = 0.8$ V.